Detailed Assessment Report

Assessing Loss and Damage from Extreme Rainfall *in* Kanchanpur District

May 2025

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Government of Nepal
Ministry of Home Affairs
National Disaster Risk Reduction and Management Authority
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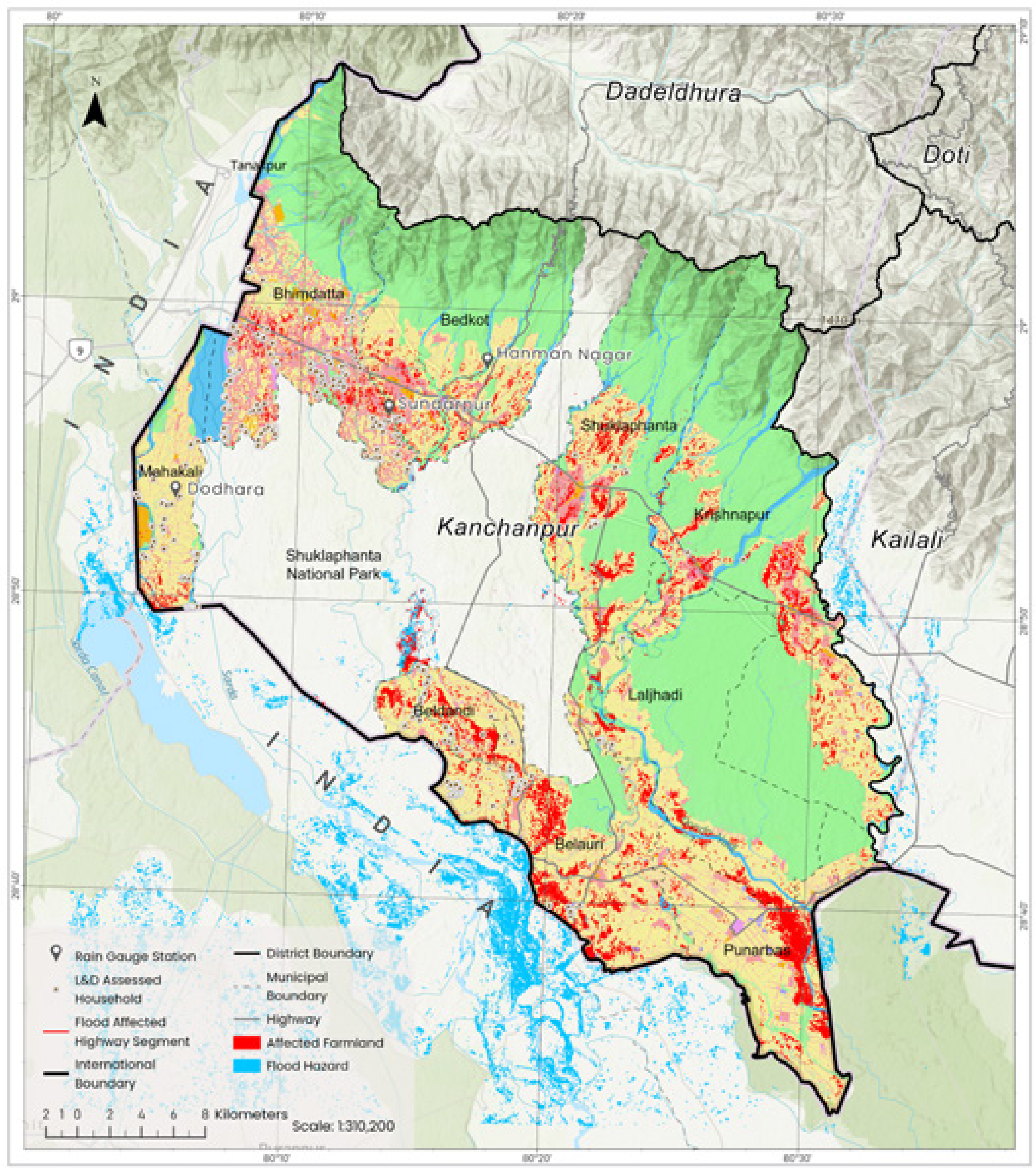

Detailed Loss and Damage Assessment Map: A Scenario of Kanchanpur Flood 2024

Description

Kanchanpur, in south-western Nepal, experienced extreme rainfall on July 7-8, 2024, with three stations recording unprecedented totals: Dodhara (624 mm), Hanuman Nagar (573.6 mm), and Sundarpur (556.4 mm). This event surpassed previous records of 516.2 mm in Hetauda (2017) and 540.0 mm in Tistung (1993).

This map depicts the flooded areas and impacts of Kanchanpur, estimated through Sentinel-1 imagery analysis on July 8, 2024, following the heavy rainfall.

Impact on Kanchanpur District

Total Inundated Area: 97.9 Sq.Km
Field Surveyed Household: 670
Affected Houses: 14, 315
Affected Agricultural Land: 85.27 Sq.Km
Affected National Highway: 165km

Data Source

Pre Images: Sentinel 1, 21 May 2024
Post Images: Sentinel 1, 08 July 2024
Land Use Map: ICIMOD 2020
Building Footprints: OpenStreetMap
Boundary: Survey Department, Nepal
Highway: Department of Road, Nepal
Precipitation: DHM, Nepal

Geospatial Analyst: Er. Prakash Gharti Magar
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Projected Coordinate System: LCC Nepal
Geographic Coordinate System: Everest 1830

BaseMaps: Land Use Zone Map (Survey Department, Nepal) Esri, NASA, NGA, USGS, Esri, © OpenStreetMap contributors, TomTom, Garmin, Foursquare, FAO, METI/NASA, USGS

Executive Summary

The Kanchanpur District in Nepal faced unprecedented devastation due to the extreme flooding event of 7-8 July 2024. Triggered by record-breaking rainfall, this disaster led to the displacement of thousands, extensive infrastructure damage, and severe economic and non-economic losses. The Mahakali and Chaudhar rivers breached their embankments, inundating farmlands, settlements, and essential services. The flood directly affected over 20,000 households, with 1,109 homes completely destroyed and 8,026 partially damaged, while causing long-term disruptions to health, education, livelihoods, and community well-being.

This detailed loss and damage assessment was conducted to comprehensively evaluate both economic and non-economic losses, integrating quantitative surveys from 670 affected households and qualitative data from community consultations. The assessment focused on property damage, livelihood and agricultural losses, livestock impacts, financial strain due to loans, and the non-economic dimensions such as mental health deterioration, social displacement, cultural heritage losses, and environmental degradation. The findings provide valuable insights into the challenges and disparities faced by different municipalities, highlighting the urgent need for inclusive disaster response frameworks that address the full spectrum of flood impacts.

Findings reveal severe disparities among municipalities in terms of reported losses and recovery support. Urban centers such as Bheemdatt and Belauri experienced the highest economic damages, particularly in lost business income and property damage. In contrast, rural areas such as Beldandi and Shuklaphanta reported lower financial losses but suffered from limited access to healthcare, psychological support, and emergency relief. The flood exacerbated pre-existing vulnerabilities, leaving already marginalized communities struggling to recover.

Compensation and external financial assistance were unevenly distributed, with Bheemdatt receiving NPR 24.41 million, while municipalities like Krishnapur and Shuklaphanta had significantly lower aid access. The reliance on loans for recovery exacerbated financial vulnerability, with many families borrowing at high-interest rates, prolonging their economic instability. Households were left financially burdened, often resorting to informal borrowing with unsustainable repayment conditions.

The non-economic losses were equally devastating, with 444 households receiving no psychological support despite experiencing mental distress. School attendance was severely affected, with 82 households reporting more than 15 days of missed education. Damage to religious and cultural heritage further impacted social cohesion, particularly in Bheemdatt, where 40 temples were affected. The loss of natural resources, soil erosion, and land degradation in 410 households highlighted the long-term environmental consequences, further endangering agriculture-dependent livelihoods.

Early warning systems played a crucial role in reducing losses and damages by providing advance notice of rising water levels and enabling timely evacuations. Households that received early flood alerts were able to relocate to safer areas, protecting lives and minimizing asset losses. However, unequal access to early warning systems resulted in disproportionate impacts, with rural areas receiving fewer timely updates. The effectiveness of early response measures was evident in Bheemdatt and Belauri, where a higher percentage of residents had access to flood forecasts, compared to remote municipalities like Laljhadi and Beldandi, where warnings were often delayed or inaccessible.

Despite these benefits, gaps in communication and infrastructure limitations hindered the full potential of early warning systems. Some households reported confusion regarding evacuation orders, while others faced logistical challenges in reaching designated shelters. This underscores the need for community-based early warning dissemination strategies, where local leaders, volunteers, and digital platforms work together to ensure that information reaches all at-risk populations, particularly vulnerable and isolated groups.

The study underscores the critical need for integrating non-economic loss and damage (NELD) into disaster risk reduction frameworks, ensuring that policies and interventions address the full spectrum of flood impacts. Without recognizing mental health, cultural disruptions, and environmental losses, disaster response efforts will remain incomplete. The following sections outline the key abbreviations, conclusions, way forward, and strategic recommendations to build resilience in Kanchanpur District.

OVERALL ECONOMIC LOSS IN NPR

NPR 2,00,25,12,211

OVERALL ECONOMIC LOSS IN USD

$ 14.35 Million

HUMAN CASUALTIES

DEATH

5
Total Death

INJURED

2
Total Injured

HOUSING AND HUMAN SETTLEMENTS

1,109
Fully Destroyed

8,026
Partially Destroyed

30,547
Affected Families

994
Displaced Families

AGRICULTURE

30,463 Hectare
Affected Areas

NPR 80,59,83,907
Standing Crops Estimated Loss

NPR 2,41,30,696
Stored Grains

NPR 83,01,14,603
Total Estimated Loss

PHYSICAL INFRASTRUCTURE

BUILDINGS

1,891
Affected

NPR 5,02,90,225
Estimated Loss

SCHOOLS

7
Affected

NPR 3,40,00,000
Estimated Loss

IRRIGATION PROJECTS

4
Affected

NPR 34,49,05,800
Estimated Loss

ROADS/BRIDGE/CULVERT

37
Affected

NPR 6,88,70,000
Estimated Loss

EMBANKMENT

NPR 54,62,58,983
Estimated Loss

TOTAL ESTIMATED LOSS

NPR 1,04,93,25,008

LIVESTOCK

2,48,023
Number Affected

NPR 4,48,71,600
Total Estimated Loss

INDUSTRIES

NPR 7,82,01,000
Total Estimated Loss

The data in this report was produced by the National Disaster Risk Reduction and Management Authority (NDRRMA) in collaboration with sectoral ministries, DAO Office Kanchanpur, line agencies and division offices Kanchanpur and provincial and local governments as of December 2024. The data in this report reflects information provided by the municipalities and does not comprehensively cover all nine municipalities of Kanchanpur District due to gaps in data collection and reporting.

HOUSHOLD ECONOMIC LOSS AND DAMAGE ANALYSIS

TOTAL ESTIMATED ECONOMIC LOSS

NPR 14,10,88,280

The overall financial impact of the flood, including property damage, livelihood disruptions, and agricultural losses.

TOTAL EXTERNAL FINANCIAL ASSISTANCE (INSURANCE AND COMPENSATION) RECEIVED

NPR 8,10,800

The total financial aid received from insurance claims and compensation—covering only a small fraction of the total economic loss.

TOTAL LOANS AND DEBTS TAKEN

NPR 21,70,000

The total amount borrowed by affected households and businesses to recover and rebuild after the disaster.

PROPERTY DAMAGE

50.30%
Partial Damage
Half of the households surveyed reported partial damage to their properties due to the flood.

49.70%
Full Damage
Nearly half of the respondents suffered complete destruction of their homes, making recovery extremely challenging.

93.58%
Household Item Damaged or Lost
A vast majority of respondents reported losing essential household items, adding to the financial and emotional burden.

31.19%
Household Infrastructure Damage
Almost one-third of the surveyed households reported damage to structural elements such as walls, roofs, and foundations.

LIVELIHOOD AND INCOME LOSS

48.66%
Disruption in Household Income
Nearly half of the affected families experienced disruptions in their primary sources of income due to the flood.

5.82%
Damage in Income Generating Assets
Some households reported damage to tools, equipment, or assets essential for their income generation.

9.25%
Loss of Savings or Financial Resources
A significant number of respondents reported depletion of their savings, reducing their ability to recover financially.

AGRICULTURAL LOSS

72.54%
Crops Destroyed
Most affected households reported severe crop losses, leading to food insecurity and income reduction.

17.31%
Agricultural Infrastructure Damage
Farmers faced additional hardships as essential agricultural infrastructure, such as irrigation systems and storage facilities, was damaged.

LIVESTOCK LOSS

33.58%
Livestock Lost
Over a third of the respondents reported the loss of livestock, impacting their food sources and income.

31.19%
Damage in Livestock Facilities
Livestock shelters and barns were damaged, making it difficult to protect and sustain remaining animals.

32.39%
Damage to Any Tools, Machinery or Vehicles
Flooding damaged essential equipment, vehicles, and tools necessary for livelihoods and recovery.

INSURANCE AND COMPENSATION

No Insurance Coverage for Flood Related Damage
None of the surveyed households had insurance coverage for flood-related damages, leaving them vulnerable to financial strain.

90%
No External Financial Assistance Received
A vast majority of affected individuals did not receive any external financial support or aid for recovery.

44.33%
Loans and Debt Taken to Recover
Nearly half of the respondents had to take out loans or incur debt to rebuild their lives after the disaster.

OTHER ECONOMIC LOSS

10.75%
Disruption in Business Operation
Small businesses faced interruptions, leading to income loss and long-term economic instability.

5.82%
Rental Income Lost
Some landlords suffered losses due to damage to rental properties and displaced tenants.

8.66%
Additional Economic Loss (Not Covered)
Several households reported other unforeseen financial burdens not captured in standard assessments.

HOUSHOLD NON-ECONOMIC LOSS AND DAMAGE ANALYSIS

TOTAL ESTIMATED NON-ECONOMIC LOSS

NPR 35,66,960

The total estimated non-economic loss reported after the flooding event, highlighting the broader societal and personal impacts beyond direct financial damage.

HEALTH IMPACT

51.49%
Post Flood Felt Sick Households
Over half reported illness due to poor sanitation and water contamination.

71.79%
Challenges Faced in Access to Health Services
Many struggled to access healthcare due to damaged infrastructure and financial issues besides that 49 households reported average access to services.

PSYCHOLOGICAL IMPACT

88.21%
Mental Stress or Trauma Experienced
Most households faced mental stress due to financial loss and displacement.

75.13%
Psychological Support Not Received
Three quarters received no mental health support post-flood.

CULTURAL IMPACT

3.43%
Cultural/Religious Items Lost
Some lost religious artifacts and family heirlooms.

6.27%
Cultural Sites Affected
Damage to community temples and cultural landmarks.

EDUCATIONAL IMPACT

68.96%
Unable to Attend School After Flood
Schools were closed or inaccessible due to flood damage for upto 15 days on average.

SOCIAL CONNECTIVITY

12.39%
Contact Lost Due to Displacement
Some lost contact with family and community members.

ENVIRONMENTAL IMPACT

47.61%
Natural Resources Affected
Nearly half reported damage to forests, rivers, and farmland.

33.58%
Noticeable Change in Quality of Water Sources
Water sources became muddy and contaminated.

61.19%
Soil Erosion and Land Degradation
Widespread land erosion reduced agricultural productivity.

25.97%
Local Wildlife or Biodiversity Impact
Loss of wildlife and damage to natural habitats.

MOBILITY AND TRANSPORTATION IMPACT

65.97%
Disruption in Access to Transportation
Travel was affected due to damaged roads and bridges for upto 8 days on average.

64.33%
Damage to Roads, Bridges or Paths
Infrastructure damage led to travel delays and higher travel costs.

24.18%
Damage to Personal Vehicle
Nearly a quarter reported damage to personal vehicles.

COMMUNITY COHESION & SUPPORT IMPACT

11.04%
Social Cohesion Affected in the Community
The flood weakened community ties.

20.31%
Created Conflicts
Disputes arose over aid distribution and resources.

93.13%
No Organized Community Effort
Most respondents saw no coordinated recovery efforts.

SOCIAL DISPLACEMENT

34.03%
Household Member Displaced Due to Flood
Many families had to temporarily relocate due to flood damage.

34.03%
Displacement of Other Community Member
One quarter of the household reported that their community members had to temporarily relocate due to flood damage.

43.28%
Challenges Faced Due to Migration or Relocation
Finding safe housing and jobs was difficult.

81.64%
Felt stressed and feared from the potential future flood Sense of Safety and Security Affected
Most felt less safe after the flood.

ACCESS TO ESSENTIAL SERVICES & GOODS

99.85%
Affected in Access to Essential Needs
Nearly all struggled to get food, water, and healthcare .

99.40%
Future Coping Mechanisms to Address or Reduce Challenges
Most stressed the need for better disaster preparedness.

74.78%
Disruption in Schooling for Children and Youth
School closures affected children's education upto 15 days on average.

21.64%
Household Response on Non-Economic Loss
Reports of community trust loss, economic insecurity, and distress.

COMMUNITY WELL-BEING PROGRAMS

75.22%
No immediate Post Flood Support Provided by Local Organizations
Three quarters received no immediate post-flood organizational aid.

Abbreviation

BIPAD	Building Information Platform Against Disaster
DAO	District Administration Office
DEOC	District Emergency Operation Center
DHM	Department of Hydrology and Meteorology
EC	Executive Council
ELD	Economic Loss and Damage
EWS	Early Warning System
FGD	Focus Group Discussion
KIIs	Key Informant Interviews
LDA	Loss and Damage Assessment
LEOC	Local Emergency Operation Center
MCN	Mercy Corps Nepal
MEWRI	Ministry of Energy, Water Resources, and Irrigation
MM	Millimeter
MoAL	Ministry of Agriculture and Livestock
MoFE	Ministry of Forests and Environment
MoHA	Ministry of Home Affairs
MoPID	Ministry of Physical Infrastructure Development
MoUD	Ministry of Urban Development
NDRRMA	National Disaster Risk Reduction and Management Authority
NEEDS	National Environment and Equity Development Society Nepal Together for Prosperity
NELD	Non-Economic Loss and Damage
NEOC	National Emergency Operation Center
NNSWA	Nepal National Social Welfare Association
NPR	Nepalese Rupees
PEOC	Provincial Emergency Operation Center
YI-Lab	Youth Innovation Lab

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> On 8 July 2024, Dodhara Chandani recorded a record-breaking 624mm (2.05 ft) of rainfall in 24 hours, the highest ever in Nepal, according to the Department of Hydrology and Meteorology (DHM)"

Introduction

Nepal's Sudurpaschim Province is highly susceptible to disasters, particularly during the monsoon season, due to its mountainous terrain and complex river systems. Among the districts in this region, Kanchanpur stands out for its vulnerability to flooding and landslides. In 2024, this vulnerability was starkly highlighted when Kanchanpur, especially Dodhara Chandani Municipality, faced one of the most severe flood events in its history due to record-breaking rainfall.

On 8 July 2024 Dodhara Chandani recorded an unprecedented 624mm (about 2.05 ft) of rainfall within 24 hours, the highest ever recorded in Nepal. Likewise, Sundarpur recorded 555.8 mm, and Hanumannagar with 573.6 mm (about 1.88 ft) of rainfall. The reports from Department of Hydrology and Meteorology (DHM) and National Disaster Risk Reduction and Management Authority (NDRRMA) concluded the event as climate induced disaster.

The torrential rain caused the Mahakali River to overflow, leading to severe flooding across large parts of the district, mostly in all municipalities of the Kanchanpur District and 3 municipalities of Kanchanpur district. Key rivers like Mahakali and Chaudhar breached their embankments, submerging farmlands, homes, and critical infrastructure. The flood impacted over 20,000 households, displacing families, damaging livelihoods, and disrupting essential services such as health, education, and transportation. Preliminary assessments indicate that the installed early warning system functioned effectively, enabling timely evacuations and saving numerous lives. However, despite this success, significant losses and damages were still encountered, revealing critical gaps in overall preparedness and disaster response mechanisms. The rapid rise in water levels forced more than 7000 residents to evacuate their homes, resulting in massive displacement, extensive property damage, and substantial losses in agriculture and infrastructure in 9 municipalities of Kanchanpur District and three municipalities of Kailali district. Among all municipalities, Dodhara Chandani has been shaken by the massive impacts from the flooding and heavy rainfall. The communities are recurrently affected by the disaster despite taking flood mitigation and adaptation measures.

As emergency repairs and relief measures are mobilized, it is essential to understand the full extent of the loss and damage caused by this natural disaster. The findings of this detailed assessment report will inform recovery needs and enhance preparedness for future climate-related events.

Objectives

Building upon the preliminary assessment conducted by NDRRMA guided by a mandate from the National Risk Reduction and Management Executive Council (EC) on August 5, 2024, and led by the Joint-Secretary of the National Disaster Risk Reduction and Management Authority (NDRRMA) of the July 2024 Kanchanpur flood, the detailed loss and damage assessment aims to provide a more systematic, scientific, and comprehensive evaluation of the disaster's impacts. These objectives reflect a continuation and expansion of the goals established in the preliminary report while addressing identified data gaps and refining methodologies for future resilience-building efforts:

1. Systematic and Scientific Loss and Damage Assessment:
To conduct a thorough assessment of the economic and non-economic losses caused by the flood, with a focus on both direct and indirect impacts across multiple sectors. This includes identifying the extent of underreporting and addressing inconsistencies in municipal and agency-level reporting.

2. Data Compilation and Validation:
To compile and validate loss and damage data from municipalities, line agencies, and other stakeholders, ensuring accuracy and reliability for effective policy-making and planning.

3. Toolkit Development and Refinement:
To utilize and further refine the pilot toolkit designed for assessing household-level economic and non-economic losses. The toolkit aims to address previously identified data gaps, ensuring a comprehensive capture of flood impacts at the community level.

4. Institutional Capacity Building and Strategic Recommendations:
To develop actionable recommendations for mitigation, preparedness, and adaptation measures. These will be provided to relevant line agencies and authorities to enhance resilience and reduce future loss and damage caused by such disasters.

5. Gap Analysis and Reporting Improvements:
To identify and analyze gaps in current reporting mechanisms, methodologies, and databases, providing strategies to standardize and improve disaster data collection and management systems.

Methodological Approach

This detailed assessment builds upon the structured and multi-sectoral methodology employed during the preliminary assessment of the July 2024 flooding in Kanchanpur district. The approach has been designed to capture a comprehensive picture of the economic and non-economic losses while incorporating the human and institutional dimensions of the flood's impact.

The preliminary assessment was guided by a mandate from the National Risk Reduction and Management Executive Council (EC) on August 5, 2024, and led by the Joint-Secretary of the National Disaster Risk Reduction and Management Authority (NDRRMA). A dedicated committee was formed, comprising representatives from key ministries, including the Ministry of Energy, Water Resources, and Irrigation (MEWRI); Ministry of Agriculture and Livestock (MoAL); Ministry of Urban Development (MoUD); Ministry of Forests and Environment (MOFE); and the Ministry of Physical Infrastructure Development (MoPID) of Sudurpashchim Province. The committee was supported by technical experts from NDRRMA, as well as development partners such as Youth Innovation Lab (YI-Lab) and Mercy Corps Nepal (MCN).

Data collection during the preliminary phase involved a collaborative effort with provincial and district administrations, leveraging datasets from nine severely affected municipalities: Dodhara Chandani, Bhimdutta, Bedkot, Belauri, Beldandi, Krishnapur, Laljhadi, Punarbas, and Shuklaphanta. Key data sources included agencies like the Mahakali Irrigation Office, Road Division Mahendranagar, Water Resource and Irrigation Development Division, and others. These datasets covered various sectors, including Physical Infrastructure, Agriculture and Livestock, and Cottage and Small Industries, providing a foundation for the subsequent detailed assessment.

Building on this foundation, the detailed assessment employed a mixed methods approach to ensure a holistic understanding of the flood's impacts. The study employed a mixed-methods approach to capture both quantitative and qualitative data. Household surveys were conducted across flood-affected municipalities to assess economic and non-economic losses and damages.

Methodological Approach for Detailed Assessment

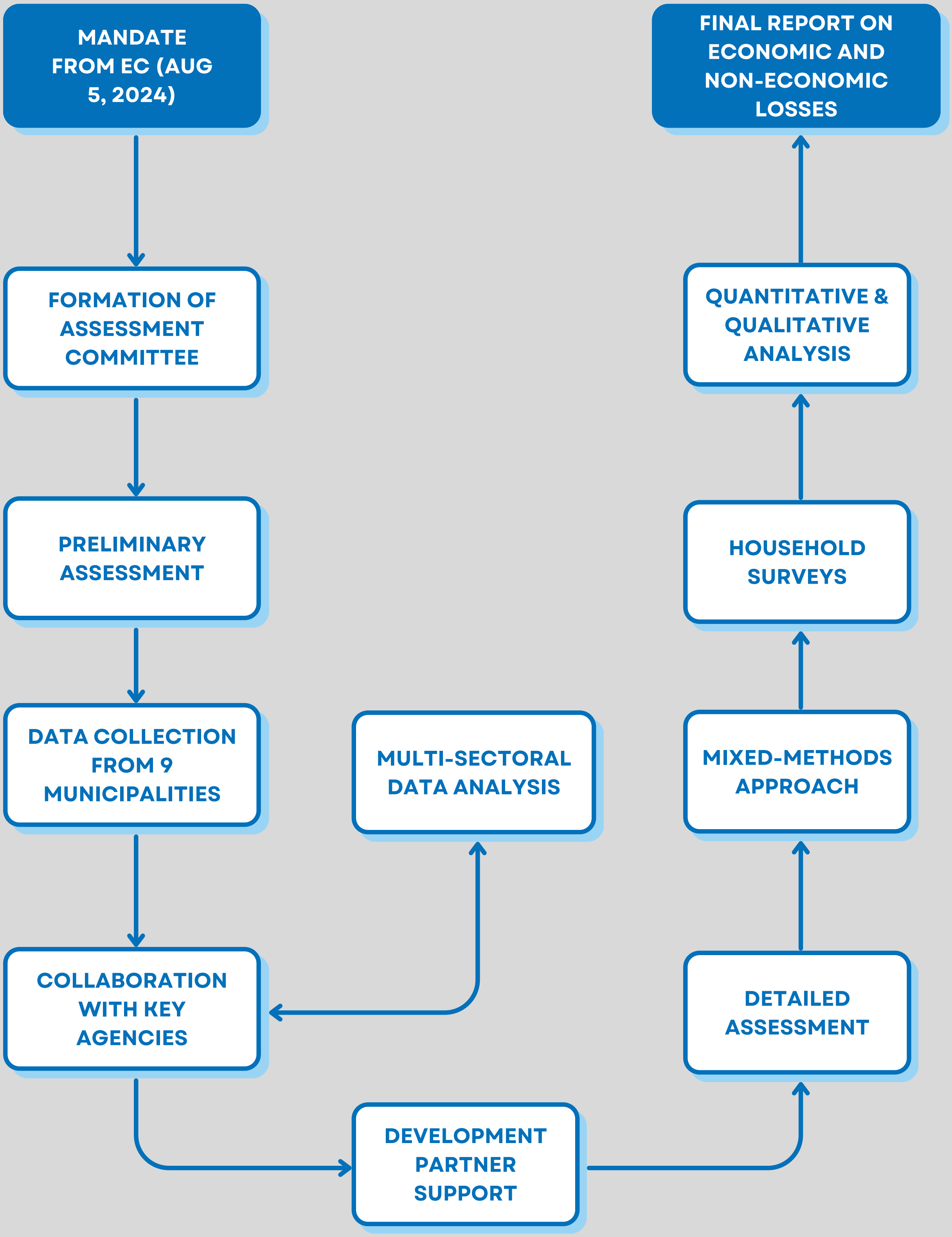


Figure 1: Methodological Approach for the Detailed Assessment

Quantitative Data Collection

The assessment utilized a stratified sampling method, focusing on both partially and fully damaged households to ensure representative insights. A total of 670 households were surveyed, representing 5% of partially damaged and 27% of fully damaged households as reported by the respective municipalities. The sample size was designed to ensure statistical reliability while capturing the diverse impacts across affected municipalities. Household surveys employed structured questionnaires to measure economic and non-economic losses and damages, with data collection facilitated by trained enumerators.

Qualitative Data Collection

To complement the quantitative data, qualitative methods provided deeper insights into community vulnerabilities and institutional responses:

Key Informant Interviews (KIIs):
Conducted with municipal representatives, line agencies, and other stakeholders, these interviews captured institutional perspectives on the flood's impact and recovery strategies.

Focus Group Discussions (FGDs):
Groups were convened, particularly focusing on women and marginalized populations, to understand community-level coping mechanisms and challenges during the recovery process.

Pilot Toolkit Development and Testing:
The questionnaire-based toolkit (See ANNEX I & II) was piloted with the 670 surveyed households to capture household-level economic and non-economic losses. Insights from this phase will guide further refinements to the toolkit during data analysis, committee meetings, and validation workshops.

Scope and Limitation of the Study

The detailed assessment aims to provide a comprehensive evaluation of the loss and damage caused by the extreme flooding in Kanchanpur District in July 2024. It focuses on quantifying both economic and non-economic losses across key sectors such as agriculture, livestock, physical infrastructure while highlighting gaps identified in the preliminary report. Utilizing contextualized toolkits and validated data from 670 stratified household samples, the study captures a nuanced understanding of the flood's impact in terms of sampled household number and does not refelect details from all the affected households within Kanchanpur district. This assessment also serves as a foundation for developing standardized methodologies, addressing unassessed components like non-economic losses, and providing actionable insights for future recovery and resilience planning.

Despite its comprehensive design, the study faced several limitations:

Incomplete Data Reporting:

Municipalities did not consistently update or report their data to the District Administration Office (DAO), creating gaps in the datasets.

Lack of Standardized Formats:

Both municipalities and line agencies lacked and differed in standardized tools and databases for collecting and managing data on losses and damages. This resulted in variability in the quality and consistency of the data.

Reliance on Secondary Data:

The assessment relied heavily on data from responsible authorities, supplemented by information from the preliminary assessment report to address missing values. This approach, while necessary, introduces the potential for inaccuracies stemming from incomplete or outdated reporting by the agencies.

These limitations highlight the need for systemic improvements in disaster data management, including the development of standardized formats and robust data collection practices at the municipal and district levels. Despite these challenges, the study provides critical insights that will inform recovery planning and future assessments.

Early Response, Rainfall Warnings and Monitoring on Flood Preparedness

Before the catastrophic flooding on 7-8 July 2024, early warning systems had been actively disseminating weather alerts. On July 6, 2024, the Department of Hydrology and Meteorology (DHM) issued a warning of heavy rainfall across the region, particularly targeting the municipalities of Dodhara Chandani, Bheemdatt, and surrounding areas. This information was relayed through various platforms, including SMS, radio broadcasts, and local announcements, ensuring that residents were informed well in advance (DHM, 2024).

On July 08, 2024, a significant rainfall event occurred, measured at key monitoring stations across the affected regions. For instance, in Dodhara, the Tipping Bucket Rain gauge recorded a staggering 624.0 mm of rainfall between July 07 at 8:45 AM and July 08 at 8:45 AM, surpassing all previously recorded rainfalls. Hanuman Nagar recorded 573.6 mm, and Sundarpur recorded 556.4 mm, both of which were also historic measurements (DHM, 2024).

The monitoring systems, including automatic rain gauges, were pivotal in tracking this exceptional rainfall. The Tipping Bucket Rain gauge provided real-time data to authorities, while manual rain gauges were used for additional validation. However, challenges were faced with overflow during intense rainfall, which led to minor discrepancies in data collection, especially at locations with inadequate infrastructure.

The rainfall recorded on July 08 was among the heaviest in recorded history, surpassing the 2017 August 13 rainfall and even the 1993 records (DHM, 2024). The early warning system had provided a crucial timeframe for the people of Dodhara Chandani, allowing them to evacuate and prepare for the floods. The rain measurements validated the concerns of local authorities about the severity of the coming floods.

In response to the situation, the communities had implemented preparedness measures, aided by the early warning systems. These efforts included community evacuations and rescue team mobilizations. While the flood was still devastating, particularly in terms of infrastructure and displacement, the timely warning and preemptive actions taken in the days prior helped prevent loss of life.

During this period, floodwaters began rising rapidly on the night of July 07, especially in flood-prone zones of Dodhara Chandani Municipality, where the community had been alerted the day before. These precautionary actions, combined with accurate weather forecasting, were instrumental in minimizing human casualties.

Despite the severity of the floods, the early warning system, along with significant rainfall measurements taken at key monitoring stations, played an essential role in mitigating the overall impact on the population.

Historical Context of Flood in Kanchanpur District

Kanchanpur District, situated in the far-western region of Nepal, has a long history of being impacted by floods due to its geographic location near the Mahakali River and its exposure to monsoon rains. These natural conditions make the district particularly vulnerable to recurrent flooding, which disrupts daily life, damages infrastructure, and leaves lasting effects on local communities. Over the years, floods have become a part of life for residents, affecting both their livelihoods and their resilience to future disasters.

In the past decade, Kanchanpur has experienced significant flooding events in the years 2013, 2017, 2019, 2021 and now in 2024. During these years, the district faced cumulative losses of NPR 8 million, with 35 people losing their lives and 52 sustaining injuries. Moreover, 129 infrastructures were damaged in a total of 110 reported incidents in Bipad Portal (https://bipadportal.gov.np/). However, these figures likely underestimate the true extent of the damage. For instance, during the unusually heavy rains in October 2021, losses in the agriculture sector alone amounted to millions of Nepalese rupees. These losses were not fully captured in official reports, indicating gaps in data collection and documentation processes.

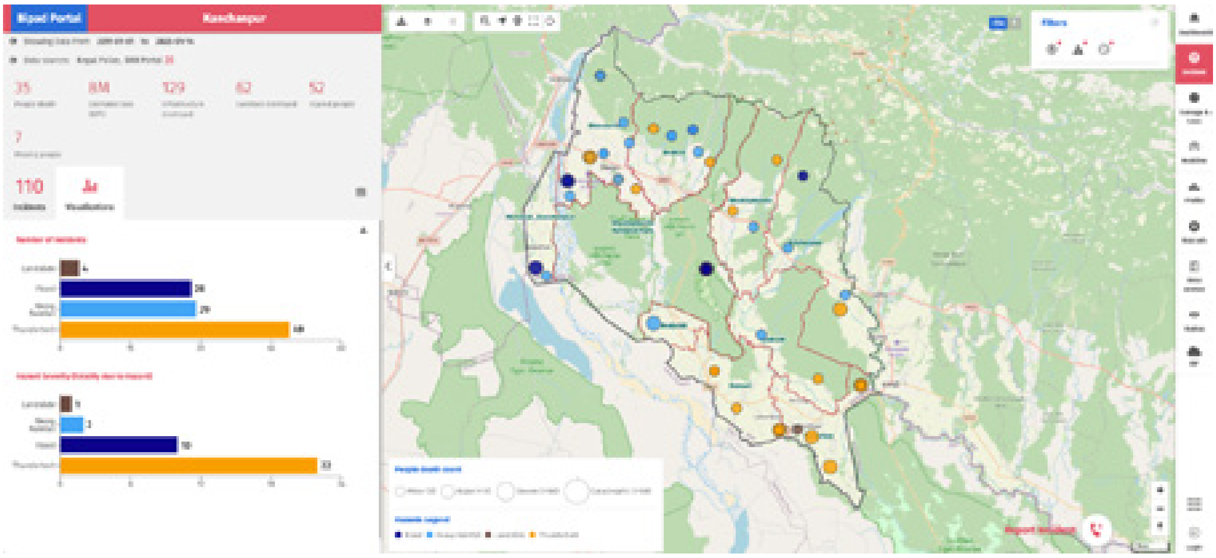


Figure 2: Showing the incident data reported by Nepal Police in Bipad Portal from 01 Jan 2011 to 01 Jan 2025.

Historical data from 1971 to 2018 offers an even broader perspective on the severity of floods in the district. According to records from DesInventar and the Ministry of Home Affairs (2018), floods during this period caused 385 deaths, with an additional 25 individuals reported missing. Furthermore, over 71,000 families were affected, and the district suffered economic losses amounting to NPR 47.6 million. The impact on housing was severe, with 1,664 private houses completely destroyed and 2,776 partially damaged. Additionally, 5,044 individuals were injured, and significant losses were reported in livestock, with 42 animals lost and 43 sheds displaced. These figures illustrate the far-reaching consequences of floods, extending beyond immediate physical destruction to long-term social and economic challenges.

Despite the availability of such historical data, gaps in reporting and sector-specific analysis persist. Existing records often fail to provide detailed insights into how floods have impacted key areas like agriculture, health, education, and infrastructure. For example, while the Bipad Portal compiles basic statistics, the underreporting of the incidents overlooks the nuanced effects of floods on livelihoods and local economies. These limitations make it difficult to design effective response and recovery strategies.

This detailed assessment builds on historical evidence while addressing these gaps. Using household surveys, interviews with stakeholders, and field-level documentation, the study provides a comprehensive understanding of flood impacts in Kanchanpur. By connecting past trends with present realities, the assessment not only highlights the district's vulnerabilities but also underscores the importance of targeted interventions to reduce future risks. This report aims to inform policies and practices that strengthen the community's capacity to respond to and recover from floods, ultimately fostering greater resilience in the face of natural disasters.

Impact of Heavy Rainfall and Flood

The Bipad Portal, which hosts various datasets on hazards, exposure, and risk, includes a flood hazard map covering the entire country. This map displays modeled water depths for flood events with different return periods, with water depths represented in meters. It is important to understand that not all flood events shown are expected to occur simultaneously. Instead, the data illustrates the maximum water depth that could be expected at a particular location during a flood event of the specified return period. In essence, the data provides the likelihood of experiencing a certain water depth in a given year. For example, the '1-in-100 year' layer represents a 1% (or 1-in-100) chance of such flooding occurring in any given year. The map below illustrates the potential extent of impacts during a large-scale flood event in the Sudurpaschim Province of Nepal.

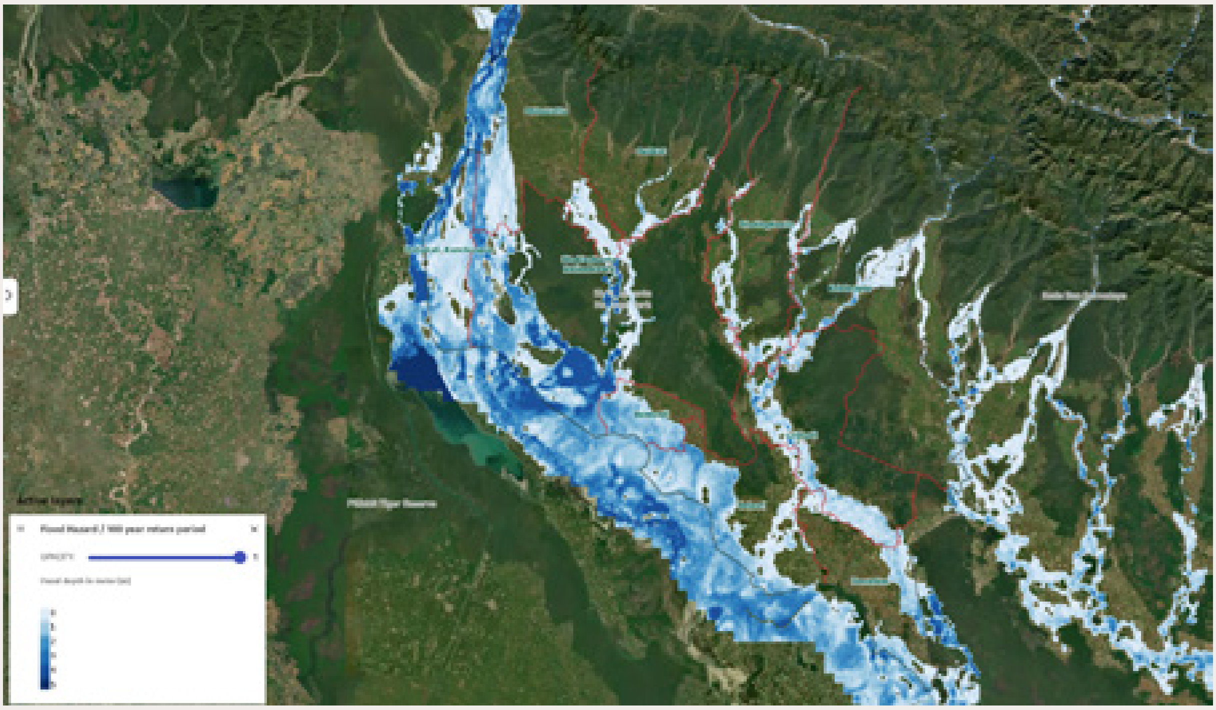


Figure 3: Inundation Level of Kanchanpur District in 100 years flood return period

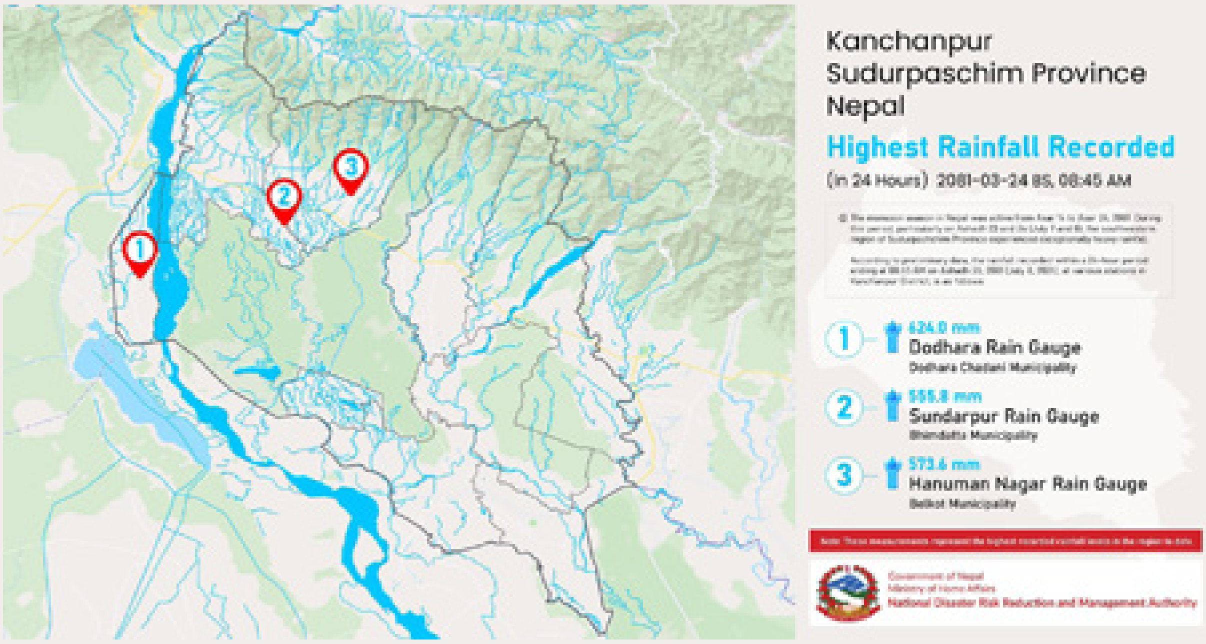


Figure 4: Highest Rainfall recorded on 8 July 2024, Kanchanpur District, Sudurpaschim Province.

On July 8, 2024, Dodhara Chandani recorded a record-breaking 624 mm (approximately 2.05 feet) of rainfall within a 24-hour period, marking the highest rainfall ever recorded in Nepal. Similarly, Sundarpur saw 555.8 mm, and Hanumannagar experienced 573.6 mm (around 1.88 feet) of rainfall. Continuous rainfall starting on July 7 resulted in widespread flooding and inundation across the entire Kanchanpur district, with some areas of Kailali district also affected. According to reports from the Department of Hydrology and Meteorology (DHM) and the National Disaster Risk Reduction and Management Authority (NDRRMA), this event was induced by cloudbrust.

The satellite imagery from Sentinel-1, Sentinel-2A, and SPOT-6, provides a comprehensive overview of the flood-affected areas in Kanchanpur District, Nepal, during the catastrophic flood event of July 2024.

The first image uses Sentinel-1 radar imagery to delineate flood-affected areas in Kanchanpur District on July 8, 2024. This map offers precise spatial boundaries of inundation, covering a total of 97.9 km². Additionally, it identifies 14,315 houses affected and 85.27 km² of agricultural land submerged, however the numbers are significantly higher as per the ground validation data provide by municipalities and several line agencies. The image integrates administrative boundaries and natural features, such as rivers, to contextualize the flooding's scope within the district. Its radar-based analysis proves essential for identifying areas requiring immediate intervention and recovery efforts, especially in regions with high population density and agricultural activity. By presenting a clear representation of flooded zones, the map aids in developing actionable strategies for disaster management and rehabilitation.

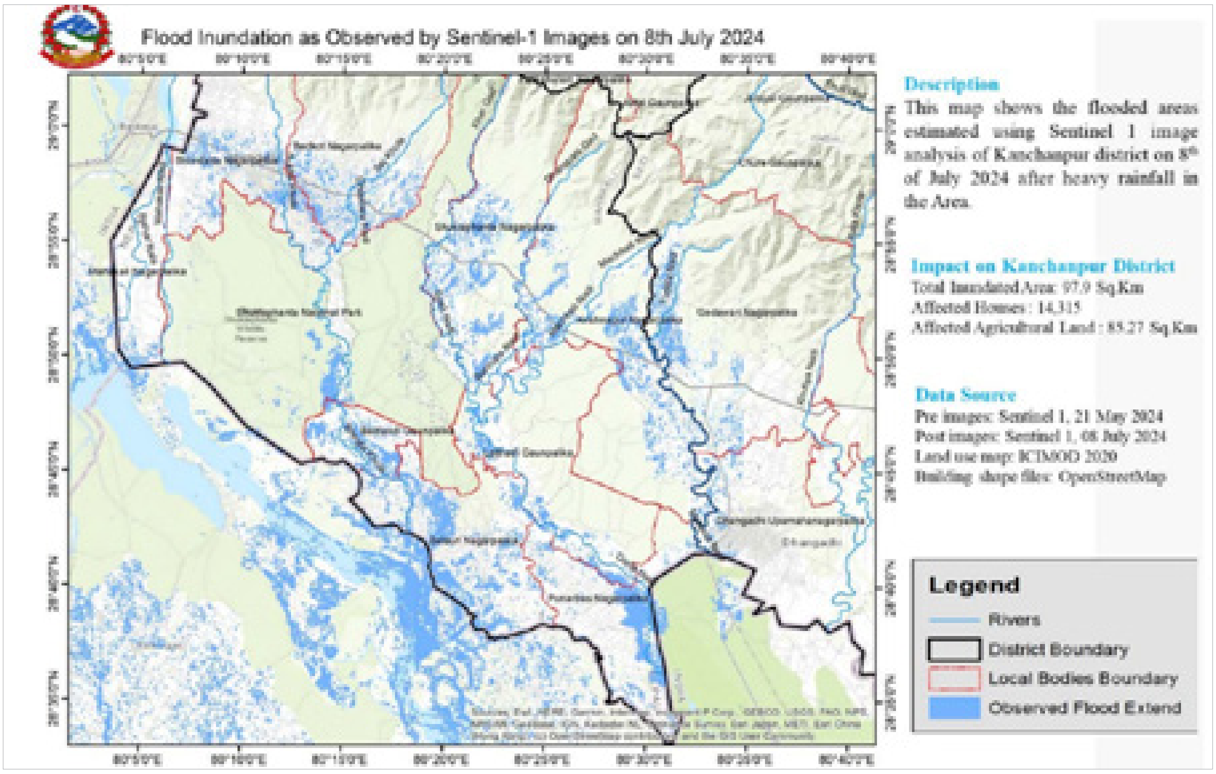


Figure 5: Flood inundation across Kanchanpur district as observed by Sentinel-1 on 8 July 2024

The image, sourced from SPOT-6 satellite imagery, highlights the flood inundation in the Beldandi region of Kanchanpur District, Nepal, as observed on July 9, 2024. The red-marked areas denote flooded zones, covering substantial portions of agricultural fields, roadways, and settlements. The image reveals extensive inundation along riverbanks and low-lying areas, emphasizing the vulnerability of these regions to seasonal flooding. The map also provides a comparative view of normal water levels versus the flood extent, offering critical insight into the magnitude of water overflow caused by heavy rainfall. The inset map further illustrates the specific inundation in fields around Beldandi, showcasing the severity of agricultural damage and potential implications for local livelihoods.

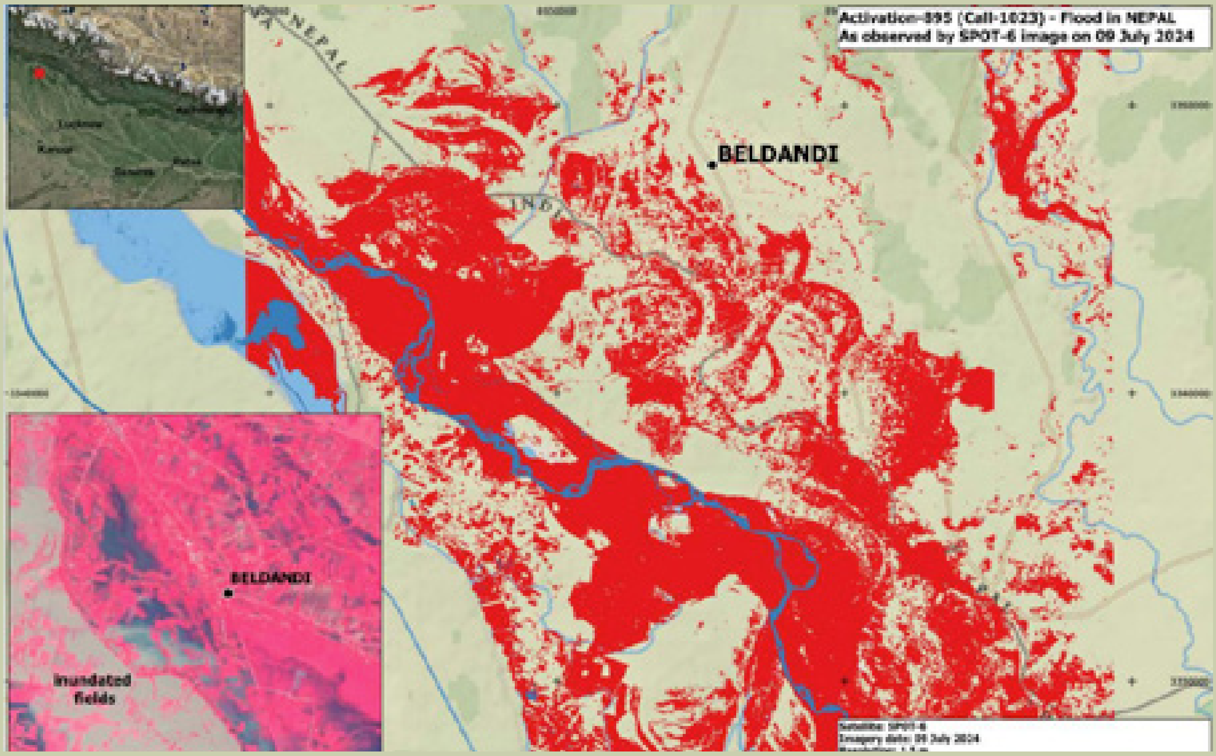


Figure 6: Flood Inundation as observed from SPOT-6 imagery in Beldandi Rural Municipality and nearby municipalities

Likewise, the image, derived from Sentinel-2A imagery, provides a comprehensive assessment of the flood's impact on infrastructure, croplands, and natural vegetation in Beldandi Rural Municipality. The map quantifies 369 km² of flooded area, including 285 km² of croplands and 251 km of roads, with affected zones marked in red and turquoise hues. This visualization is particularly valuable for identifying specific areas of agricultural damage and road disruptions. Additionally, the inset provides a closer view of flooded fields in Beldandi, emphasizing the localized effects of the disaster. The image serves as a critical resource for understanding the extensive socio-economic impacts of the flood and supports evidence-based planning for future flood resilience in the region.

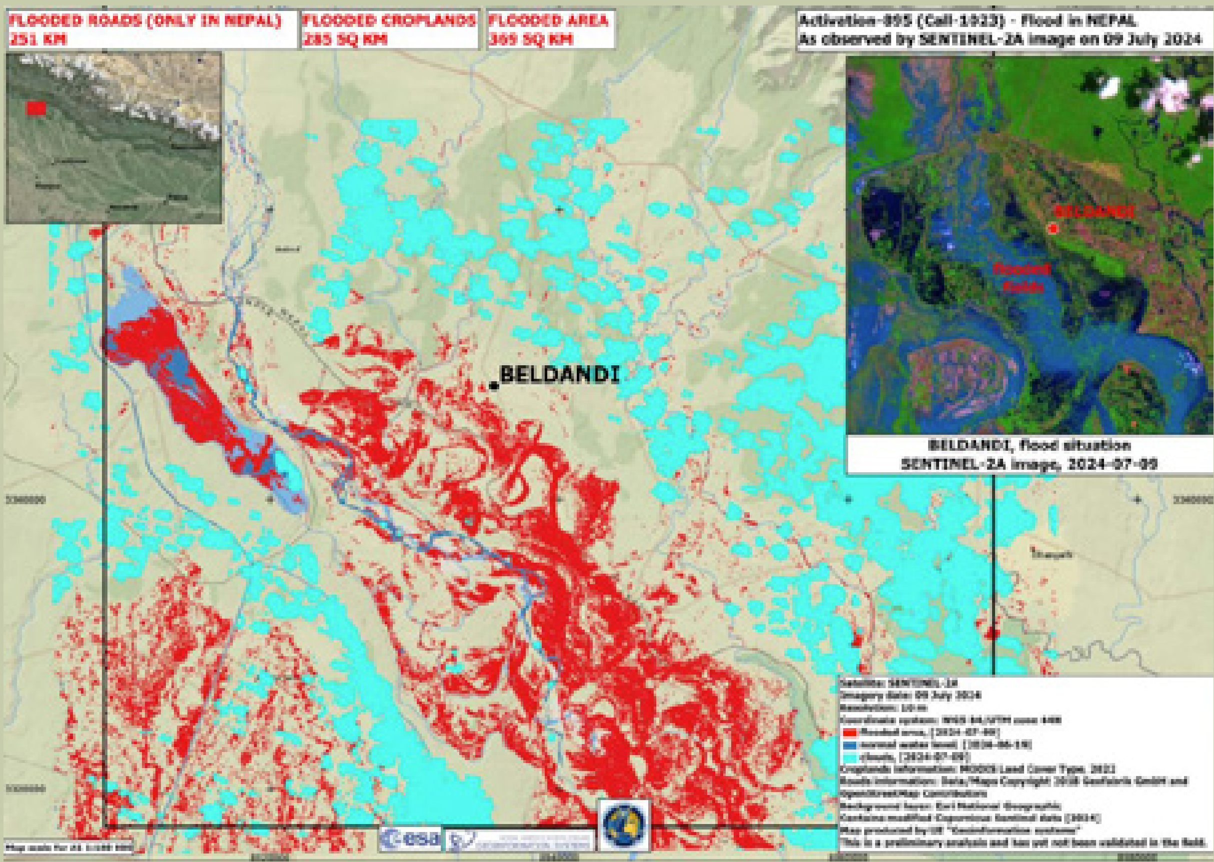


Figure 7: Flood innundation as observed from SENTINEL-2A Imagery in Beldandi Rural Municipality and nearby municipalities.

All these imageries confirm heavy rainfall as the primary cause of water overflow in the region's rivers, particularly impacting areas near water bodies, such as Beldandi Rural Municipality. Overall, the SPOT-6 imagery emphasizes the inundation of low-lying fields and settlements, while the Sentinel-1 radar analysis confirms precise flood boundaries and local infrastructure damages. The Sentinel-2A visual representation further enhances this assessment by mapping the extent of cropland and natural vegetation loss.

The discrepancy between satellite-derived flood extents and ground validation data highlights the need for integrating remote sensing with on-the-ground assessments. While satellite imagery provides precise flood mapping, it may underestimate impacts due to cloud cover, vegetation, or misclassification. Ground validation captures localized damage, offering a more accurate picture. This finding underscores the importance of combining both methods to improve disaster assessments and response strategies. Future efforts should prioritize validation mechanisms like crowdsourced reports, drone surveys, and hydrological models to enhance flood risk mapping and resilience planning.

Housing, Human Life and Settlements

The 2024 flood in Kanchanpur District resulted in varying impacts on human life across different municipalities. Beldandi Rural Municipality reported one fatality, while Krishnapur Municipality recorded one death and Punarbas Municipality experienced three fatalities, but no injuries were reported. In Dodhara Chandani Municipality, Bedkot Municipality, and Bheemdatt Municipality, no fatalities or injuries were reporte. Shuklaphanta Municipality, despite facing severe flood damage, fortunately did not experience any loss of life.

Similarly, the flood caused widespread damage to housing and settlements, with varying levels of destruction across nine municipalities. According to data from the District Administration Office, Kanchanpur, and official reports from the municipalities, the damage was extensive as 8026 houseds were partially damaged and 1109 were fully damaged as Bheemdatt Municipality, which reported the highest number of destroyed households. A total of 4,692 houses were impacted, with 4,117 partially damaged and 575 completely destroyed. This municipality experienced severe flooding, resulting in significant loss of housing and infrastructure.

Table 1: Partially and fully damaged household across nine municipalities in Kanchanpur district

S.N	Municipality	Partially Damaged	Fully Damaged
1	Dodhara Chandani Municipality	877	102
2	Bheemdatt Municipality	4117	575
3	Bedkot Municipality	94	26
4	Belauri Municipality	1019	290
5	Beldandi Rural Municipality	337	9
6	Krishnapur Municipality	91	2
7	Laljhadi Rural Municipality	1153	50
8	Punarbas Municipality	153	16
9	Shuklaphanta Municipality	185	39
	Overall Damage	8026	1109

Dodhara Chandani Municipality also suffered considerable damage, with 877 homes partially damaged and 102 completely destroyed, totaling 979 affected houses. Similarly, Beldandi Rural Municipality faced flooding, with 337 homes partially damaged and 9 fully destroyed, amounting to 346 affected houses. Laljhadi Rural Municipality saw 1,153 partially damaged homes and 50 fully destroyed houses, totaling 1,203 affected homes, highlighting the widespread impact of the floods in this region.

In Belauri Municipality, 1,309 homes were affected, with 1,019 partially damaged and 290 completely destroyed. Shuklaphanta Municipality recorded 224 affected houses, including 185 partially damaged and 39 fully destroyed. Bedkot and Krishnapur were less affected, with 120 and 93 total cases of damage, respectively. Bedkot reported 94 homes with partial damage and 26 with full destruction, while Krishnapur saw minimal damage, with 91 partially damaged homes and only 2 fully destroyed.

Household Damage from Torrential Rainfall in Kanchanpur

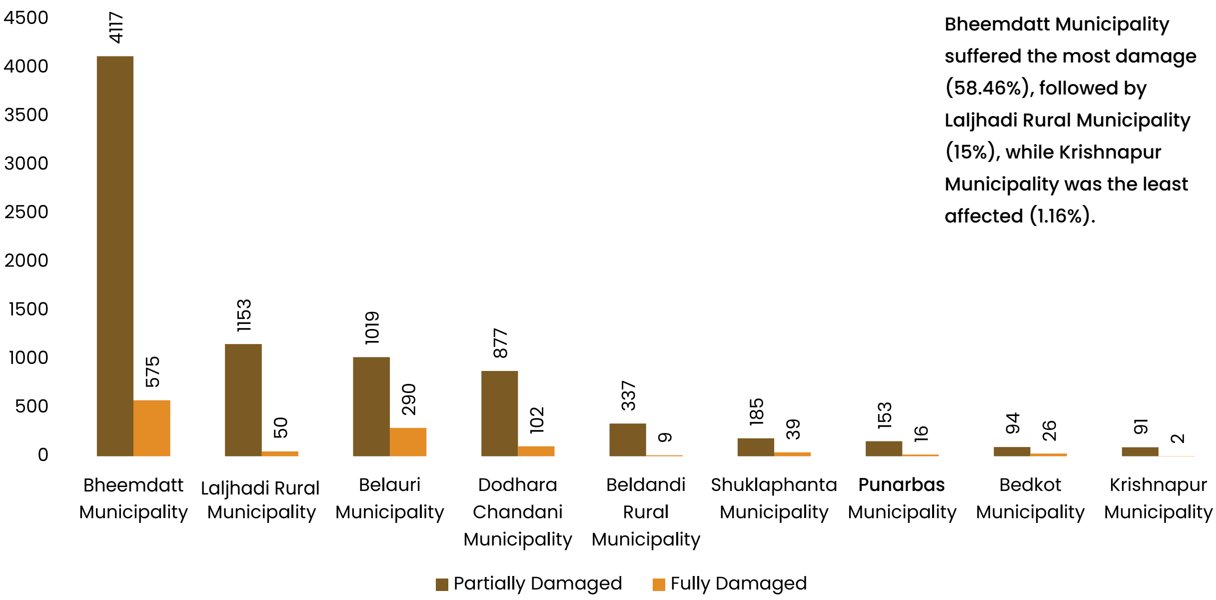


Figure 8: Number of households damaged by July Flood across nine nunicipalities of Kanchanpur District

Punarbas Municipality was also affected, with 153 homes partially damaged and 16 fully destroyed, totaling 169 impacted houses. In total, the floods left 8,026 houses partially damaged and 1,109 homes fully destroyed across the district.

The floods led to extensive damage to housing structures across multiple municipalities, with varying degrees of impact. Bheemdatt Municipality reported the highest number of affected families, with 17,929 households impacted, underscoring the region's vulnerability due to its population density and proximity to flood-prone areas. Laljhadi Rural Municipality also faced significant destruction, with 3,196 affected families, indicating widespread damage to homes and livelihoods.

Table 2: Overall flood impact on Human life across nine municipalities in Kanchanpur district

SN	Municipality	Affected Families	Dead	Injured	Displaced Families
1	Dodhara Chandani Municipality	1,626	-	-	830
2	Bheemdatt Municipality	17,929	-	-	-
3	Bedkot Municipality	234			22
4	Belauri Municipality	1,401	-	-	-
5	Beldandi Rural Municipality	3,035	1	-	-
6	Krishnapur Municipality	291	1		-
7	Laljhadi Rural Municipality	3,196			
8	Punarbas Municipality	200	3		
9	Shuklaphanta Municipality	2,635	-	2	142
	Total	**30,547**	**5**	**2**	**994**

In Dodhara Chandani Municipality, 1,626 families were affected, with 830 families displaced due to severe inundation. Similarly, Beldandi Rural Municipality witnessed damage to 346 households, which suggests partial inundation in most cases.

In Bedkot Municipality, 234 families were impacted, with 22 experiencing displacement. Krishnapur Municipality reported 291 affected families, but displacement was minimal, with only one individual recorded as displaced. Shuklaphanta Municipality suffered substantial losses, with 2,635 families affected and 142 families displaced, reflecting the widespread nature of the flood in this area.

In Belauri Municipality 1401 families were affected whereas Punarbas Municipality experienced relatively lower impacts, with 169 families affected but no recorded displacement.

Across all municipalities, approximately 994 families were displaced, with Dodhara Chandani and Shuklaphanta Municipalities experiencing the highest levels of displacement. Dodhara Chandani Municipality recorded 830 displaced families, accounting for more than 50% of its affected population, while Shuklaphanta Municipality reported 142 displaced families. These figures highlight the immediate need for temporary shelters and relocation support.

The total number of displaced individuals was comparatively low, with 167 individuals displaced in Dodhara Chandani Municipality and 306 in Shuklaphanta Municipality. The relatively small numbers of displaced individuals in contrast to displaced families may reflect the availability of nearby safer locations for temporary shelter or underreporting of individual level data from the concerned local government.

The floods caused significant disruption to livelihoods, particularly in Bheemdatt, Dodhara Chandani, and Laljhadi Municipalities, which bore the brunt of the disaster. The displacement of families necessitated the establishment of temporary shelters, the provision of food, water, and medical aid, and psychological support for those affected. Several local partners such as NEEDS Nepal,and NNSWA provided food, medical aid and other support throughout the flood affected municipalities in Kanchanpur district.

In municipalities like Bedkot, where displacement numbers were relatively low despite high levels of housing damage, the focus should be on repairing partially damaged homes and providing financial aid to support reconstruction efforts. Similarly, for municipalities such as Belauri and Punarbas, where data gaps still exist even after several months from the disaster, comprehensive assessment and data collection and coordination mechanishms are essential to address potential overlooked impacts.

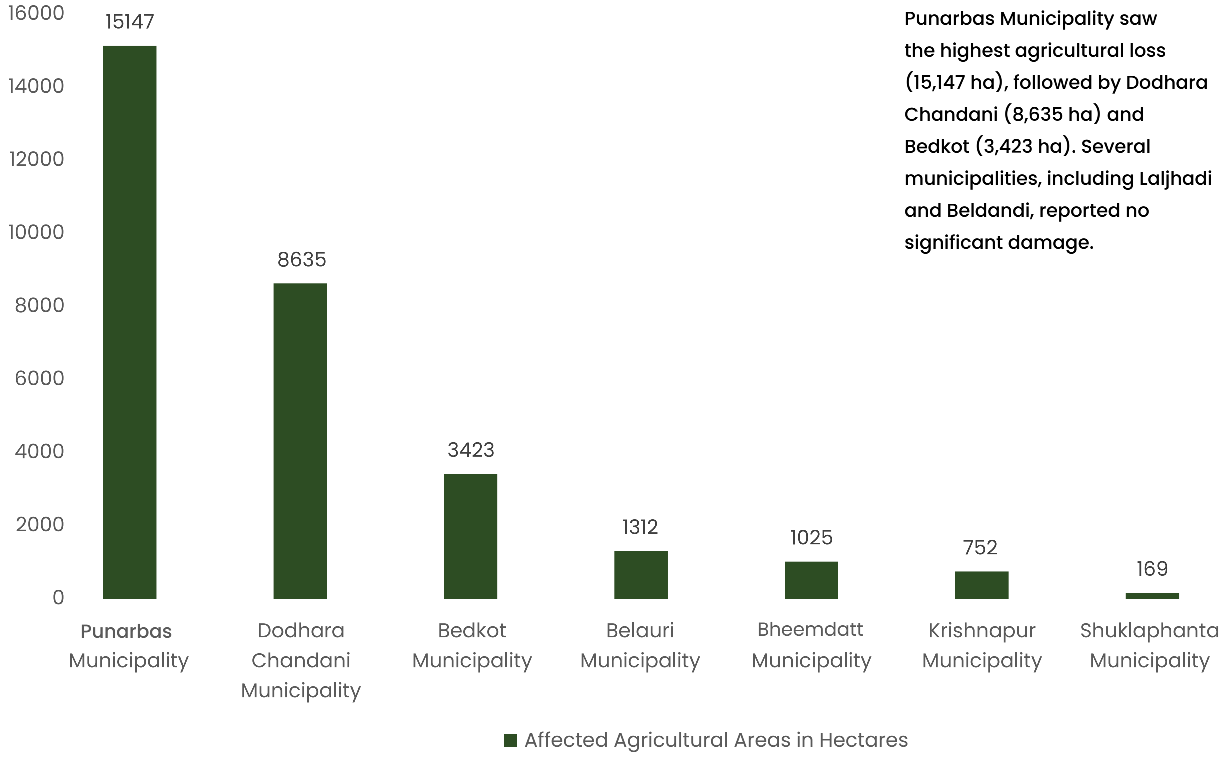
The housing damage highlights the urgent need for recovery efforts focusing on rebuilding and repairing homes and infrastructures, particularly in highly affected municipalities. Resources should prioritize both immediate shelter needs and long-term housing solutions to ensure the resilience of affected communities.

Agriculture

The recent flooding of 7–8 July 2024, in the Sudurpaschim region led to varying levels of agricultural and grain losses across its nine municipalities. This analysis examines the extent of damage in terms of agricultural land affected, standing crop losses, stored grain losses, and the cumulative financial impact, highlighting key comparisons among municipalities.

Dodhara Chandani Municipality experienced significant damage, with 8,635 hectares of agricultural land affected. This resulted in standing crop losses valued at NPR 31,823,611 and stored grain losses of 3,505 quintals, amounting to NPR 12,369,900. Combined, the total financial loss for the municipality stood at NPR 44,193,511, making it one of the worst-hit areas. Bheemdatt Municipality, while experiencing lesser agricultural land damage (1,025 hectares), reported a notable financial loss of NPR 13,302,852 due to a combination of standing crop and stored grain losses.

Flood Affected Agricultural Areas in Hectares



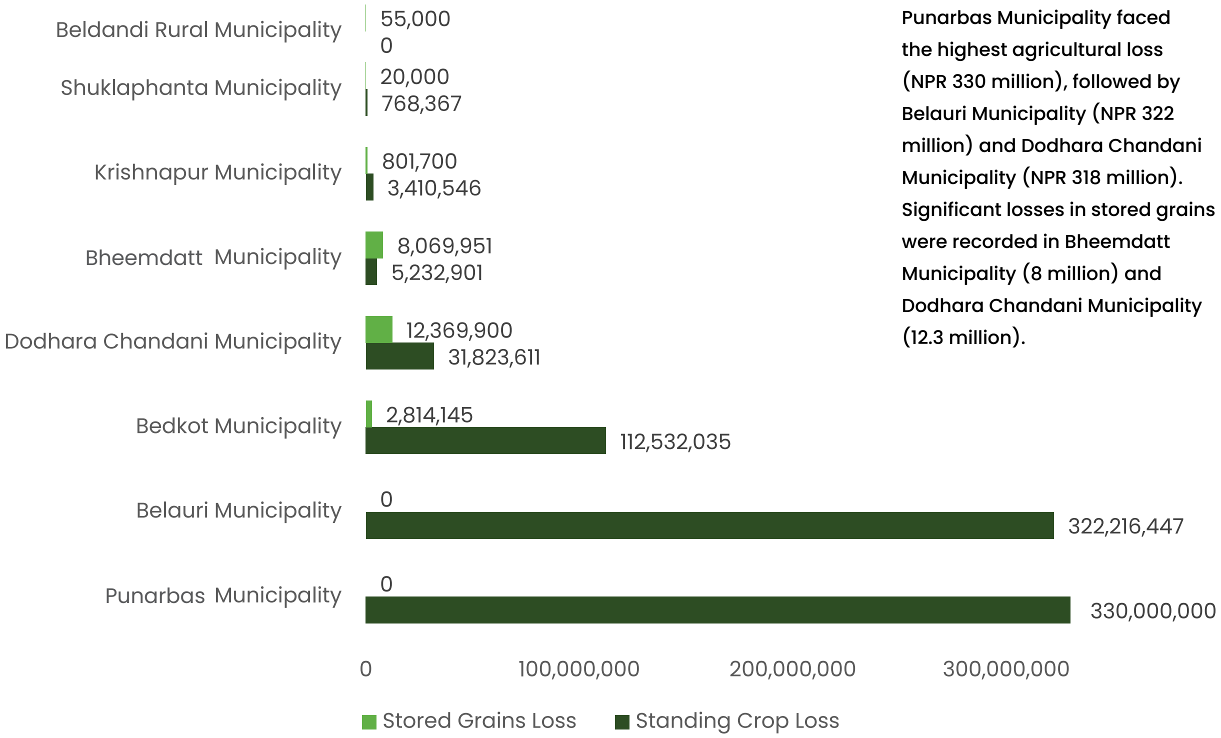
Punarbas Municipality saw the highest agricultural loss (15,147 ha), followed by Dodhara Chandani (8,635 ha) and Bedkot (3,423 ha). Several municipalities, including Laljhadi and Beldandi, reported no significant damage.

Figure 9: Flood affected agricultural areas in hectares

In contrast, Belauri Municipality reported substantial standing crop losses valued at NPR 322,216,447, which accounted for a significant share of the total regional losses. However, the municipality reported no losses in stored grains, which may suggest focused damage on active agricultural fields rather than storage facilities. Similarly, Punarbas Municipality incurred the highest financial loss in standing crops at NPR 330,000,000, affecting 15,147 hectares of land, the largest area impacted among all municipalities. Notably, like Belauri, Punarbas recorded no stored grain losses.

Shuklaphanta Municipality faced minimal impact, with only 169 hectares of agricultural land affected. This resulted in standing crop losses of NPR 768,367 and stored grain losses valued at NPR 20,000, bringing the total financial loss to NPR 788,367. Laljhadi Rural Municipality reported no damage in either category, potentially reflecting effective flood prevention measures or incomplete reporting.

Estimated Agricultural Loss in NPR



Punarbas Municipality faced the highest agricultural loss (NPR 330 million), followed by Belauri Municipality (NPR 322 million) and Dodhara Chandani Municipality (NPR 318 million). Significant losses in stored grains were recorded in Bheemdatt Municipality (8 million) and Dodhara Chandani Municipality (12.3 million).

Figure 10: Estimated agricultural loss across nine municipalities of Kanchanpur District

The cumulative losses across all municipalities amounted to NPR 830,114,603. Belauri and Punarbas municipalities together accounted for a staggering 78.5% of this total, emphasizing their vulnerability during the flooding. Dodhara Chandani, despite reporting a relatively lower financial loss, still incurred significant damage in both standing crops and stored grains. Conversely, municipalities like Shuklaphanta and Laljhadi contributed minimally to the overall loss, with Shuklaphanta accounting for less than 0.1% of the total damages.

This comparative analysis highlights the disparities in the impact of flooding across municipalities. The data suggests that high-impact areas, such as Punarbas and Belauri, require urgent disaster response and recovery measures. Additionally, the lack of reported losses in Laljhadi and minimal damage in Shuklaphanta underscore the potential benefits of effective flood prevention and mitigation strategies. These findings stress the importance of tailored interventions to address the specific needs of the most affected areas while reinforcing preventive measures in relatively less impacted municipalities.

Livestock

The assessment of livestock and aquaculture in the flood-affected municipalities highlights substantial impacts on fish and chicken populations, with notable differences across regions. Bheemdatt Municipality reported the highest impact, with 56,800 affected fish resulting in a financial loss of NPR 7,045,000, and 54,329 affected chickens, leading to a loss of NPR 17,406,600.

Shuklaphanta Municipality experienced relatively minor losses, with 4,500 affected fish amounting to NPR 450,000 and 1,200 affected chickens leading to a loss of NPR 300,000. Similarly, Dodhara Chandani Municipality reported 4,650 affected chickens with an associated loss of NPR 1,900,000, alongside 98 cattle fatalities.

Belauri Municipality recorded eight injured cattle, 12,500 affected fish with a loss of NPR 2,500,000, and 2,200 affected chickens resulting in NPR 660,000 in losses. Other municipalities like Bedkot, Laljhadi, and Beldandi faced smaller but still significant losses. Bedkot Municipality reported losses of NPR 6,100,000 from 70,000 affected fish and NPR 345,000 from 2,200 affected chickens. Laljhadi and Beldandi Rural Municipalities reported cumulative losses of NPR 450,000 and NPR 3,000,000, respectively, from fish impacts, and NPR 540,000 and NPR 1,000,000, respectively, from affected chickens.

In total, across all municipalities, the reported financial loss due to affected fish amounts to NPR 22,700,000, while losses from chicken populations total NPR 22,171,600 resulting in total loss of NPR 4,4,871,600. These figures emphasize the significant economic strain on communities reliant on aquaculture and poultry for their livelihoods, necessitating immediate recovery initiatives and sustainable risk management strategies.

Table 3: Livestock, Aquaculture, and Poultry Losses Due to Flooding in Affected Municipalities of Kanchanpur District

S.N	Municipality	Cattle			Fish		Chicken	
		Dead	Injured	Missing	Number Affected	Loss in NPR	Number Affected	Loss in NPR
1	Bheemdatt Municipality	105	0	66	56800	7045000	54329	17406600
2	Shuklaphanta Municipality	4	1	0	4500	450000	1200	300000
3	Dodhara Chandani Municipality	98	0	0	0	2455000	4650	1900000
4	Belauri Municipality	0	8	0	12500	2500000	2200	660000
5	Punarbas Municipality	2	0	1	0	0	36	20000
6	Krishnapur Municipality	1	1	0	14000	700000	0	0
7	Bedkot Municipality	16	5	0	70000	6100000	2200	345000
8	Laljhadi Rural Municipality	0	0	0	1500	450000	1800	540000
9	Beldandi Rural Municipality	0	0	0	20000	3000000	2000	1000000
	Total	**226**	**15**	**67**	**179300**	**22700000**	**68415**	**22171600**

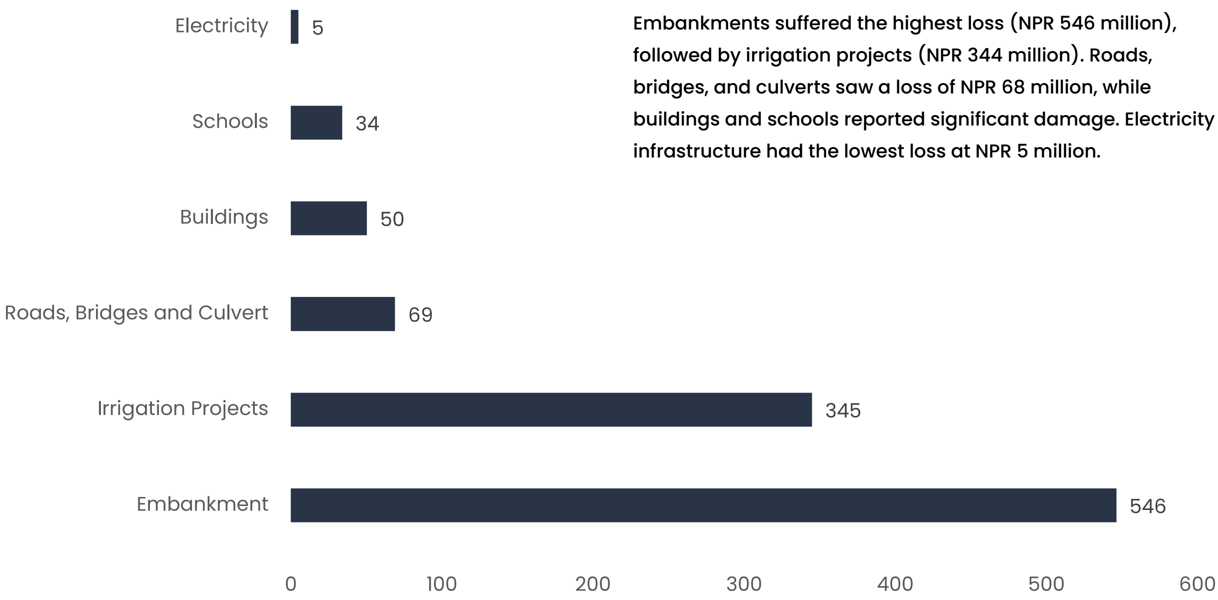
Physical Infrastructures

The assessment of physical infrastructure losses was based on the data reported primarily by line agencies overseeing specific infrastructure sectors, as municipal-level reporting was either unavailable or incomplete. This limitation necessitated reliance on estimates provided by these agencies, supplemented by data from municipalities where available.

The total reported loss for physical infrastructure amounts to NPR 1,049,325,008, with detailed breakdowns as follows: Buildings experienced damages estimated at NPR 50,290,225, while schools reported losses of NPR 34,000,000. Irrigation projects, a critical component for agricultural productivity, suffered the most significant loss, estimated at NPR 344,905,800. The electricity sector reported damages amounting to NPR 5,000,000, reflecting impacts on energy access for affected communities. Roads, bridges, and culverts, essential for mobility and connectivity, incurred losses estimated at NPR 68,870,000. Embankments, crucial for flood protection, reported losses of NPR 546,258,983, underscoring the severity of damage to flood mitigation infrastructure.

These figures highlight the extensive impact of the flood on the district's physical infrastructure, affecting sectors vital for the well-being and functionality of communities. It should be noted that this report reflects only the losses reported by line agencies and available municipal data. Comprehensive assessment and validation will require further collaboration with municipalities to address the existing data gaps and provide a more exhaustive account of infrastructure damage.

Total Physical Infrastructure Loss in NPR (In Million)

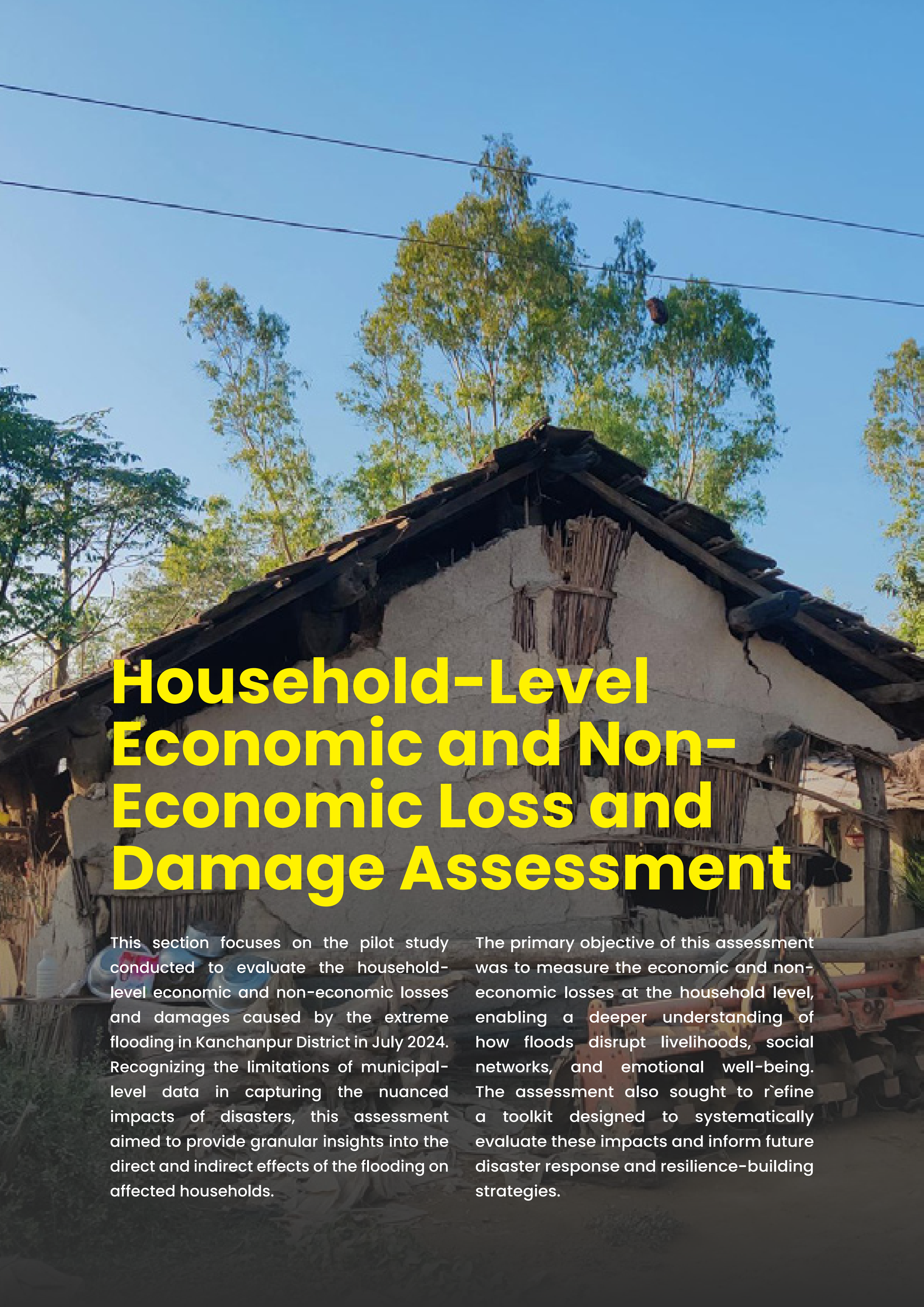


Embankments suffered the highest loss (NPR 546 million), followed by irrigation projects (NPR 344 million). Roads, bridges, and culverts saw a loss of NPR 68 million, while buildings and schools reported significant damage. Electricity infrastructure had the lowest loss at NPR 5 million.

Figure 11: Total estimated physical infrastructure loss across nine municipalities of Kanchanpur District

The physical infrastructure damages caused by the flood in Kanchanpur District have been substantial, with losses reported by various line agencies amounting to NPR 1049325008 The Mahakali Irrigation Management Office reported damages of NPR 292,200,000, highlighting the flood's impact on vital irrigation systems. Similarly, the Mahakali Irrigation Project (Third Phase) estimated losses at NPR 47,252,900, indicating vulnerabilities in ongoing projects. The Road Division Office in Mahendranagar reported damage to roads and related infrastructure worth NPR 68,870,000, while the Infrastructure Development Office recorded damages of NPR 71,975,000, covering multiple development projects. The Water Resource and Irrigation Development Office assessed losses at NPR 2,200,000, emphasizing the need for resilient water resource management systems. The People's Embankment Program suffered the most significant loss, with damages amounting to NPR 474,283,983 with overall loss of embankments resulted in an estimated loss of NPR 546258983, reflecting extensive damage to flood mitigation infrastructure

It is important to note that municipal-level reporting on infrastructure losses was either incomplete or unavailable, which limits the comprehensiveness of the data. Therefore, this report primarily relies on estimates provided by the line agencies concerned, supplemented by municipal data where available. These findings underline the critical need for enhanced disaster preparedness and resilient infrastructure development to reduce vulnerabilities and mitigate future risks in the region.



Household-Level Economic and Non-Economic Loss and Damage Assessment

This section focuses on the pilot study conducted to evaluate the household-level economic and non-economic losses and damages caused by the extreme flooding in Kanchanpur District in July 2024. Recognizing the limitations of municipal-level data in capturing the nuanced impacts of disasters, this assessment aimed to provide granular insights into the direct and indirect effects of the flooding on affected households.

The primary objective of this assessment was to measure the economic and non-economic losses at the household level, enabling a deeper understanding of how floods disrupt livelihoods, social networks, and emotional well-being. The assessment also sought to r`efine a toolkit designed to systematically evaluate these impacts and inform future disaster response and resilience-building strategies.

Methodology

The study employed a stratified sampling approach to ensure representative coverage of affected households. A total of 670 households were surveyed, representing 5% of the partially damaged and 27% of the fully damaged households as reported by municipalities. This stratification allowed for a balanced analysis of varying degrees of flood impacts across the district.

Data collection was facilitated using a pilot L&D Assessment toolkit comprising structured questionnaires designed to capture both economic and non-economic dimensions of loss and damage. Economic losses included direct costs, such as damage to property and livelihoods, while non-economic losses encompassed intangible effects, such as disruptions to education, loss of cultural heritage, and mental health impacts.

The assessment was supplemented by qualitative methods, including focus group discussions (FGDs) and key informant interviews (KIIs), to contextualize the quantitative findings. These qualitative inputs highlighted community-level coping mechanisms, the challenges faced during recovery, and the broader socio-cultural impacts of the disaster.

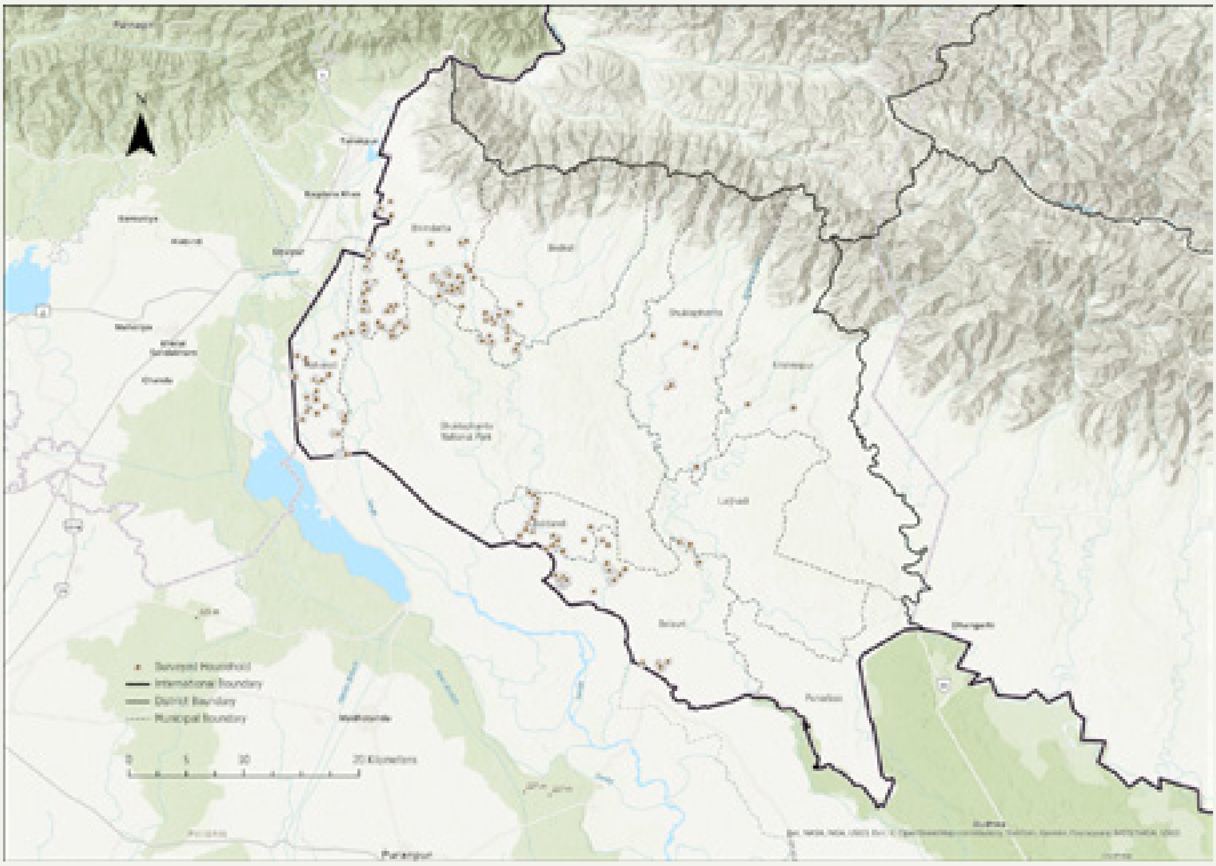


Figure 12: Map of Kanchanpur District Nine Municipalities with Surveyed Household for Economic and Non-economic loss and damage assessment

Scope and Focus

This assessment serves as a pilot initiative, with findings expected to guide further refinement of the toolkit. While comprehensive in its design, the study acknowledges limitations, such as incomplete data from some municipalities and the reliance on self-reported information from surveyed households. These limitations highlight the need for standardized reporting mechanisms and enhanced data collection processes at the municipal level.

The results of this assessment will be discussed in the following sections, offering valuable insights into the diverse and profound ways in which households were impacted by the flooding. These findings will also inform recommendations for mitigating future risks and enhancing disaster resilience at the local level.

Economic Loss and Damage Assessment Findings

The following analysis presents the findings from the household survey, focusing on the economic losses and damage caused by the flood. The analysis covers several key areas, including the extent of property damage, disruptions to livelihoods, and losses in agriculture and livestock. It also addresses damage to tools, machinery, and vehicles. In addition, the financial impact on households will be examined, specifically income losses, the depletion of savings, and the support received through external assistance or insurance claims. The analysis also considers whether households took loans or used other financial resources to recover. Each section will provide a detailed examination of the damage and challenges faced by households and discuss the strategies they used to regain financial stability and recovery after the flood.

Overall Household Level Economic Loss and Damage Analysis

The 2024 floods in Kanchanpur have caused significant economic damage across multiple municipalities, with losses categorized into property damage, agricultural loss, livelihood and income loss, livestock loss, and other setbacks. External financial assistance, such as insurance, compensation, and loans, has played a crucial role in post-disaster recovery. Among the affected areas, Bheemdatt Municipality suffered the highest economic losses, with property damage alone reaching NPR 30.44 million. Additional losses in agriculture of NPR 8.86 million, livelihoods and income of NPR 6.20 million, and livestock by NPR 7.91 million further exacerbated the situation. The municipality received NPR 24.41 million in external assistance, highlighting its reliance on recovery support. Similarly, Belauri Municipality recorded substantial losses, with property damage at NPR 12.77 million and the highest agricultural loss at NPR 13.50 million. The livelihood and income loss in Belauri stood at NPR 18.20 million, indicating a severe impact on household economies.

In contrast, municipalities such as Laljhadi, Krishnapur, and Bedkot reported relatively lower economic losses, though their recovery remains challenging due to limited financial resources. For instance, Krishnapur faced property damage worth NPR 1.06 million and an agricultural loss of NPR 0.13 million but received minimal external assistance, making reconstruction efforts more difficult.

While Bheemdatt and Belauri bore the brunt of the disaster, their high property and agricultural losses suggest long-term economic instability. Belauri's high agricultural losses may lead to food insecurity and employment challenges, requiring swift intervention. Meanwhile, municipalities like Shuklaphanta and Beldandi, despite lower losses, struggle with slow recovery due to insufficient financial aid. The widespread livelihood and income loss, particularly in Belauri (NPR 18.20 million), signals a deeper crisis that could lead to increased poverty, migration, and social instability.

Livestock loss, though smaller in proportion, is critical for rural households reliant on animal husbandry. Bheemdatt and Beldandi reported significant losses in this sector, further straining economies already affected by property and livelihood damages. The financial burden imposed by these floods will likely have long-term socio-economic repercussions, especially for families who lost homes and income sources. With extensive agricultural damage, it may take multiple planting seasons for communities to recover, worsening food insecurity and economic instability. Additionally, asset loss and displacement could drive rural-to-urban migration, straining city resources while depopulating villages.

As municipalities work toward recovery, key challenges include limited financial aid, slow agricultural rehabilitation, and infrastructure rebuilding. Municipalities like Bedkot, Laljhadi, and Krishnapur, which received minimal support, face particularly slow rehabilitation. Belauri's high agricultural losses require immediate action to restore farming infrastructure and support affected farmers. Meanwhile, rebuilding homes, roads, and irrigation systems in Bheemdatt and Belauri is critical for restoring economic activity. Restoring livelihoods is another priority. High income losses across multiple municipalities call for employment programs, vocational training, and small business support to prevent prolonged economic decline. Future disaster preparedness must also become a priority, particularly in Bheemdatt and Belauri, where large-scale damage highlights the urgent need for improved flood mitigation measures. Strengthening embankments, upgrading drainage systems, and investing in early warning systems will be key to reducing future flood risks. Community-based adaptation should be integrated into municipal planning to enhance local governance and disaster preparedness. By prioritizing inclusive recovery strategies that support the most vulnerable populations, municipalities can build resilience and better withstand future climate-related disasters.

The 2024 Kanchanpur floods have caused extensive economic damage, disproportionately affecting municipalities with weaker financial resilience. While some areas have received substantial aid, others remain vulnerable due to insufficient recovery funds and infrastructure damage. Addressing these challenges requires a multi-pronged approach, including immediate relief, long-term economic rehabilitation, and investments in climate-resilient infrastructure. By implementing effective recovery strategies and improving disaster preparedness, municipalities can rebuild stronger and create a more sustainable, disaster-resilient future.

Household Level Loss and Damage Analysis (In Millions-NPR)

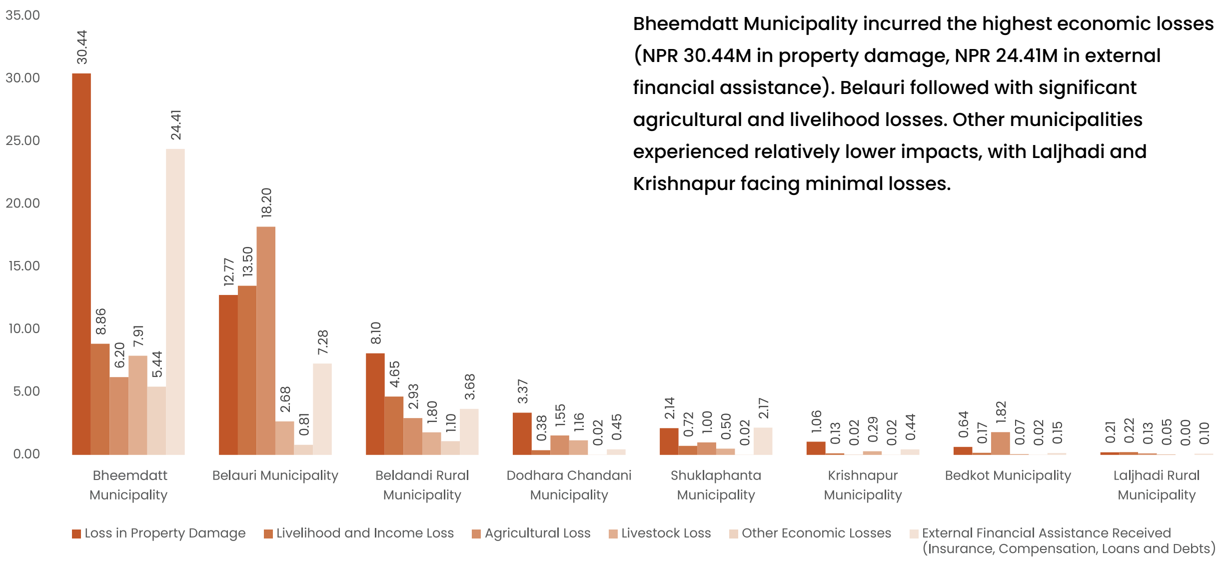


Figure 13: Breakdown of Economic Losses: Property, Agriculture, Livelihood, and Livestock Damages Across Nine Municipalities

Household Damage Assessment

Municipality Wise Household Damage Assessment

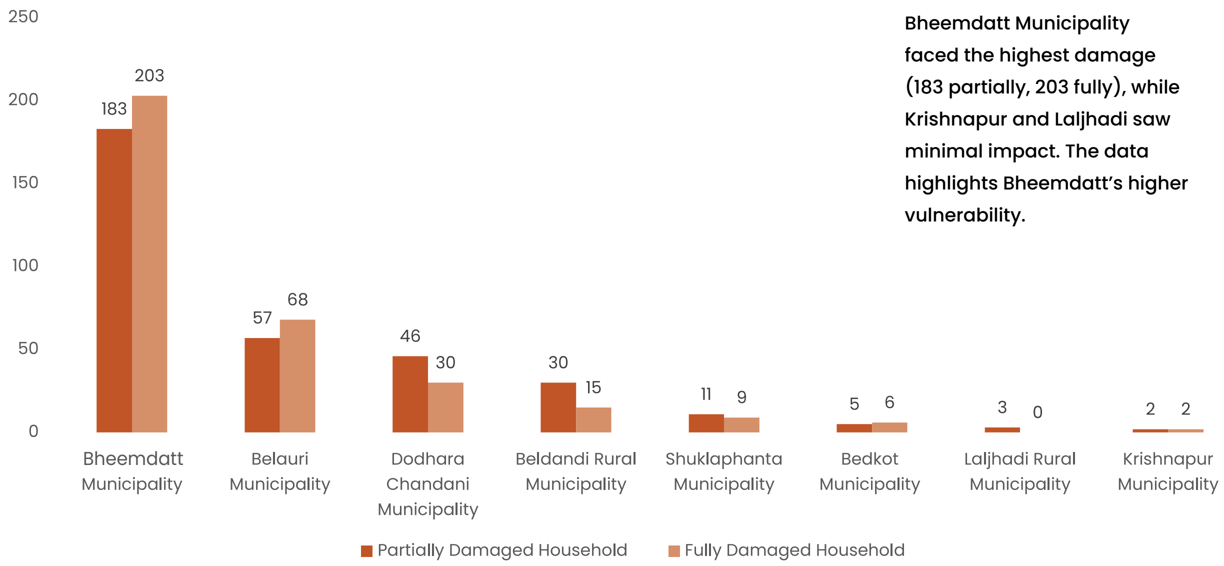


Figure 14: Household Damage Assessment in Kanchanpur

This chart summarizes the number of partially and fully damaged households across Kanchanpur municipalities. Bheemdatt Municipality reported the highest impact, with 183 partially damaged and 203 fully damaged households. Belauri Municipality follows with 57 partially damaged and 68 fully damaged households. Dodhara Chandani Municipality recorded 46 partially damaged and 30 fully damaged households. Beldandi Rural Municipality had 30 partially damaged and 15 fully damaged households. Shuklaphanta Municipality reported 11 partially damaged and 9 fully damaged households. Bedkot Municipality had 5 partially damaged and 6 fully damaged households, while Krishnapur Municipality recorded 2 partially and fully damaged households each. Laljhadi Rural Municipality reported only 3 partially damaged households with no fully damaged ones.

External Financial Assistance Received and Total Economic Loss

Municipality Wise ELD Assessment

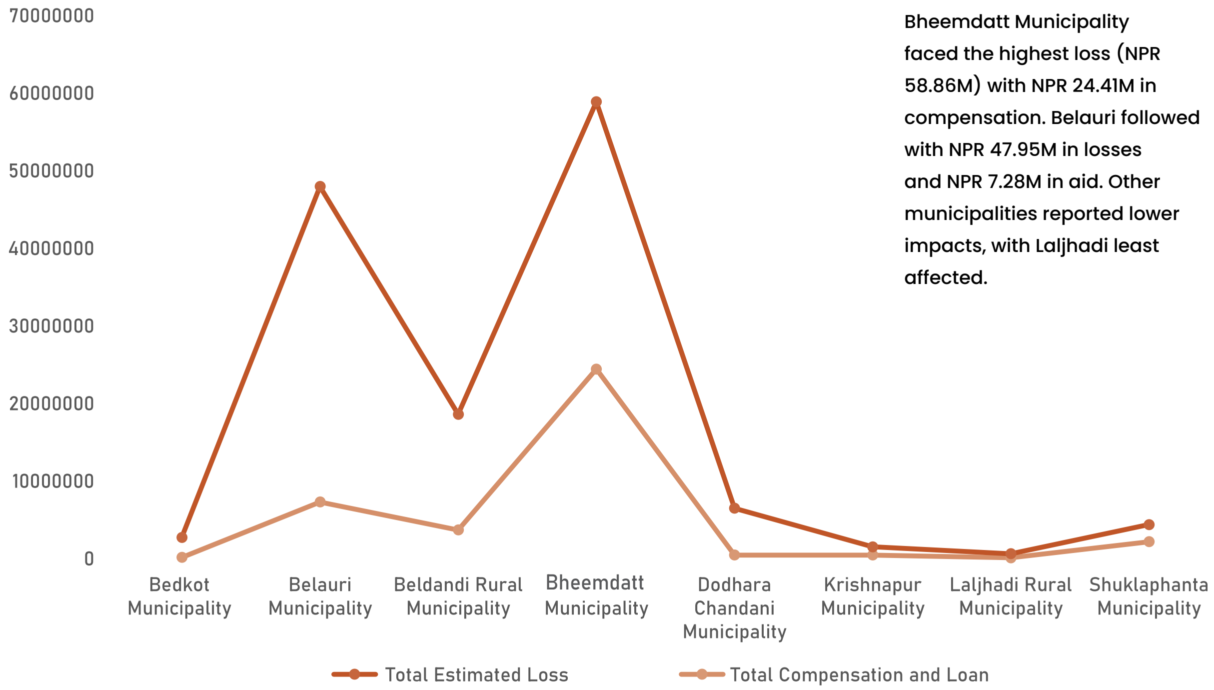


Figure 15: Comparision between compensation, loan and total estimated loss in Kanchanpur Municipalities

This chart analyzes the compensation and loans received alongside the total estimated losses across Kanchanpur municipalities. Bheemdatt Municipality recorded the highest total loss of NPR 58.86 million, with compensation and loans totaling NPR 24.41 million, reflecting both significant damage and a substantial reliance on external financial support. Belauri Municipality had a total estimated loss of NPR 47.95 million, with compensation and loans amounting to NPR 7.28 million, indicating a considerable gap between losses and recovery support.

Beldandi Rural Municipality reported a total loss of NPR 18.58 million, with compensation and loans of NPR 3.68 million, showing a moderate level of damage and assistance. Dodhara Chandani had a total loss of NPR 6.48 million, receiving NPR 449,300 in compensation and loans, which represents a more limited support response relative to its losses.

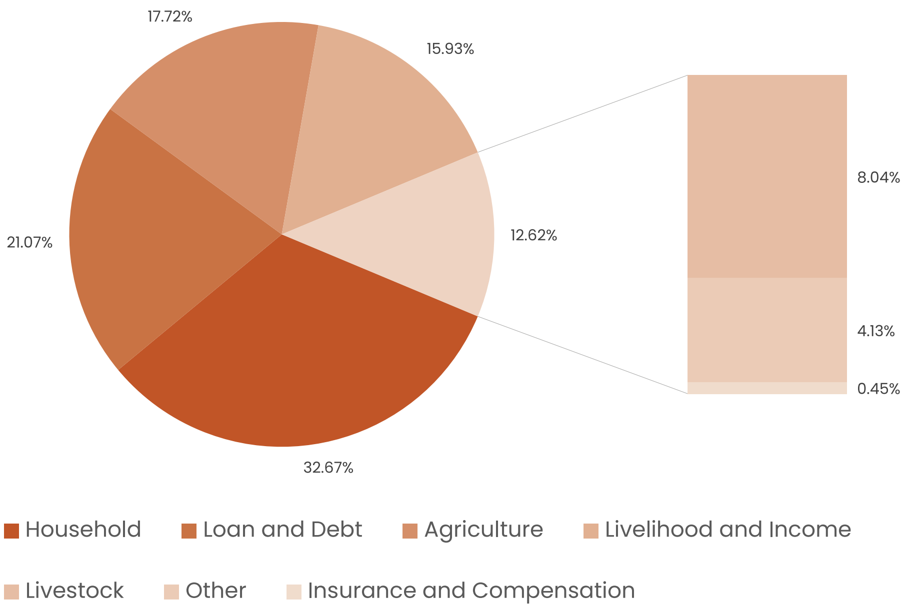
In Krishnapur Municipality, the total loss was NPR 1.52 million, with compensation and loans of NPR 440,000, demonstrating a relatively smaller scale of damage and financial recovery assistance. Bedkot Municipality reported a total loss of NPR 2.73 million, receiving NPR 150,000 in compensation and loans, while Laljhadi Rural Municipality experienced the smallest loss of NPR 610,000, with compensation and loans totaling NPR 100,000.

Finally, Shuklaphanta Municipality faced a total loss of NPR 4.37 million, with compensation and loans amounting to NPR 2.17 million, indicating moderate damage and financial assistance.

Overall, the data highlights the uneven distribution of compensation and loans in relation to the total losses, with larger municipalities such as Bheemdatt and Belauri receiving more financial support compared to smaller municipalities with lower reported losses.

Sectoral Economic Loss Analysis

Sectoral Loss in Percentage



Household losses (32.67%) account for the highest economic burden, followed by loan and debt obligations (21.07%), indicating financial distress. Agricultural (17.72%) and livelihood losses (15.93%) highlight significant disruptions in income and food security. Livestock losses (8.04%) further contribute to economic strain, while minimal insurance and compensation (0.45%) suggest limited financial recovery support.

Figure 16: Sectoral Loss Distribution in Kanchanpur

This chart presents the distribution of sectoral losses in Kanchanpur, shown as percentages of total losses and arranged in descending order. The household sector accounts for the largest share at approximately 32.67%, indicating a significant impact on residential stability. The loan and debt sector follows at 21.07%, reflecting the financial burdens incurred to cope with the flooding's effects. Agricultural losses represent about 17.72% of total losses, highlighting adverse impacts on food production. The livelihood and income sector accounts for approximately 15.93%, emphasizing economic challenges faced by individuals and families. Livestock losses comprise around 8.04%, while other losses account for about 4.13%. Insurance and compensation claims represent a minimal share at around 0.45%, indicating limited recovery through these channels.

Household Damage Analysis

Municipality Wise Damage Analysis

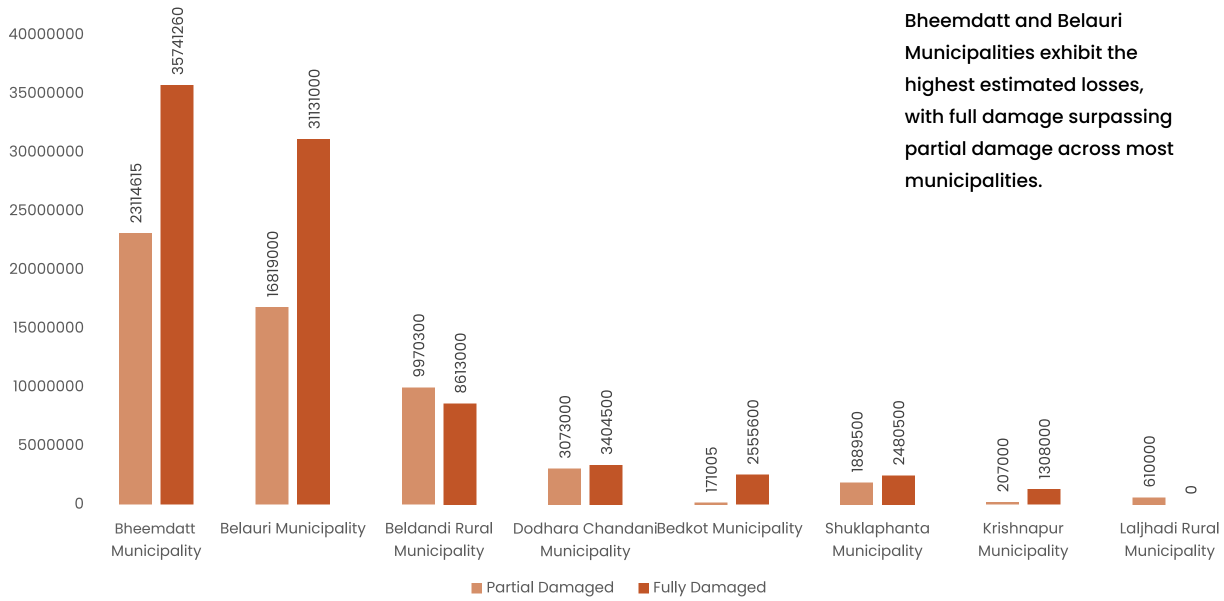


Figure 17: Household Damage Assessment in Kanchanpur

This chart presents the extent of household damage across Kanchanpur municipalities, with total loss. Bheemdatt Municipality reported the highest losses at NPR 58.85 million, with NPR 23.11 million in partial damage and NPR 35.74 million in fully damaged houses. Belauri Municipality follows with total losses of NPR 47.95 million, including NPR 16.82 million in partial damage and NPR 31.13 million in fully damaged houses. Beldandi Rural Municipality assessed losses of NPR 18.58 million, comprising NPR 9.97 million in partial damage and NPR 8.61 million in fully damaged houses. Dodhara Chandani reported total losses of NPR 6.48 million, with NPR 3.07 million in partial damage and NPR 3.40 million in fully damaged houses. Shuklaphanta Municipality had total losses of NPR 4.37 million, including NPR 1.89 million in partial damage and NPR 2.48 million in fully damaged houses. Bedkot Municipality reported total losses of NPR 2.73 million, with NPR 0.17 million in partial damage and NPR 2.56 million in fully damaged houses. Krishnapur Municipality assessed total losses of NPR 1.52 million, consisting of NPR 0.21 million in partial damage and NPR 1.31 million in fully damaged houses, while Laljhadi Rural Municipality reported total losses of NPR 0.61 million, with no fully damaged houses recorded.

Comparison of Estimated Loss in Partial Damage, External Financial Assistance and Loan Taken

Municipality Wise Estimated Loss in Partial Damage, Loan Taken and External Financial Assistance

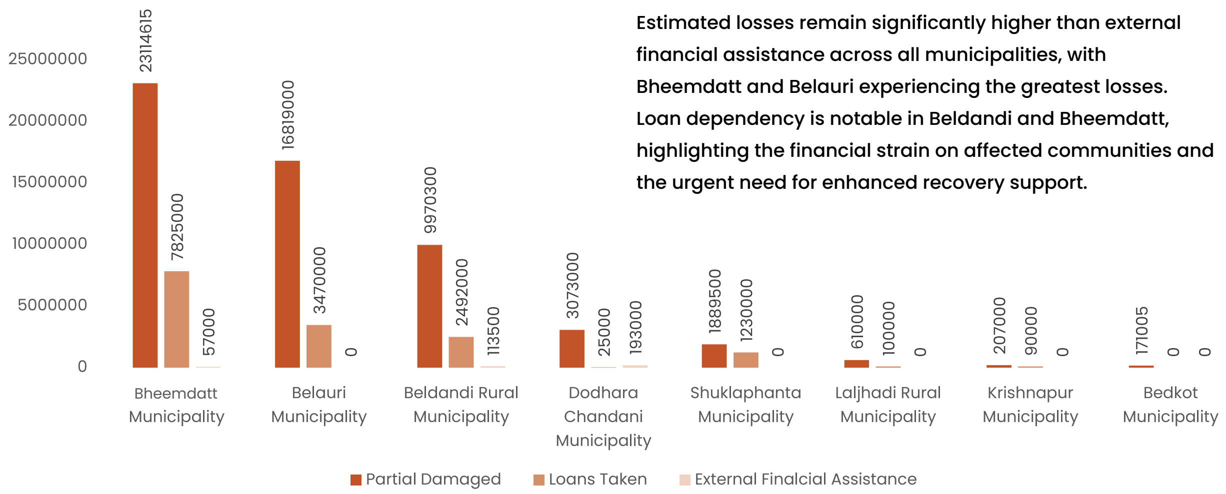


Figure 18: Comparison of Estimated Loss in Partial Damage, External Financial Assistance Received and Loans Taken

This chart details the financial assistance and loans taken by Kanchanpur municipalities, alongside reported partial damage as monetary losses. Bheemdatt Municipality experienced the highest partial damage at NPR 23.11 million, receiving NPR 57,000 in external assistance and taking loans of NPR 7.83 million. Belauri Municipality follows with NPR 16.82 million in partial damage, with no external assistance but loans of NPR 3.47 million. Beldandi Rural Municipality reported NPR 9.97 million in partial damage, receiving NPR 113,500 in assistance and taking loans of NPR 2.49 million. Dodhara Chandani had partial damage of NPR 3.07 million, with NPR 193,000 in assistance and a loan of NPR 25,000. Shuklaphanta Municipality reported NPR 1.89 million in partial damage, with no external assistance but loans of NPR 1.23 million. Bedkot Municipality recorded partial damage of NPR 171,005 with no financial support or loans taken. Krishnapur Municipality reported NPR 207,000 in partial damage, with loans of NPR 90,000 and no external assistance. Laljhadi Rural Municipality experienced partial damage of NPR 610,000, receiving a loan of NPR 100,000 but no external assistance.

Comparison of Estimated Loss in Full Damage, External Financial Assistance and Loan Taken

Municipality Wise Estimated Loss in Full Damage, Loan Taken and External Financial Assistance

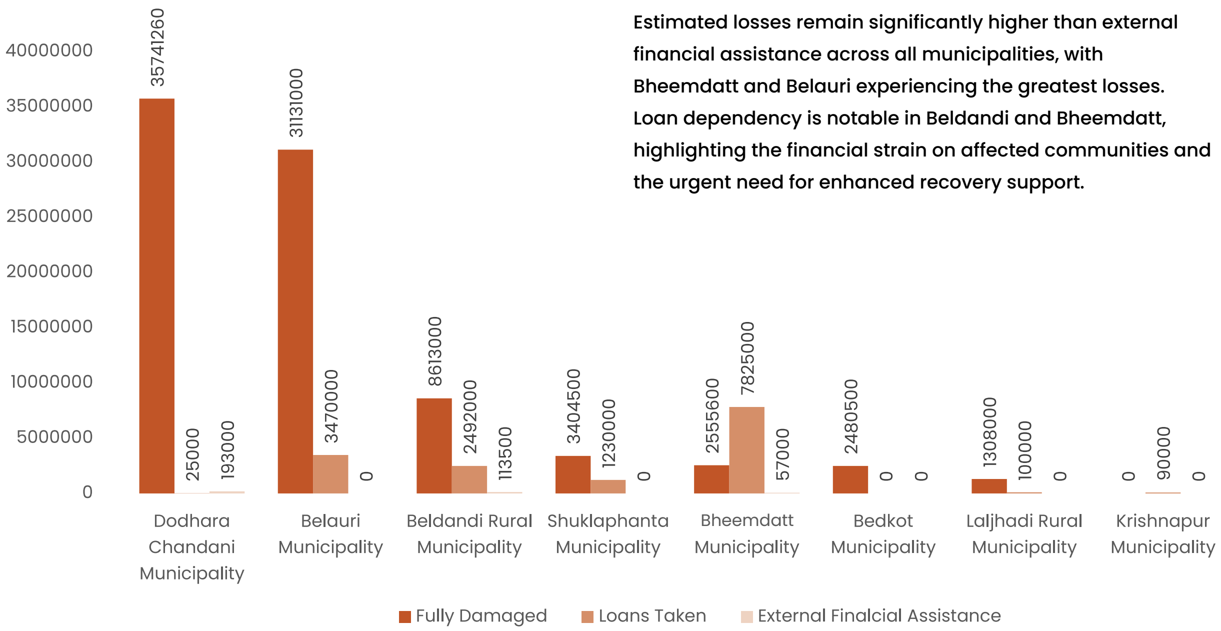


Figure 19: Comparison of Estimated Loss in Full Damage, External Financial Assistance Received and Loans Taken

The chart provides a detailed analysis of the extent of damage to fully destroyed houses across Kanchanpur municipalities, alongside external financial assistance and loans taken for recovery.

Bheemdatt Municipality reported the highest damage, NPR 35.74 million, with minimal assistance of NPR 189,000 and loans of NPR 16.34 million. This indicates substantial damage and a heavy reliance on loans for recovery. Belauri Municipality followed with NPR 31.13 million in damage, receiving only NPR 2,000 in aid and loans of NPR 3.81 million, showing a significant reliance on credit despite limited support.

In Beldandi Rural Municipality, damage amounted to NPR 8.61 million, with assistance of NPR 65,000 and loans of NPR 1.01 million, reflecting less severe financial strain compared to Bheemdatt and Belauri. Dodhara Chandani had damage of NPR 3.40 million, with higher external assistance of NPR 191,300 and a smaller loan of NPR 40,000, indicating better external support.

Shuklaphanta Municipality had NPR 2.48 million in damage but received no external assistance, relying on loans of NPR 0.94 million. Bedkot Municipality reported NPR 2.56 million in damage with no aid, taking loans of NPR 150,000. Krishnapur Municipality experienced the least damage at NPR 1.31 million, with no external assistance and loans of NPR 350,000.

Lastly, Laljhadi Rural Municipality recorded no damage and did not require assistance or loans, suggesting either minimal impact or self-recovery.

Overall, municipalities with higher damage, like Bheemdatt and Belauri, took out larger loans, while those with less damage or better aid, like Dodhara Chandani, required fewer financial resources. This highlights the crucial role of loans and external assistance in disaster recovery.

Non-Economic Loss and Damage Assessment Findings

The pilot study of 670 flood-affected households in Kanchanpur highlights significant non-economic loss and damage (NELD) from the 2024 floods, affecting health, psychosocial well-being, cultural heritage, social ties, education, and the environment.

Health impacts include increased waterborne diseases and mental health struggles, while displacement weakened community bonds. Cultural losses stemmed from damage to religious sites, and school disruptions affected children's education. Environmental damage, such as soil erosion and water contamination, further worsened recovery.

Bheemdatt and Belauri saw severe psychosocial and cultural impacts due to higher displacement, while other areas faced healthcare and social recovery challenges. Addressing NELD requires mental health support, cultural restoration, education aid, and environmental rehabilitation to ensure a holistic and resilient recovery.

Post-flood Household-level Health Impacts

The July 2024 flood in Kanchanpur District caused significant health and well-being challenges for the affected communities. The data visualizations illustrate the extent of post-flood health impacts, revealing widespread illness, barriers to healthcare access, and the financial burden of medical treatment. These findings provide insight into the non-economic losses and damages experienced by the population, which extend beyond immediate financial costs to long-term consequences for public health, social stability, and livelihoods.

One of the most striking observations from the data is the prevalence of post-flood illnesses, with 215 cases of waterborne diseases being the most commonly reported, followed by 85 cases of respiratory infections and 99 cases of skin diseases This trend underscores the impact of flood-induced contamination of water sources, inadequate sanitation, and overcrowded living conditions in affected areas. Additionally, 71 reported mental health concerns suggest the psychological toll of displacement, loss of property, and stress associated with rebuilding lives. The long-term implications of these health impacts are severe, as untreated illnesses can lead to chronic health conditions, further exacerbating vulnerability among already disadvantaged populations.

The financial burden of medical treatment is another significant concern, particularly for municipalities like Bhimdutta and Beldandi, where the highest medical expenses were recorded. Households in these areas faced significant costs for both injury and illness treatment, highlighting the strain on families who may already struggle with economic recovery. Without sufficient financial assistance or accessible healthcare services, many individuals may delay or forego necessary medical treatment, leading to prolonged suffering and increased mortality rates. Over time, this could deepen poverty levels as affected families are forced to divert resources from other essential needs such as food, education, and housing repairs.

In addition to direct health impacts, challenges in accessing healthcare services further compounded the crisis. The data show that damage to health posts (401 cases) was the most significant barrier, followed by disruptions in mobility (113 cases) and psychological stress related to travel (51 cases). These factors indicate that even where medical services were available, physical and mental barriers prevented people from seeking timely care. In the long term, the disruption of healthcare services can weaken overall health resilience in the region, making communities more susceptible to future disease outbreaks and reducing their ability to cope with subsequent disasters.

The broader social and economic consequences of these health impacts are profound. With high numbers of households reporting illness, productivity losses are inevitable. Many flood-affected families depend on daily-wage labor, farming, and small businesses for survival, and when household members are sick or injured, their ability to work is diminished. This loss of productivity not only affects individual income but also slows economic recovery at the community level. Moreover, the psychological stress experienced by affected populations can lead to long-term trauma, increasing the need for mental health support services that are often unavailable in rural and disaster-prone areas.

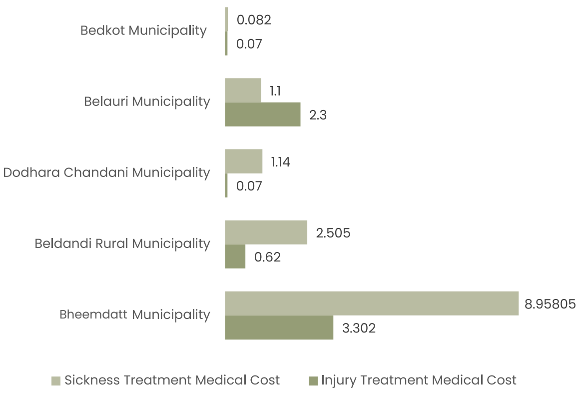
Children and elderly individuals are particularly vulnerable to the long-term effects of post-flood health crises. Children suffering from illnesses such as malnutrition or chronic infections due to unsafe living conditions may experience developmental setbacks, impacting their education and future opportunities. Likewise, elderly individuals with pre-existing health conditions are at a higher risk of severe complications from untreated infections or prolonged exposure to poor living conditions. Without targeted intervention, these vulnerable groups will face worsening health outcomes, further straining local healthcare systems.

Given these findings, a strengthened approach to disaster preparedness and response is essential. Investments in resilient healthcare infrastructure, mobile health services, and mental health support systems are crucial to mitigating the impacts of future floods. Additionally, improved sanitation measures, early disease detection, and community-based health education initiatives can help reduce the prevalence of waterborne and vector-borne diseases. Without such interventions, the cycle of health vulnerability and economic hardship in flood-affected communities will continue, making long-term recovery more challenging.

Addressing the health impacts of the July 2024 Kanchanpur flood presents both challenges and opportunities for the local government. One of the key challenges is the limited financial and technical capacity to rebuild damaged health infrastructure while simultaneously addressing the immediate medical needs of affected populations. The government must navigate budget constraints, coordinate with multiple stakeholders, and ensure efficient resource allocation, all while maintaining routine public health services. Another major challenge is the integration of mental health and psychosocial support into disaster response, as such services are often overlooked in favor of physical healthcare. However, this crisis also provides an opportunity to strengthen disaster preparedness by investing in resilient healthcare systems, mobile medical units, and improved early warning systems. It is also an opportunity to foster collaboration with national and international agencies, leveraging external funding and expertise to build a more robust response framework. Moreover, by engaging communities in recovery efforts, local governments can promote decentralized, community-driven solutions such as localized health monitoring and awareness campaigns, ensuring long-term sustainability in disaster risk reduction and public health resilience.

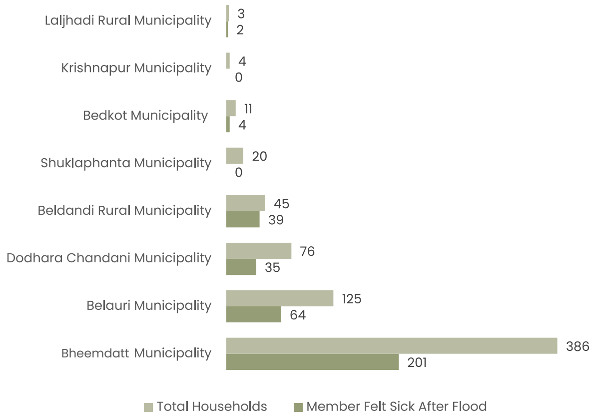
The July 2024 Kanchanpur flood not only caused immediate health crises but also triggered a series of interconnected social and economic consequences that will persist for years. The data clearly highlight the need for comprehensive flood response strategies that go beyond financial relief and infrastructure repair to include health-centered interventions. Addressing these non-economic losses is crucial for ensuring the long-term resilience of communities and preventing future disasters from having similarly devastating impacts.

Household Level Post Flood Health Impact in Kanchanpur (In Lakhs-NPR)



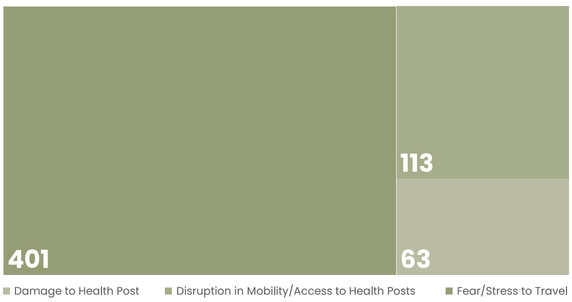
Bheemdatt Municipality impacted the highest medical costs (12.25 Lakhs) for both injury and sickness treatment, followed by Belauri and Beldandi. Other municipalities reported minimal but Shuklaphanta, Laljhadi and Krishnapur reported no expenses.

Post-Flood Health Impacts Analysis



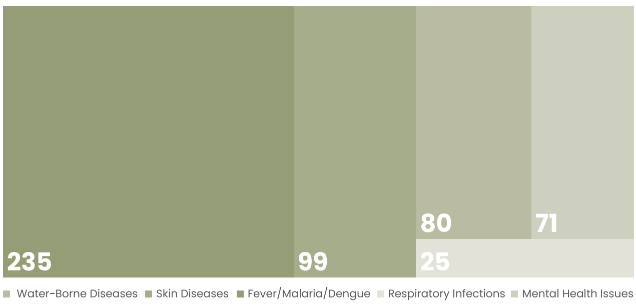
Bheemdatt Municipality had the highest health impact where members of 201 households felt sick in 386 surveyed households.

Challenges Faced in Access to Health Services After Flood



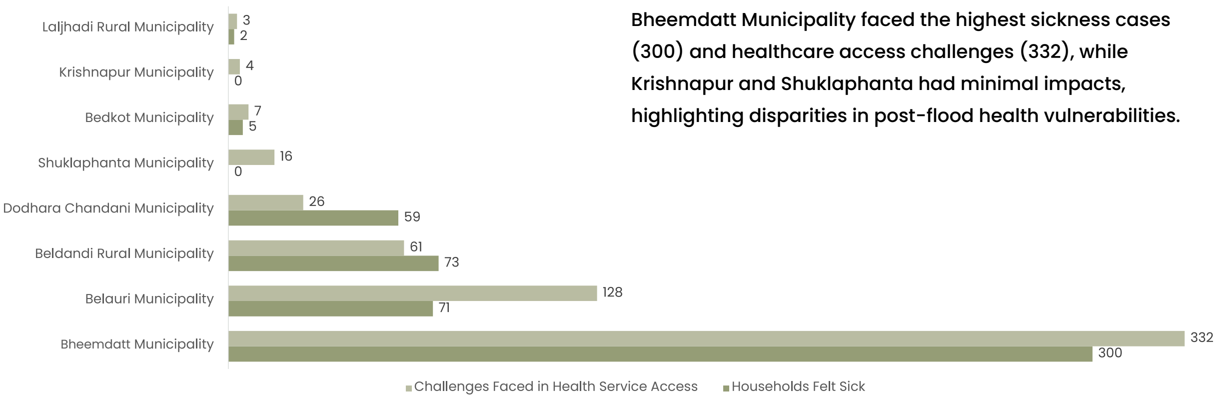
About 70% of households reported fear/stress to travel.

Health Impact After Flood



Fever, Malaria and Dengue were the most prevalent illness with a total of 235 household cases, followed by skin diseases and the least reported illness was respiratory infection.

Household Level Post-Flood Health Impact



Bheemdatt Municipality faced the highest sickness cases (300) and healthcare access challenges (332), while Krishnapur and Shuklaphanta had minimal impacts, highlighting disparities in post-flood health vulnerabilities.

Figure 20: Municipality Wise Post-Flood Health Impacts Analysis

Post Flood Household Level Psychological Impacts

Disasters not only cause physical destruction but also leave lasting psychological scars on affected communities. The July 2024 floods in Kanchanpur district disrupted lives in multiple ways, including widespread mental distress among households. Beyond the immediate economic losses, many families faced emotional trauma, anxiety, and uncertainty about the future. The psychological toll of disasters is often overlooked in recovery efforts, yet it plays a crucial role in shaping long-term resilience and social stability. The following analysis explores the extent of mental stress experienced by affected households, the availability of psychological support, and the disparities in access to such services across different municipalities.

As illustrated in Figure, 440 households reported experiencing psychological distress due to the flood, whereas only 107 households received any form of psychological support. This stark disparity highlights a critical gap in post-disaster mental health interventions, leaving a vast majority of affected individuals without necessary psychosocial assistance. The long-term consequences of unaddressed psychological trauma could manifest in prolonged anxiety, depression, and post-traumatic stress disorder (PTSD), ultimately affecting productivity, community interactions, and overall well-being.

The illustration further details the sources of psychological support received post-flood. The majority of affected individuals in 109 households relied on community-based support systems, emphasizing the importance of informal networks in coping with trauma. However, institutional responses were significantly lower, with only 41 households receiving assistance through community facilitators, 24 from Asian Service Nepal, and just three through other sources. The limited reach of formal mental health interventions underscores a lack of structured psychosocial support mechanisms within disaster response frameworks. Given the critical role of mental health in long-term recovery, the absence of widespread, professionally guided psychological support may hinder affected households' ability to rebuild resilience.

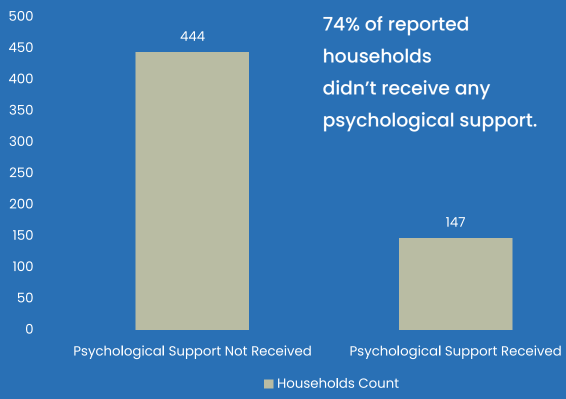
Similary, the graph presents a municipality-wise distribution of psychological support received. The trend indicates an uneven distribution of support, with some municipalities benefiting more than others. The highest levels of psychological support were recorded in the municipality with the greatest outreach efforts, while other areas, such as Shuklaphanta Municipality, received little to no assistance. This uneven provision of psychological aid suggests disparities in service availability, possibly due to logistical challenges, resource constraints, or a lack of trained professionals. The neglect of certain areas raises concerns about the inclusivity and effectiveness of mental health responses in disaster management.

The long-term implications of insufficient psychological support extend beyond individual suffering. Persistent mental distress can weaken social cohesion, reduce economic participation, and increase dependency on external aid. Households experiencing unaddressed trauma may struggle to reintegrate into normal life, affecting their ability to engage in livelihood activities, education, and community rebuilding efforts. Furthermore, children and vulnerable populations, such as the elderly and marginalized groups, may face heightened risks of emotional distress, potentially leading to generational cycles of psychological hardship.

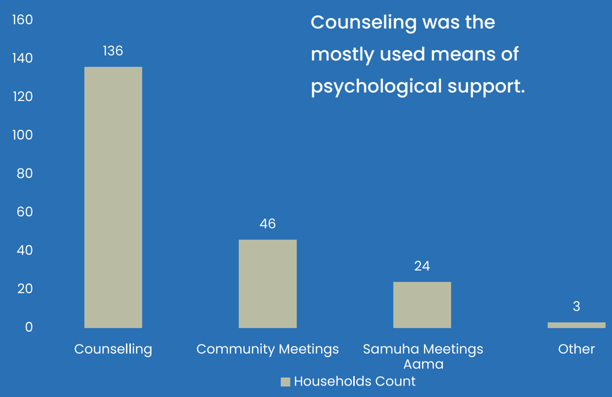
For local governments, addressing these challenges requires an integrated approach that prioritizes mental health alongside physical reconstruction. Strengthening mental health infrastructure, training local facilitators in psychosocial support, and ensuring equitable access to mental health services across municipalities are essential steps in fostering long-term recovery. Additionally, community-based interventions, peer support programs, and culturally sensitive mental health initiatives can enhance resilience and mitigate the long-term consequences of disaster-induced trauma.

While physical reconstruction often takes priority in disaster recovery, the findings highlight the urgent need for greater attention to mental health support. The psychological distress experienced by affected households, coupled with limited institutional interventions, underscores the gaps in disaster response frameworks. If left unaddressed, prolonged emotional suffering can hinder social cohesion, economic recovery, and overall community resilience. Moving forward, integrating mental health support into disaster management strategies, expanding access to psychosocial services, and ensuring equitable distribution of aid will be critical in fostering a more comprehensive and sustainable recovery for affected populations.

Mental Stress or Trauma Experienced Households



Psychological Support Received Post Flood



PSYCHOLOGICAL SUPPORT RECEIVED

The affected households of Dodhara Chandani received the highest psychological support while Bedkot municipality received the least.

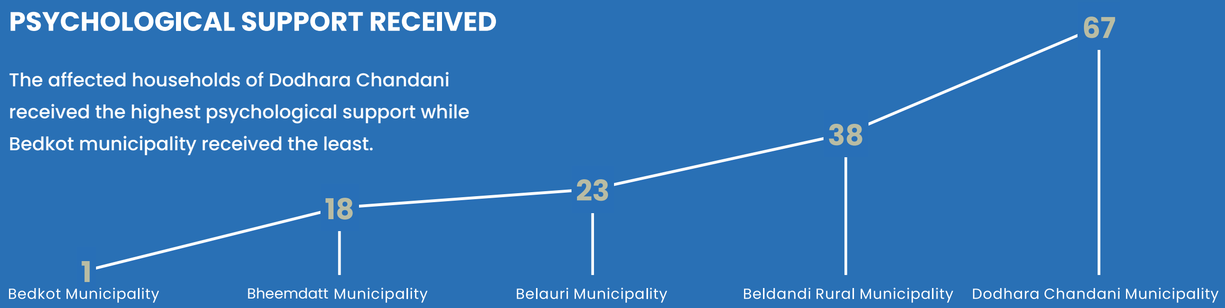


Figure 21: Post-flood psychological impacts and support received

Post Flood Cultural Impacts

The July 2024 floods in Kanchanpur district caused significant non-economic losses to households, particularly in cultural and religious aspects. The cultural loss assessment reveals that 23 households reported the loss of cultural or religious items, while nine households indicated disruptions to community-wide cultural or religious practices. These disruptions reflect a broader impact beyond material loss, affecting emotional well-being and social cohesion.

Community-wide cultural activities were also severely affected, as illustrated in the figure. Festivals experienced the highest impact, with six households reporting disruptions, followed by daily prayers in three households and cultural events such as Bratabandha and marriage ceremonies in two households. The absence of reported impacts on death rituals suggests that essential rites were prioritized despite the disaster. However, the disruption of routine and communal religious practices could weaken social bonds, leading to psychological distress and a loss of shared cultural identity among affected families.

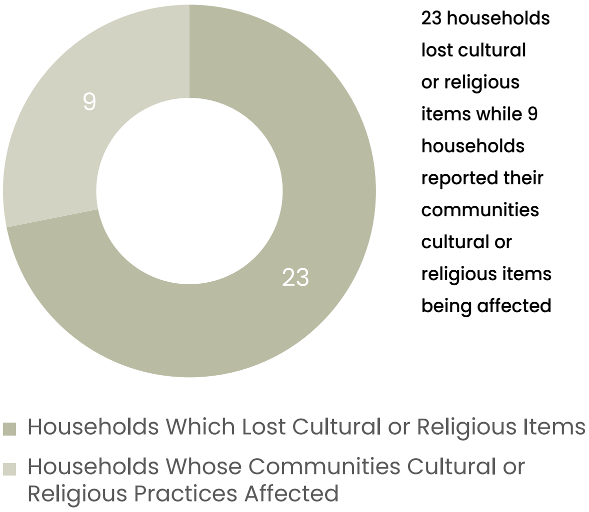
Furthermore, Figure highlights the destruction of cultural sites, with 42 sites affected, including 40 temples. The complete absence of damage to stupas, mosques, churches, or cremation sites suggests that Hindu places of worship bore the brunt of the destruction. This selective damage could lead to disparities in recovery efforts and exacerbate vulnerabilities within specific communities, particularly those reliant on temples for spiritual and social gatherings.

The long-term implications of these cultural losses extend beyond the immediate aftermath of the flood. The loss of religious artifacts and damage to temples can erode cultural heritage, diminishing intergenerational knowledge transfer and weakening community resilience. Additionally, disrupted festivals and religious practices may reduce social interaction and collective coping mechanisms, potentially increasing isolation and mental health struggles among affected individuals. For many, religious and cultural spaces serve as sources of emotional support, and their destruction can lead to feelings of displacement even within one's own community.

For local governments, addressing these impacts poses significant challenges. Reconstruction efforts often prioritize physical infrastructure and economic recovery over cultural restoration, leaving communities to bear the burden of re-establishing their traditions. Limited resources, competing priorities, and the difficulty of quantifying non-economic losses make it challenging to advocate for policies that support cultural rehabilitation. Without targeted interventions, there is a risk of cultural erosion, weakening the social fabric essential for long-term community resilience. Integrating cultural preservation into disaster recovery planning, allocating resources for temple reconstruction, and supporting affected households in reviving disrupted practices will be critical to fostering long-term recovery and well-being.

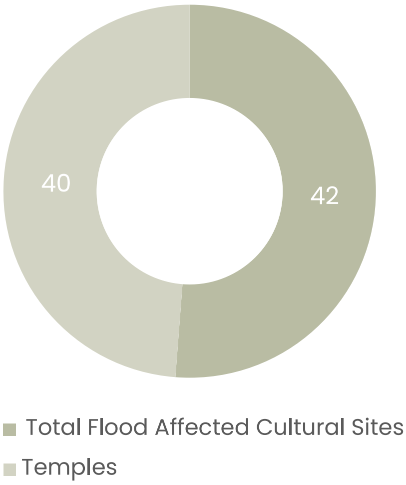
The floods in Kanchanpur district resulted in significant non-economic losses related to cultural and religious life, disrupting household and community practices while damaging key cultural sites. These impacts extend beyond material loss, affecting mental health, social cohesion, and intergenerational traditions. Addressing such losses requires a comprehensive approach that acknowledges the role of cultural identity in disaster resilience and ensures that recovery efforts restore not just infrastructure but also the social and cultural well-being of affected communities.

Cultural Loss Assessment



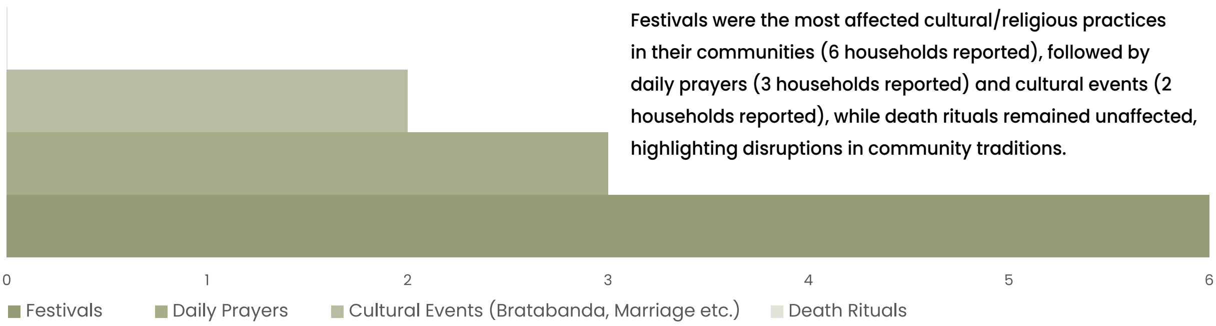
23 households lost cultural or religious items while 9 households reported their communities cultural or religious items being affected

Flood Affected Cultural Sites



42 cultural sites were affected including 40 temples.

Sectoral Flood Affected in the Community



Festivals were the most affected cultural/religious practices in their communities (6 households reported), followed by daily prayers (3 households reported) and cultural events (2 households reported), while death rituals remained unaffected, highlighting disruptions in community traditions.

Figure 22: Post-flood impacts on Culture, Religion and Traditions

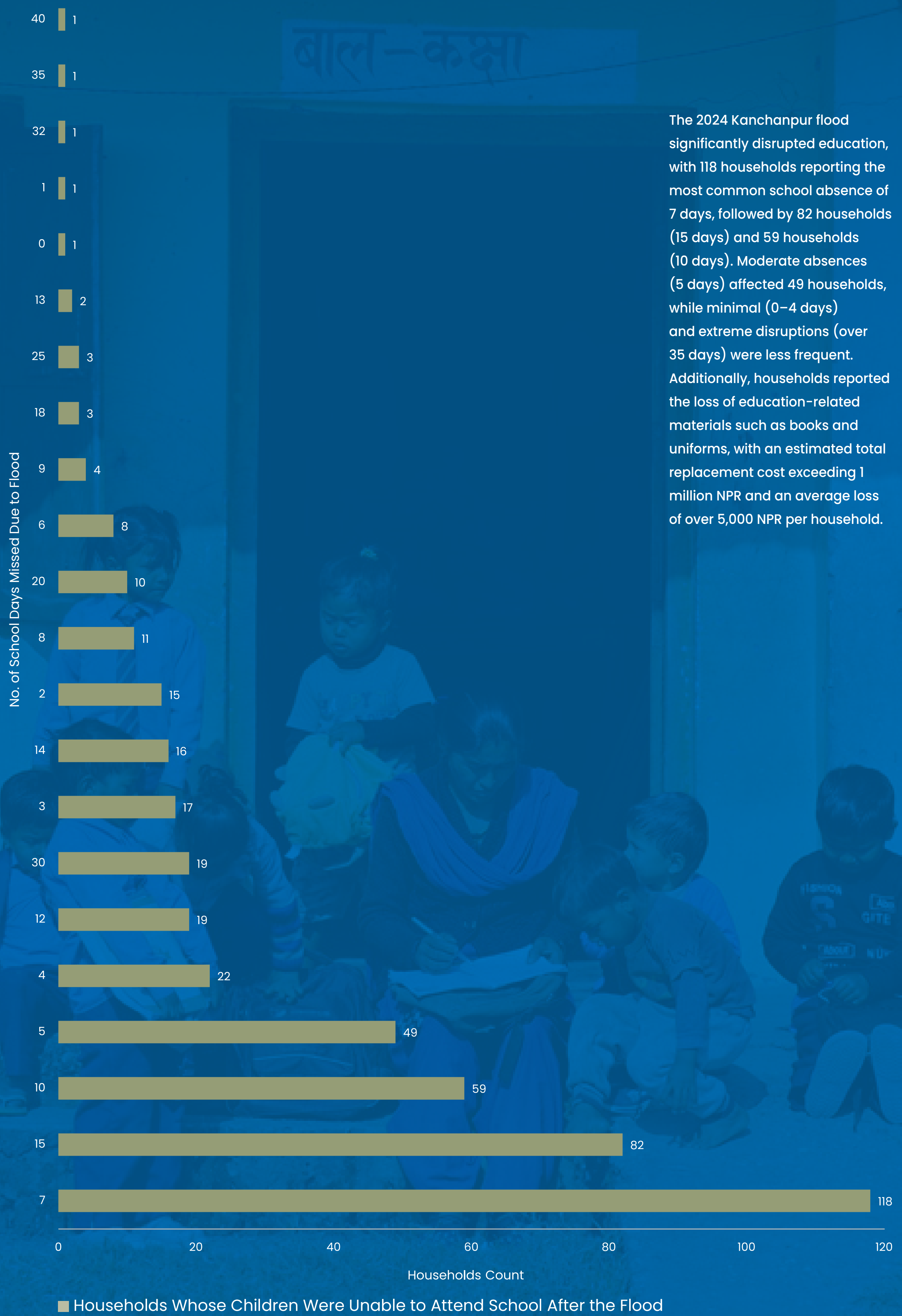
Post Flood Household Level Educational Impacts

Flooding is a recurring natural disaster that significantly disrupts daily life, particularly affecting children's education. The figure illustrates the number of households whose children were unable to attend school due to flooding, categorized by the number of school days missed. The data reveal that a substantial number of students missed multiple days of schooling, with the highest concentration at seven days as reported by 118 households, followed by 15 days by 82 households and 10 days by 59 households. These figures highlight the extent to which natural disasters can disrupt children's academic continuity, creating long-term educational setbacks.

For community members, the disruption in schooling extends beyond lost classroom hours. Many households face difficulties ensuring their children return to school after prolonged absences, as families may prioritize survival over education. Flooding often damages school infrastructure, displaces families, and limits access to educational materials. In rural or low-income communities, children missing school due to floods may be pulled into household labor or economic activities, further reducing their likelihood of returning to formal education. Additionally, families dealing with economic losses may struggle to afford school-related expenses such as uniforms, books, and transportation, increasing the risk of dropouts. For local authorities, addressing these educational disruptions poses a significant challenge. Restoring school infrastructure, ensuring safe transportation, and providing support to affected families require coordinated disaster response strategies. Stronger policies on disaster preparedness in education, such as temporary learning centers and psychological support programs, are essential. Without intervention, recurring floods could weaken the educational system and reduce literacy rates in affected regions.

The psychological impact on students is another major concern. The stress of displacement, witnessing property loss, and experiencing their families' economic struggles can contribute to emotional distress, reducing motivation and engagement in learning. Over time, these challenges can lead to lower academic performance, increased dropout rates, and limited future opportunities. Addressing these issues requires an integrated approach involving community resilience planning, disaster-responsive education policies, and mental health support for affected children. Without proactive measures, the long-term impact of flooding on education could deepen social inequalities and hinder community development.

Impact in Education After Flood



The 2024 Kanchanpur flood significantly disrupted education, with 118 households reporting the most common school absence of 7 days, followed by 82 households (15 days) and 59 households (10 days). Moderate absences (5 days) affected 49 households, while minimal (0–4 days) and extreme disruptions (over 35 days) were less frequent. Additionally, households reported the loss of education-related materials such as books and uniforms, with an estimated total replacement cost exceeding 1 million NPR and an average loss of over 5,000 NPR per household.

Figure 23: Reported cases of students unable to attend school after the flood

Post-flood Environmental Impacts

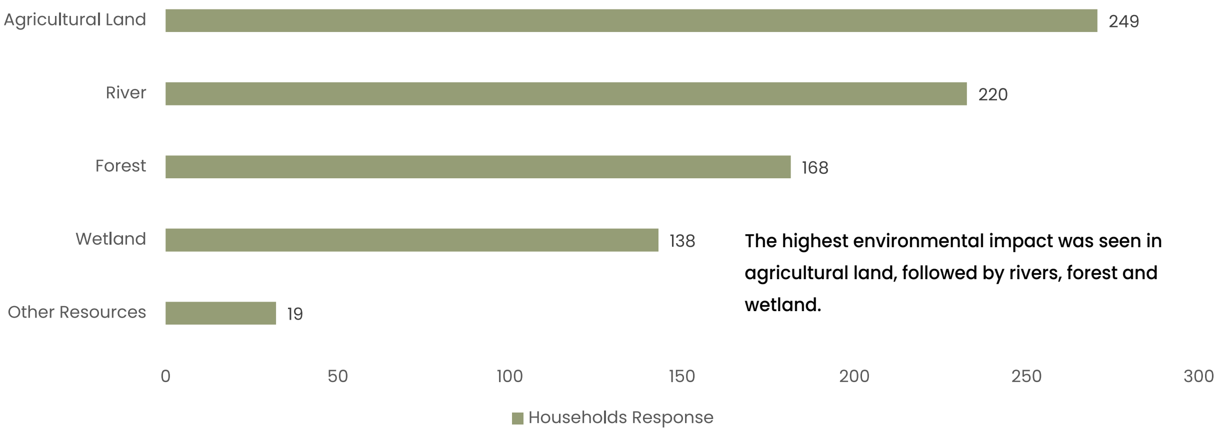
The post-flood environmental impacts in Kanchanpur have affected various natural resources, with significant consequences for agriculture, water quality, and biodiversity. The data highlights that agricultural land in 249 households, rivers in 220 households, and forests in 168 households were among the most affected sectors, followed by wetlands in 138 households. This indicates that flooding had a profound impact on both terrestrial and aquatic ecosystems, leading to soil erosion, water contamination, and habitat degradation.

In terms of water resource quality, the most reported impacts included vegetation loss reported y 164 households, reduced water levels in lakes or rivers by 157 households, and wildlife displacement or mortality by 78 households. Additionally, human-wildlife conflicts by 59 households and increased cases of forest fires by 28 households were noted, possibly due to habitat disruption and changes in ecosystem dynamics. The introduction of invasive plant species was a lesser but still notable impact as reported by 11 households, indicating a potential shift in biodiversity patterns post-flooding.

Households also reported broader environmental changes, with 410 households citing soil erosion and land degradation, 319 mentioning damage to natural resources, and 225 observing noticeable changes in water quality. Additionally, 174 households noted an impact on local wildlife and biodiversity, reinforcing concerns about ecosystem stability and the long-term effects of flooding on habitat sustainability.

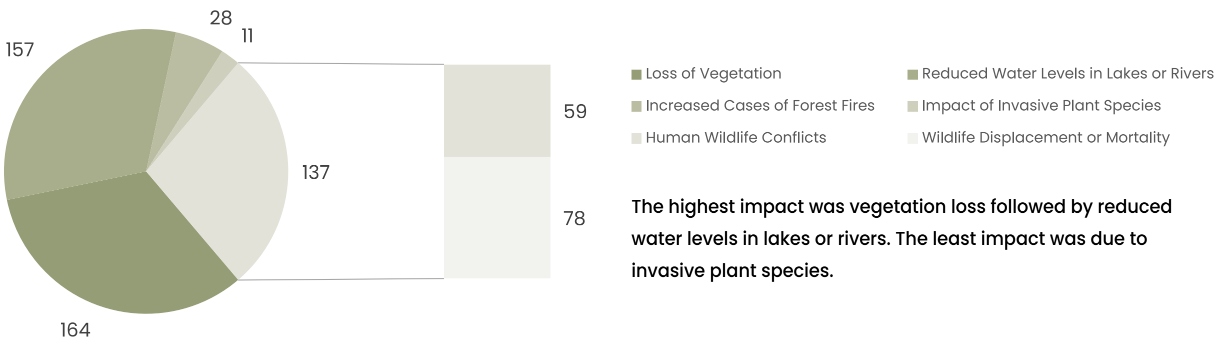
These findings underscore the urgent need for post-flood ecological restoration efforts, including soil conservation practices, reforestation, and wetland rehabilitation. Strengthening riverbanks, implementing sustainable land management strategies, and enhancing floodplain ecosystem resilience could help mitigate future environmental risks and support biodiversity conservation.

Sectoral Post Flood Environmental Impact



The highest environmental impact was seen in agricultural land, followed by rivers, forest and wetland.

Post Flood Sectoral Impact in the Quality of Water Resources



The highest impact was vegetation loss followed by reduced water levels in lakes or rivers. The least impact was due to invasive plant species.

Household Response on Post Flood Environmental Impact

Soil erosion and land degradation was reported to be the highest environmental impact followed by impact in natural resources.

Figure 24: Post-flood impacts on Environmental, Ecosystem Services and Biodiversity

Post Flood Mobility and Transportation Impacts

The July 2024 floods in Kanchanpur significantly disrupted mobility and transportation, affecting access to essential services and daily commuting patterns. The data indicates that 224 households experienced disruptions in mobility and transportation, while 446 households remained unaffected. This suggests that although a majority of respondents did not face mobility challenges, a substantial number struggled with transportation issues, highlighting the flood's uneven impact across different communities.

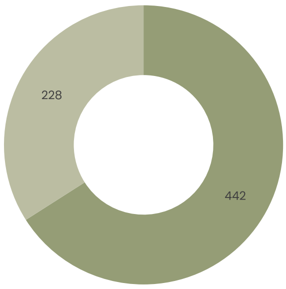
The use of transportation services was heavily impacted, with noticeable spikes in disruptions across various modes. The most affected mode appears to be public transport and personal vehicles, with a high number of respondents reporting difficulties. This could be attributed to damaged road infrastructure, submerged streets, or washed-out bridges. The accessibility of essential services was also severely impacted, with 41 households reporting difficulty accessing medical facilities, 61 struggling to reach markets, and 55 facing challenges in traveling to workplaces. These figures underscore the cascading effects of transportation disruptions on livelihood, healthcare access, and daily necessities.

Different aspects of transportation were affected in varying degrees. The primary concerns were road damage and flooding of key transport routes, which hindered mobility and prolonged recovery efforts. The data also shows that the most affected mode of transportation was walking, with nearly 300 households reporting difficulties.

This suggests that flooded roads and pathways created significant barriers, even for pedestrians. Public transportation was the second-most affected mode, followed by private vehicles and bicycles, indicating that both urban and rural commuters faced substantial mobility challenges.

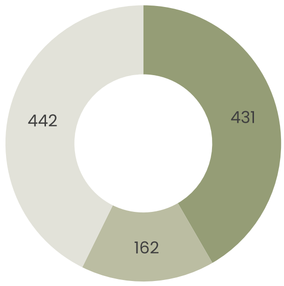
These findings highlight the urgent need for improving transportation resilience in flood-prone areas. Strengthening road infrastructure, ensuring efficient drainage systems, and implementing early warning mechanisms can help mitigate mobility disruptions during future flood events. Additionally, establishing emergency transportation services and alternative commuting routes can ensure that essential services remain accessible, reducing the broader socioeconomic impact of such disasters.

Post Flood Impact in Mobility Post Flood Impact in Mobility



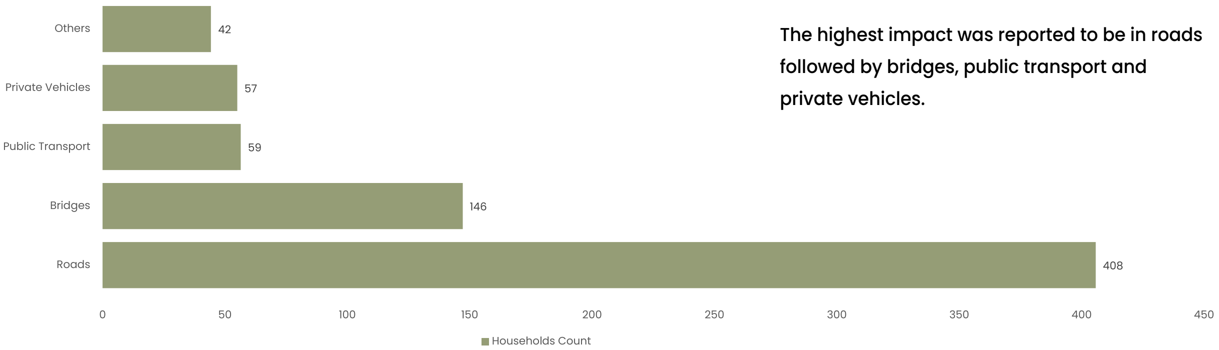
70% of reported households faced disruption in mobility and transportation.

Post Flood Impact in Different Aspects of Transportation



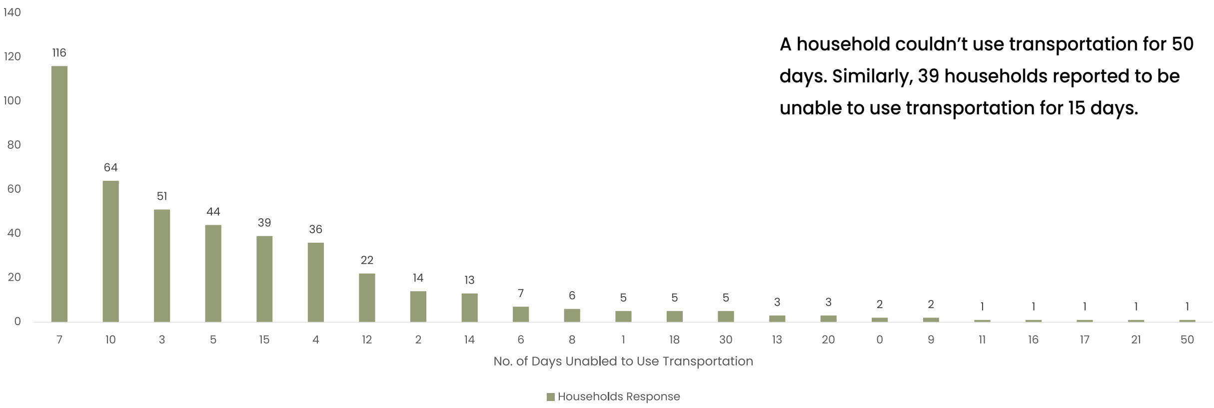
43% of households reported to face significant challenges in accessing essential services. About 42% had their personal vehicle damaged.

Primarily Affected Different Transportation



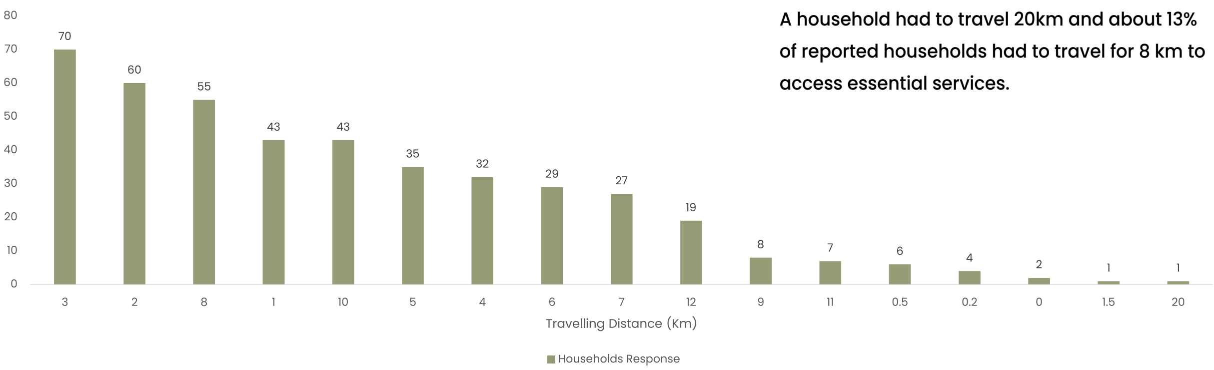
The highest impact was reported to be in roads followed by bridges, public transport and private vehicles.

Post Flood Impact on Use of Transportation Services



A household couldn't use transportation for 50 days. Similarly, 39 households reported to be unable to use transportation for 15 days.

Post Flood Essential Services Accessibility



A household had to travel 20km and about 13% of reported households had to travel for 8 km to access essential services.

Figure 25: Post-flood impacts on mobility and transportation services

Post Flood Impact on Community Cohesion and Support

The July 2024 floods in Kanchanpur had significant implications for community social cohesion, support networks, and local conflict dynamics. The data indicates that 74 households reported disruptions to social cohesion, whereas 596 stated that their sense of community remained unaffected. While the majority of the population retained social ties, the fact that nearly 11% of households experienced social cohesion challenges suggests that flooding has the potential to erode community bonds, particularly in areas where displacement, resource scarcity, and stress-induced tensions occur. The physical destruction of community spaces further exacerbated this issue, as 43 households reported damage to community buildings, 40 noted damage to parks, and 8 observed harm to community shelters or safe spaces. These structural damages contribute to a reduced sense of belonging and weaken collective support systems crucial for disaster recovery.

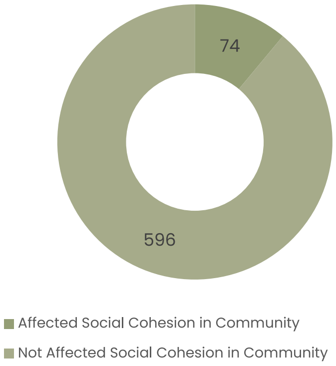
Post-flood conflicts in affected communities were primarily centered around disputes over natural resources such as land, water, and grazing areas, with 4 households identifying such tensions. Additionally, 3 households reported conflicts related to displacement or migration, and an equal number indicated a breakdown in social support networks. While these figures may seem relatively low, they highlight underlying vulnerabilities that can escalate if recovery efforts fail to address equitable resource distribution and social integration. Encouragingly, no households explicitly reported increased crime or violence, which suggests that while stress-induced tensions exist, they have not yet led to significant law and order challenges. However, ensuring that such conflicts do not escalate remains a key concern for local governance and community leaders.

Interestingly, the flood had mixed effects on social cohesion, with 278 households stating that the disaster brought the community together, while 136 reported that it created conflicts. Meanwhile, 256 households indicated that the flood had no major impact on their social ties. This finding underscores that disasters have the potential to both unite and divide communities, depending on how recovery efforts are managed and whether local institutions can foster cooperation and mutual support. Organized community recovery efforts played a crucial role in strengthening bonds, as seen in the 624 households that participated in collective flood recovery actions, compared to only 46 households that did not. Community efforts such as participation in meetings (183 households), spending time in shelters (199 households), and involvement in post-flood sediment clearing (11 households) were notable indicators of resilience and solidarity.

The long-term implications of these social disruptions and recovery efforts are critical. If community spaces and social support networks are not restored promptly, there is a risk of prolonged disintegration of trust, increased social fragmentation, and the marginalization of vulnerable groups. Additionally, conflicts over natural resources may intensify as recovery progresses, particularly if land disputes arise or migration patterns shift. However, the data also suggests a strong foundation for resilience, as many households engaged in recovery actions and found solidarity through shared experiences. By capitalizing on these strengths, local governments and civil society organizations can foster long-term community resilience by investing in the reconstruction of communal spaces, supporting social programs, and facilitating inclusive decision-making processes.

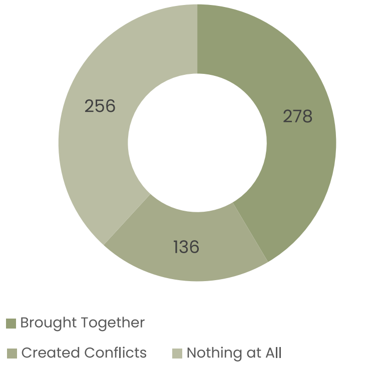
For the local government, addressing these social cohesion challenges presents both obstacles and opportunities. A major challenge lies in ensuring that recovery efforts are equitable and do not inadvertently deepen pre-existing divisions, particularly in resource allocation and land tenure disputes. Additionally, the strain on public services may hinder swift restoration efforts, particularly in rebuilding community infrastructure and providing psychosocial support. However, this crisis also provides a unique opportunity to strengthen local governance by fostering participatory recovery processes that actively involve community members. Investing in disaster preparedness, reinforcing social safety nets, and supporting initiatives that promote local leadership in crisis management can help build more cohesive and resilient communities capable of withstanding future shocks.

Post Flood Impact in Community Social Cohesion and Support



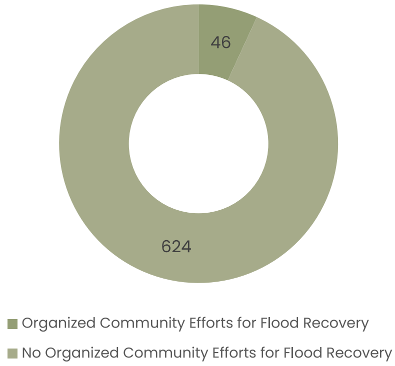
11% of reported households expressed the impact in social cohesion of community.

Post Flood Impact in Communities Social Cohesion



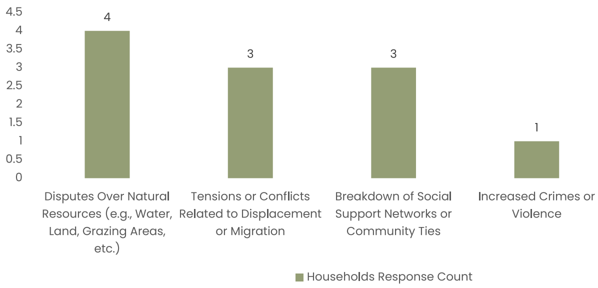
41% of reported households expressed that the society was brought together post flood event. 20% reported to have conflicts. 41% of reported households expressed that the society was brought together post flood event. 20% reported to have conflicts.

Post Flood Community Effort for Recovery



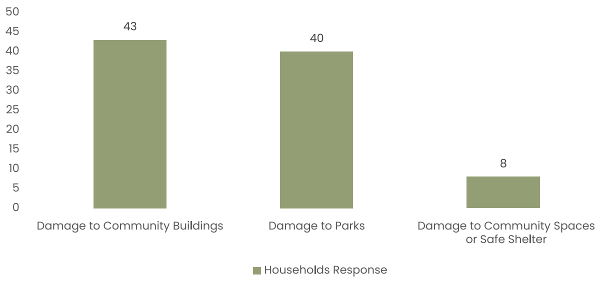
About 7% of households expressed that there was organized community events for flood recovery

Post Flood Community Conflict



4 households reported to have disputes over natural resources.

Post Flood Sectoral Impact in Community Social Cohesion and Support



Significant damage to community buildings and parks were reported.

Post Flood Community Togetherness

Majority of responders (199 households) reported that they stayed and spent time in community shelters and safe spaces followed by community or samuha meetings after flood.

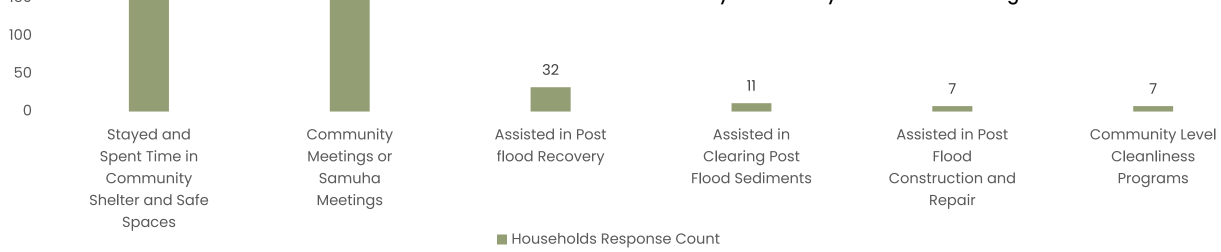


Figure 26: Post-flood impacts on social cohesion, conflict and recovery efforts

Post Flood Impact on Social Displacement

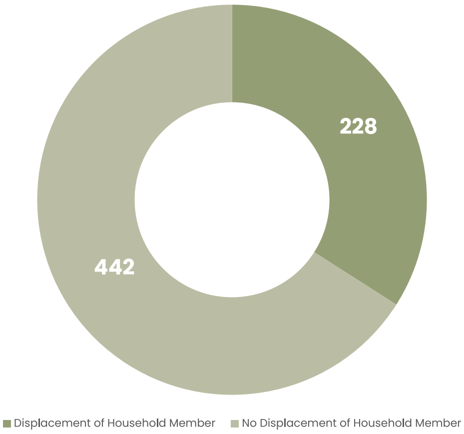
The July 2024 floods in Kanchanpur significantly impacted social displacement, safety, and security in affected communities. Displacement varied across municipalities, with Bheemdatt Municipality experiencing the highest number of displaced families with 2,191, followed by Beldandi Rural Municipality with 620 and Belauri Municipality with 549. These numbers suggest that urban and peri-urban areas bore the brunt of displacement due to higher population density and flood vulnerability. In contrast, Shuklaphanta, Krishnapur, and Dodhara Chandani reported fewer displaced families, possibly due to better resilience or lower exposure to extreme flooding.

At the household level, 228 households reported the displacement of at least one member, while 442 households remained intact. This separation likely led to social and economic burdens. Safety and security concerns were also prevalent, with 123 households reporting impacts during displacement, underscoring the need for better evacuation and protection measures for vulnerable populations.

Migration and relocation decisions were driven by multiple factors, with 260 households citing the increased frequency of floods as the primary reason for relocation. Water contamination affected 170 households, while loss of assets influenced 131 households. Only three households reported conflict or insecurity, suggesting limited resource disputes. Interestingly, one household chose not to migrate, possibly due to strong community ties or economic constraints.

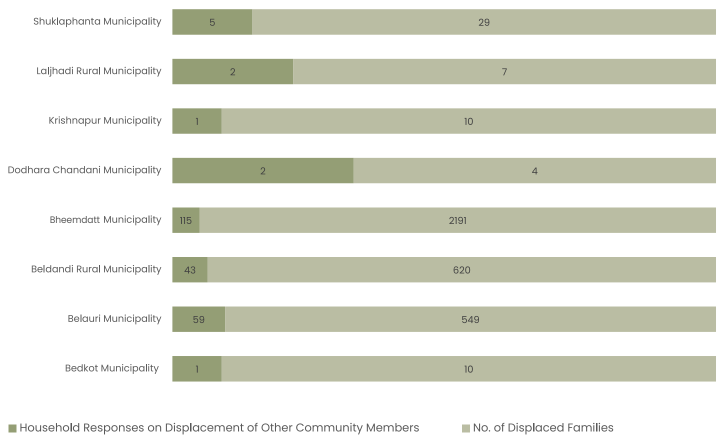
The overall findings highlight the multifaceted challenges of flood-induced displacement, encompassing physical, emotional, and social hardships. Local governments and humanitarian organizations must prioritize holistic recovery strategies, including restoring access to basic services, providing psychological support, and strengthening community networks. Long-term resilience planning should focus on safer housing, improved drainage infrastructure, and proactive relocation strategies for high-risk communities to foster stability and recovery.

Post Flood Social Displacement Assessment



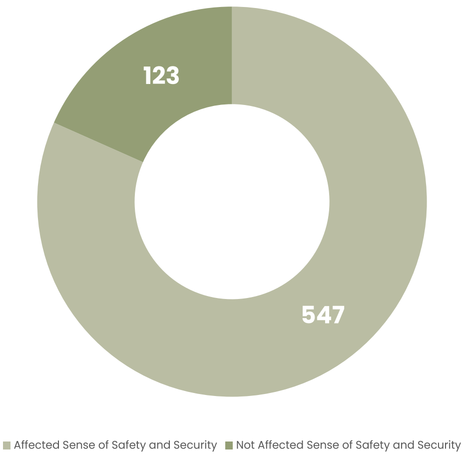
Members from 34% of reported households got displaced.

Municipality Wise Social Displacement Assessment



2191 families were temporarily displaced in Bheemdatt municipality, followed by 620 in Beldandi and 549 in Belauri municipalities. Dodhara Chandani had the least families displaced.

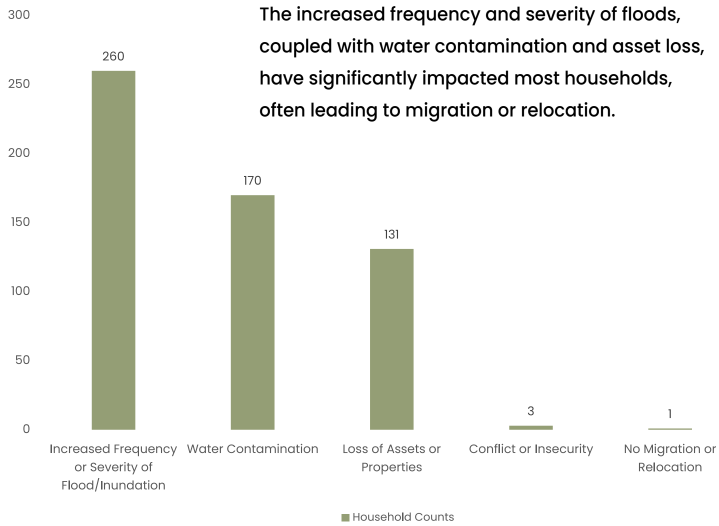
Post Flood Sense of Safety and Security During Social Displacement



A majority of responders (123 households) reported feeling a low sense of safety and security due to fears of future floods and wildlife attacks after flood.

Post Flood Migration or Relocation Assessment

The increased frequency and severity of floods, coupled with water contamination and asset loss, have significantly impacted most households, often leading to migration or relocation.



Post Flood Challenges Faced During Migration or Relocation

About 40% of reported households had difficulty in accessing basic needs and services while 30% suffered from mental stress during migration or relocation.

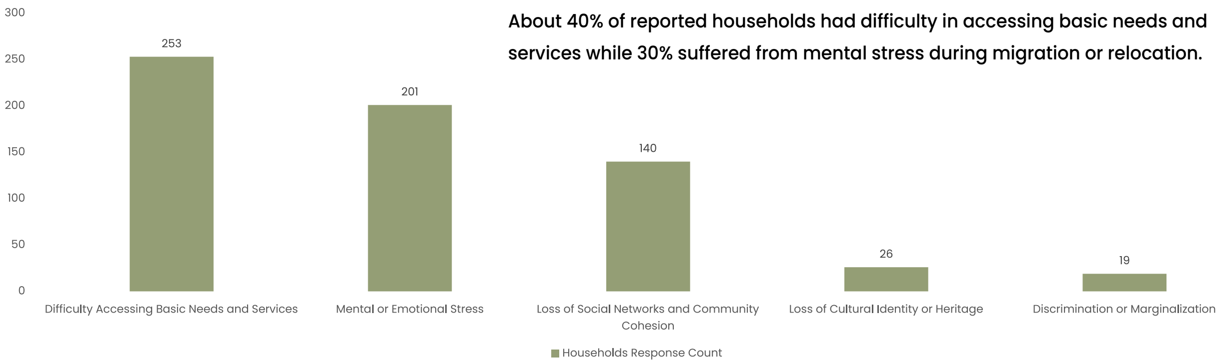


Figure 27: Post-flood induced challenges on social displacement, migration and relocation

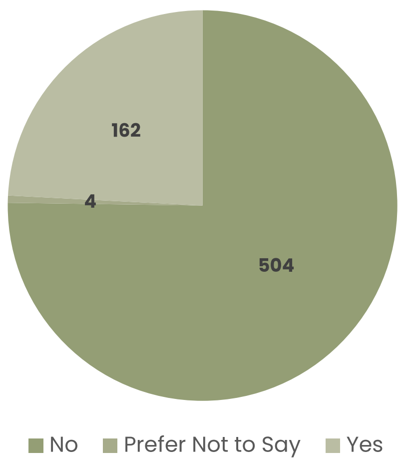
Post Flood Impact on Community Well Being Programs

The July 2024 floods in Kanchanpur district caused extensive damage, leaving many households without adequate institutional support. Data indicates that 504 households did not receive any assistance from local organizations or the government, while only 162 received some form of help. This lack of support likely worsened the struggles of affected families, making recovery more difficult.

Among post-flood well-being programs, counseling was the most widely accessed service, with 125 households receiving psychological support. However, critical programs such as rebuilding as reported by 27 households and livelihood support by 22 households had limited reach. This suggests that while mental health concerns were addressed to some extent, long-term economic and social recovery efforts were inadequate.

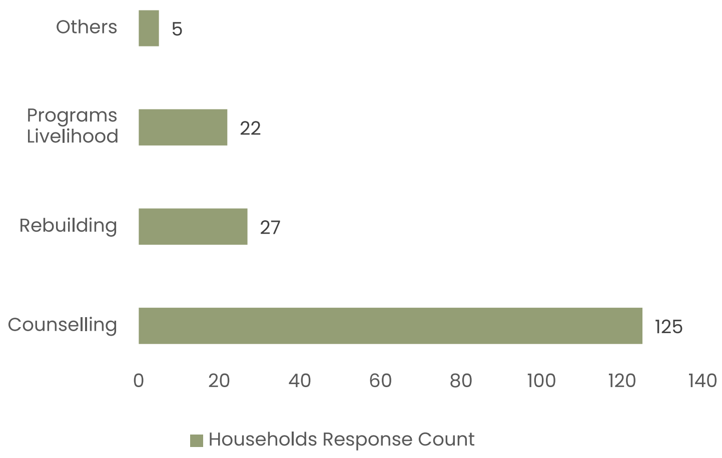
The limited assistance and insufficient recovery programs highlight key gaps in disaster response. Affected households need more than just psychological support—they require rebuilding efforts, financial stability, and sustained institutional backing. Strengthening disaster preparedness and post-flood recovery measures is crucial to preventing prolonged hardships and ensuring resilience against future disasters.

Post Flood Support From Local Organizations / Government



75% of reported households didn't get support from local government

Post Flood Sectoral Community Well-Being Programs



About 70% of reported households participated in counselling programs, followed by rebuilding and livelihood programs

Figure 28: Post-Flood Support and Community Well-Being Programs in Kanchanpur District

Post Flood Impact on Access to Essential Services and Goods

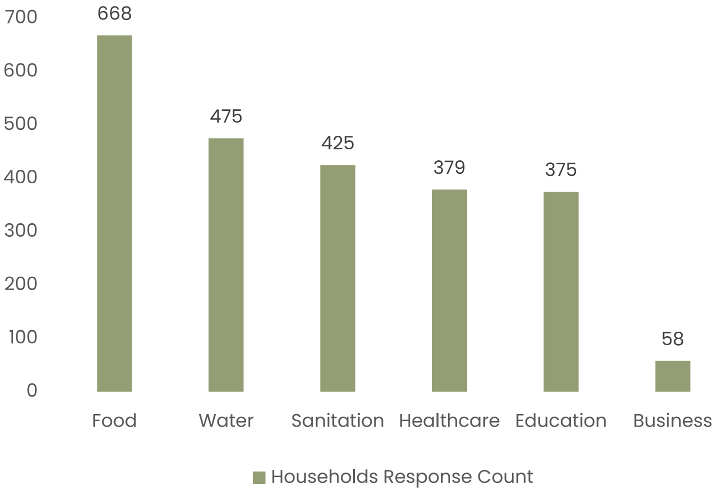
The aftermath of the July 2024 floods in Kanchanpur district brought significant disruptions to the accessibility of essential services and goods, further exacerbating the hardships faced by affected households. Beyond immediate economic losses, disruptions in access to food, water, healthcare, and education had profound non-economic implications, affecting physical and mental well-being, social stability, and long-term resilience. Households not only struggled to meet their basic needs but also faced challenges in maintaining their livelihoods, education, and overall quality of life. Understanding the extent of these impacts and the coping mechanisms employed by affected families is crucial in developing more effective disaster response and recovery strategies.

As shown in the data, food accessibility was the most impacted sector, with 666 households reporting difficulties in securing adequate nutrition. Water supply and sanitation services were also severely affected, impacting 451 and 443 households, respectively. These disruptions increased the risk of waterborne diseases and other health complications in the aftermath of the flood. Healthcare services were also significantly impacted, with 375 households struggling to access medical care, further compounding health risks. Additionally, education was disrupted for 310 households, creating long-term consequences for children's learning and development. Although 36 households reported business-related impacts, the overall economic instability caused by these disruptions likely had a cascading effect on financial security and daily survival.

The primary reasons behind these accessibility challenges included damage to key infrastructure, such as bridges, schools, hospitals, and transportation networks. Damage to bridges was reported by 176 households and roads by 176 households severely hindered mobility, restricting access to essential services, markets, and medical care. Similarly, damage to hospitals by 130 households) further strained an already limited healthcare system, while destruction of schools by 74 households disrupted education, delaying students' return to classrooms. These factors collectively created a situation where recovery was not only delayed but also uneven, with marginalized communities facing greater difficulties in accessing relief and rebuilding their lives.

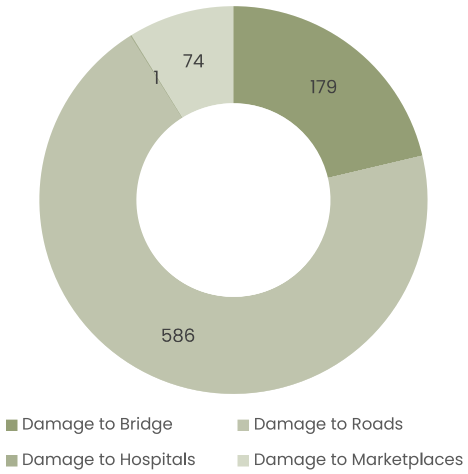
To mitigate future risks and enhance resilience, affected households employed

Post Flood Sectoral Impact in Assessing Essential Services and Goods



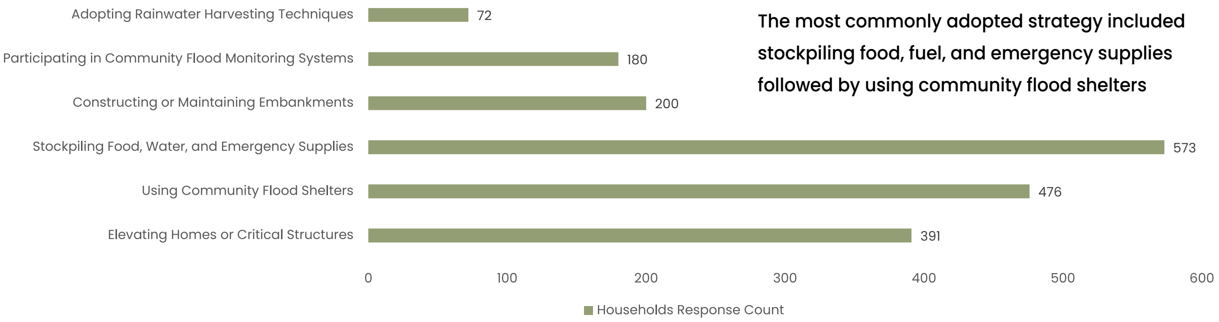
The highest impact was seen in access to food while the least was seen in business

Post Flood Reasons of Assessing Essential Services and Goods Affected



Most of the responders reported that they pointed out the reason for damage to roads followed by damage to bridges, marketplaces and hospitals in accessing essential services and goods after flood.

Coping Mechanisms Availability to Reduce Future Challenges



The most commonly adopted strategy included stockpiling food, fuel, and emergency supplies followed by using community flood shelters

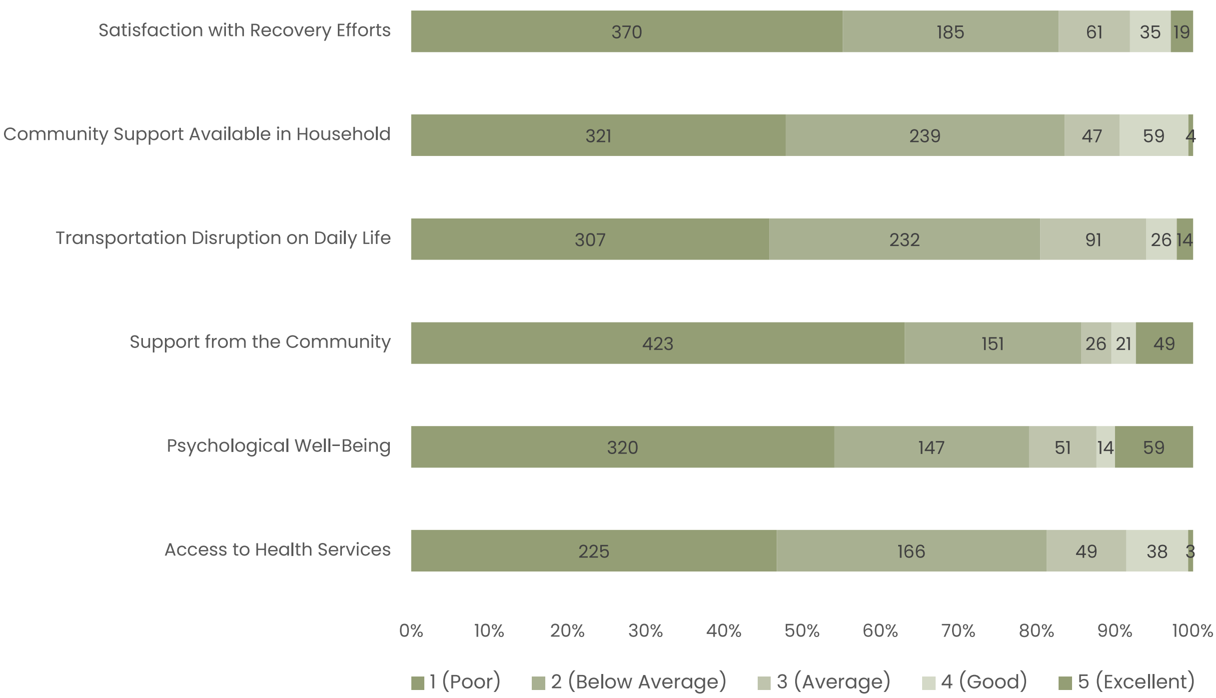
Figure 29: Post-flood impacts on accessing essential services, goods and coping mechanisms

various coping mechanisms. The most commonly adopted strategies included stockpiling food, fuel, and emergency supplies was reported by 575 households and using community flood shelters by 515 households. Additionally, many households took proactive steps to strengthen their homes against future floods, with 200 households constructing or retrofitting their dwellings. Participation in community-led flood monitoring systems was from 112 households and disaster preparedness training from 72 households further indicated an increasing awareness of disaster risks and the importance of early warning systems. Some households also invested in elevating farms or critical structures to reduce vulnerability. While these measures demonstrate community resilience and adaptation, they also highlight the persistent threat of future flooding and the urgent need for systemic improvements in disaster management.

The findings underscore the multifaceted challenges that households faced in accessing essential services following the July 2024 floods, with food, water, sanitation, healthcare, and education being the most severely impacted sectors. The destruction of infrastructure such as bridges, hospitals, and transportation networks compounded these difficulties, delaying recovery efforts and increasing long-term vulnerabilities. While communities have taken proactive steps to enhance resilience, relying solely on individual and community-level coping mechanisms may not be sufficient in mitigating future flood risks. Strengthening disaster preparedness, investing in resilient infrastructure, and ensuring equitable access to essential services must be key priorities for local governments and disaster response agencies. Without sustained efforts to improve flood resilience, affected communities may continue to face recurrent disruptions, deepening socio-economic inequalities and hindering long-term recovery.

Community Perceptions on Post-Flood Impact and Response in Kanchanpur

Post-Flood Household Impact and Response: Ratings



	Access to Health Services	Psychological Well-Being	Support from the Community	Transportation Disruption on Daily Life	Community Support Available in Household	Satisfaction with Recovery Efforts
1 (Poor)	225	320	423	307	321	370
2 (Below Average)	166	147	151	232	239	185
3 (Average)	49	51	26	91	47	61
4 (Good)	38	14	21	26	59	35
5 (Excellent)	3	59	49	14	4	19

Figure 30: Household and community perception on post-flood impact and response in Kanchanpur District

Floods are among the most devastating natural disasters, affecting not only infrastructure but also the well-being of individuals and communities. The given data illustrates how households in Kanchanpur perceived post-flood recovery efforts across various dimensions, including recovery satisfaction, community support, transportation disruptions, psychological well-being, and healthcare access. The findings suggest that a significant portion of the population faced severe challenges in the aftermath of the disaster, with most respondents rating key aspects of recovery as poor or below average. One of the most concerning findings is the widespread dissatisfaction with recovery efforts, with 370 households rating it as poor. This indicates that flood victims felt that relief and reconstruction measures were either inadequate or poorly executed. Additionally, community support within households was also rated poorly, suggesting that many families lacked the necessary social and economic resources to cope with the aftermath. This lack of internal household support could be due to financial constraints, displacement, or the overwhelming nature of the disaster.

Transportation disruption was another major concern, as a majority of households reported that daily life was significantly impacted. Floods often damage roads, bridges, and transportation infrastructure, limiting access to essential services such as markets, schools, and workplaces. The inability to commute freely can have cascading effects on livelihoods, delaying economic recovery and increasing dependency on external aid. Moreover, community support was perceived as weak, with 423 respondents rating it as poor. This suggests that either local networks were stretched too thin or that there were gaps in mobilizing resources at the community level. Beyond physical infrastructure, psychological well-being emerged as a crucial concern. Floods not only displace individuals but also leave lasting emotional and mental health challenges. The fact that 320 households rated their psychological well-being as poor underscores the mental strain caused by displacement, loss of property, and uncertainty about the future. Mental health challenges, if left unaddressed, can have long-term effects on productivity, social cohesion, and overall community resilience.

Access to health services was another critical challenge, with 225 households expressing dissatisfaction. Flood-affected regions often experience a surge in waterborne diseases, malnutrition, and injuries, yet healthcare services may be disrupted due to damaged facilities or resource constraints. Limited access to health services further compounds the vulnerabilities of affected populations, particularly pregnant women, children, and the elderly. The flood's aftermath has far-reaching consequences on individuals and the broader community. Households that experienced poor recovery support, disrupted transportation, and psychological stress are likely to face prolonged hardship. Economic activities, particularly for daily wage workers and small-scale farmers, are severely affected due to loss of income sources and restricted mobility. Many families may have exhausted their savings, pushing them into cycles of debt and long-term financial insecurity. On a broader scale, community resilience is weakened when disaster response mechanisms are inadequate. Weak social networks and insufficient community support make it difficult for families to rebuild their lives. In disaster-prone regions like Kanchanpur, a lack of coordinated and inclusive disaster management strategies exacerbates vulnerabilities, making communities more susceptible to future calamities.

The long-term impact of the flood extends beyond immediate economic and infrastructural damage. Environmental degradation resulting from soil erosion and water contamination can have lasting effects on agriculture, potentially reducing productivity for years. If local governments fail to implement sustainable rehabilitation strategies, communities may continue to struggle with food security, inadequate housing, and unstable income sources. Additionally, psychosocial impacts may persist for generations if mental health interventions are not prioritized. Children who grow up in disaster-affected areas often experience disruptions in education, leading to reduced future opportunities. Similarly, migration patterns may shift, with families relocating permanently to seek safer and more stable environments, potentially depopulating rural areas and increasing urban pressures elsewhere.

The local government faces several institutional and logistical challenges in managing post-flood recovery. Limited financial resources constrain the ability to provide adequate relief, rebuild infrastructure, and support affected populations. Additionally, bureaucratic inefficiencies and coordination gaps among different agencies may slow down response efforts, leaving many families without timely assistance. Another significant challenge is the lack of disaster preparedness and resilience-building programs at the local level. If authorities do not invest in early warning systems, sustainable land-use planning, and climate-adaptive infrastructure, similar disasters will continue to have devastating impacts. The absence of robust mental health and social protection programs further complicates recovery, as many individuals struggle in silence without access to counseling or emotional support. For affected people, the high cost of rebuilding, loss of employment opportunities, and persistent health risks make it difficult to return to normalcy. Without adequate governmental and non-governmental support, recovery becomes a slow and inequitable process, with the poorest households suffering the most.

The post-flood household impact assessment in Kanchanpur reveals deep-seated challenges in disaster recovery and response. While floods are natural occurrences, their long-term consequences are shaped by how effectively governments, communities, and individuals respond. Addressing the gaps in disaster preparedness, community resilience, and mental health support is crucial to preventing long-term economic and social instability. Moving forward, a more inclusive and well-coordinated disaster response system is necessary to ensure that flood-affected populations receive timely assistance and have the means to rebuild their lives sustainably.

Conclusion

The findings of this assessment reveal that the July 2024 Kanchanpur flood had devastating consequences beyond immediate economic damages, deeply affecting social structures, environmental stability, and mental well-being. Despite existing disaster risk management strategies, the severity of losses highlights critical gaps in preparedness, equitable relief distribution, and long-term recovery mechanisms. Many communities remain trapped in cycles of vulnerability, facing slow recovery and insufficient support.

Municipalities like Bheemdatt and Belauri bore the highest financial burden due to their economic density, while rural areas such as Laljhadi and Beldandi struggled with limited access to essential services, particularly healthcare, education, and infrastructure recovery. The psychological impact, particularly the lack of mental health interventions, exacerbated post-disaster vulnerabilities, leaving many communities in distress. With 444 households lacking psychological support, the emotional toll remains an overlooked but pressing concern.

Moreover, the underreporting of non-economic losses such as cultural damage, environmental degradation, and social displacement underscores the need for inclusive disaster documentation. Without recognizing and incorporating these dimensions into policy frameworks, recovery will remain incomplete, and future resilience-building efforts will be hindered. The need for comprehensive data collection mechanisms is crucial to ensure equitable recovery efforts.

The flood also exposed financial vulnerabilities, as many affected households relied on high-interest loans to cover damages. This reflects the absence of robust insurance mechanisms and equitable compensation strategies, leaving already struggling families burdened with debt. Long-term economic stability requires affordable recovery financing options and government-backed relief mechanisms.

To address these challenges, a multi-pronged approach is necessary—one that enhances data collection, equitable disaster relief, mental health support, infrastructure resilience, and sustainable financial mechanisms. The way forward and recommendations below provide a roadmap for effective disaster response and preparedness, ensuring that both economic and non-economic losses are accounted for in policy and planning.

Way Forward and Recommendations

Building on the assessment's findings, future flood response strategies must prioritize integrated disaster resilience, equitable assistance distribution, and institutional strengthening.

1. **Strengthening Disaster Preparedness and Early Warning Systems:**
   Expand localized real-time monitoring and forecasting mechanisms to improve anticipatory action and community preparedness.
2. **Enhancing Non-Economic Loss and Damage (NELD) Documentation:**
   Establish standardized methodologies for tracking non-economic losses, particularly in mental health, education, cultural damage, and social cohesion.
3. **Improving Financial Recovery Mechanisms:**
   Develop low-interest recovery loans and insurance schemes to reduce dependency on high-interest borrowing, ensuring financial security for affected households.
4. **Mental Health and Psychosocial Support Integration:**
   Incorporate mental health services into disaster response plans, ensuring that trauma support is available to affected individuals.
5. **Equitable Distribution of Relief and Assistance:**
   Address disparities in compensation and aid allocation, prioritizing rural municipalities and vulnerable populations often excluded from mainstream recovery programs.

6. **Sustainable Infrastructure and Housing Rehabilitation:**
   Implement flood-resilient infrastructure projects, such as elevated housing, reinforced embankments, and improved drainage systems, to mitigate future risks.
7. **Community-Based Disaster Recovery Approaches:**
   Strengthen local governance participation in recovery planning, ensuring that disaster management strategies are inclusive and community-driven.
8. **Environmental Protection and Ecosystem-Based Adaptation:**
   Prioritize River embankment reinforcement, reforestation, and soil erosion control to safeguard natural resources and mitigate long-term ecological damage.

Based on the findings of this assessment, the following key recommendations should be implemented to enhance resilience, preparedness, and equitable recovery in Kanchanpur District:

- Adopt a holistic disaster response framework that incorporates both economic and non-economic loss and damage into planning, policy-making, and relief efforts.
- Expand targeted financial aid and loan support programs to reduce the burden on affected households, ensuring access to affordable credit and compensation for damages.
- Enhance emergency healthcare accessibility by investing in mobile clinics, telemedicine services, and community-based health support systems.
- Integrate flood-resilient agricultural practices, including crop diversification, climate-resilient seeds, and improved irrigation techniques to reduce food security risks.
- Increase investments in school disaster preparedness, including safe school infrastructure, emergency education kits, and remote learning options to prevent educational disruptions.
- Strengthen disaster risk reduction governance, ensuring better coordination between local, provincial, and national authorities to streamline response and preparedness efforts.
- Develop long-term community-based resilience programs, empowering local leaders, youth, and marginalized groups to actively participate in disaster management planning.

By implementing these recommendations, Kanchanpur can move toward a more resilient and equitable disaster management system, ensuring that both economic and non-economic losses are adequately addressed in future preparedness and response efforts.

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Annexes

Annex-I:
Household Level Economic Loss and Damage Assessment Questionnaires

A. Demographic Information and Indicators of Vulnerability	
1. Name of the Municipality	Dropdown (Nine municipalities of Kanchanpur District)
2. Ward Number	
3. Name of the Tole	
4. House Number	
5. GPS Coordinates	
6. House Photograph	
7. Do I have your consent to conduct this interview?	• Yes • No
8. Age	• Below 18 • 18 – 25 • 26 – 40 • 41 – 60 • 60+ • Prefer not to say
9. Marital Status	• Married • Single • Separated/divorced • Widow • Prefer not to say
10. How many people currently live in your household including you?	Describe
11. Describe your current household composition	Select any that apply • Male-headed • Female-headed • Elderly-headed • Not applicable (e.g. other household type) • Prefer not to say
12. Total number of female members	Number
13. Total number of male members	Number

Health Vulnerability	
14. Is there any person living with disability in your household?	• Yes • No
If yes, specify the gender of the person with disability If yes, how many members in the family are person with disability?	Describe
15. If people with disability, specify the type of disability?	Describe
16. Do the people with disability family member have identity revealing document provided by municipality?	• Yes • No
17. Which type of card have the people with disability member taken?	• Yes • No
18. Does you or your family have any or suffered from any of the following diseases?	Select all that apply • Malaria • Kala-azar • Dengue • Acute gastroenteritis • Cholera • Severe Acute Respiratory Infection • Scrub typhus • Skin infections • Covid-19 • Cardiac Diseases • Hyper/Hypotension • Diabetes • Prefer not to say
19. Does the household have access to healthcare services?	• Yes • No
20. How far is the nearest healthcare facility (in km)?	Number
21. Does the household have health insurance?	• Yes • No
22. Has the household experienced any infant mortality?	• Yes • No

Education Status Literacy	
23. Total literate female members in the house	
24. Total literate male members in the house	
25. What is the highest education level within your household?	Select one • No formal education • Lower secondary education • Upper secondary education • Vocational education • University education • Prefer not to say
26. Total number of employed	
27. Total number of unemployed	
Economic Vulnerability	
28. What is the primary occupation of your family?	• Agriculture • Business • Foreign Employment
If others, please specify	Describe
29. Do you or your family member's own land?	• Yes • No
30. If yes, How much land do you or your family hold? (Local Measurement unit)	Number
31. Does the household own livestock?	• Yes • No
32. If yes, type of livestock	Dropdown (cattle, goats, chickens, etc.)
33. Does the household cultivate crops?	• Yes • No
34. If yes, what are the primary crops grown?	Dropdown (Cash crops, Food crops, etc.)
35. What is your perception of your household income compared to the rest of the city?	
36. Does the household have any debt?	• Yes • No
37. If yes, source of loan:	Dropdown: Bank/ Cooperative/Informal/Other
38. How much does the household spend monthly on food?	NPR
39. How much does the household spend monthly on health?	NPR

Social Vulnerability	
40. Does the household receive any social protection or assistance (elderly allowance, disability allowance) from the municipality?	• Yes • No
41. Does the household receive support from family or community members during times of crisis or disasters?	• Yes • No
42. Has any household member migrated for work?	• Yes • No
43. Does the household receive remittances?	• Yes • No
44. Has any member of the household been a victim of crime or violence?	• Yes • No
Hazard Exposure	
45. Has the household experienced any disaster in the past 5 years?	• Yes • No
46. If yes, what type of disaster was it?	Dropdown: Flood/Earthquake/ Landslide/Heat/Cold waves
47. Did the household lose any assets during the previous disaster?	• Yes • No
48. If yes, what assets were lost?	Dropdown ((House, Land, Livestock, Crops, etc.)
49. Did the household lose income during the previous disaster?	• Yes • No
50. Has the household ever been forced to relocate due to disaster?	• Yes • No
51. What mechanisms did the household use to cope/ adapt with the impacts of the disaster	Dropdown: Self/ Government Aid, NGO Support, etc.

Hazard Understanding and Details	
52. Do you know about the recent flood in Kanchanpur District?	• Yes • No
53. How Severe was the flood?	• Mild • Moderate • Severe
54. Were your household affected by the recent flooding event?	• Yes • No
55. Do you have any accounted or unaccounted loss and damage due to the flooding event?	• Yes • No
56. What kinds of loss and damage to assets did you face due to the flood?	Dropdown: House / Land / Livestock / Crops / Business Assets / Other Properties
57. Are you aware about the date of incident?	Date Field
58. Are you aware about the time of incident?	Time field
59. Do you know the cause of the incident?	Description
60. In your opinion, was the disaster natural or anthropogenic?	• Natural • Man-made
61. Do you know about the impacted areas?	Description

Disaster Preparedness	
62. Does the household receive disaster early warning messages?	• Yes • No
63. Does the household have an emergency plan for disasters?	• Yes • No
64. Does the household know of safe areas for evacuation during disasters?	• Yes • No
65. If yes, how far safe space is from your house?	Number (in Km)
66. Does the household have any form of disaster insurance?	• Yes • No

B. Household Level Economic Loss and Damage	
B1. Property Damage	
1. Was your house damaged by the flood?	• Yes • No
2. If yes, describe the type and extent of damage:	Text
3. Estimated repair costs (in NPR):	Number
4. Was any household item damaged or lost?	• Yes • No
5. If yes, specify the items damaged/lost:	Text
6. Estimated value of lost/damaged items (in NPR):	Number
7. Was there damage to household infrastructure (e.g., fencing, wells, toilets, etc.)?	• Yes • No
8. If yes, describe the damage and estimated repair cost:	Text
B2. Livelihood and Income Loss	
9. Did the flood disrupt your household's income?	• Yes • No
10. If yes, what was the primary source of income affected?	Text
11. Estimated income loss during and after the flood (in NPR):	Number
12. Did the flood cause damage to income-generating assets?	• Yes • No
13. If yes, specify the assets and their estimated value:	Text
14. Did the flood cause any loss of savings or financial resources?	• Yes • No
15. If yes, estimate the amount lost:	Number
B3. Agricultural Loss	
16. Did the flood destroy any crops?	• Yes • No
17. If yes, what type of crops were affected?	Text
18. Total area of agricultural land affected (hectares/ropani):	Number
19. Estimated quantity of crops lost (in quintals):	Number
20. Estimated value of lost crops (in NPR):	Number
21. Was agricultural infrastructure damaged?	• Yes • No
22. If yes, describe the damage and estimated repair cost:	Text

B4. Livestock Loss	
23. Did the flood cause loss of livestock?	• Yes • No
24. If yes, specify the type of livestock lost:	Text
25. Number of livestock lost:	Number
26. Estimated value of lost livestock (in NPR):	Number
27. Were there any damages to livestock facilities?	• Yes • No
28. If yes, describe the damage and estimated repair cost:	Text
29. Damage to Tools, Machinery, and Vehicles	
30. Did the flood cause damage to any tools, machinery, or vehicles?	• Yes • No
31. If yes, specify the damaged items:	Text
32. Estimated repair or replacement cost (in NPR):	Number
B5. Insurance and Compensation	
33. Does your household have any insurance coverage for flood-related damage?	• Yes • No
34. If yes, what type of insurance?	Text
35. Did you file an insurance claim for the damage?	• Yes • No
36. If yes, how much compensation was received (in NPR)?	Number
37. Assistance and Aid Received	
38. Did you receive any external financial assistance post-flood?	• Yes • No
39. If yes, specify the source of assistance:	Text
40. Total amount of financial assistance received (in NPR):	Number
B6. Loans and Debts	
41. Did you need to take any loans to recover from the flood's impact?	• Yes • No
42. If yes, specify the source of the loan:	Text
43. Amount of loan taken (in NPR):	Number
44. Interest rate of the loan:	Number
45. How long do you anticipate it will take to repay the loan?	Text

B8. Other Economic Losses	
46. Did the flood cause disruption to any business you operate?	• Yes • No
47. If yes, provide details about the business and estimated loss of income:	Text
48. Did the flood cause loss of any rental income?	• Yes • No
49. If yes, estimate the loss of rental income (in NPR):	Number
50. Were there any additional economic losses not covered above?	• Yes • No
51. If yes, specify the nature of the loss and estimated value:	Text

Annex-II:
Household Level Non- Economic Loss and Damage Assessment Questionnaires

C. Non-economic Loss and Damage		
C1. Health Impact		
1.	Did any household members suffer from physical injuries during the flood?	• Yes • No
2.	If yes, provide details:	
3.	Name	Text
4.	Age	Number
5.	Type of injury	Text
6.	Medical Costs	Number (NPR)
7.	Did anyone in your household fall sick after the flood?	• Yes • No
8.	If yes, specify the illness	Text (e.g., waterborne/ vector-borne diseases)
9.	Medical costs incurred for treatment	Number (NPR)
10.	Are there any members with chronic illnesses whose treatment was disrupted due to the flood?	• Yes • No
11.	On a scale of 1-5, how would you rate your access to health services after the flood?	Scale (1-5)
C2. Psychosocial Impact		
12.	Has anyone in the household experienced mental stress or trauma?	• Yes • No
13.	Has anyone received psychological support?	• Yes • No
14.	If yes, what kind of support?	Select one: • Counseling, • Community meetings • Aama Samuha meetings • Other
15.	On a scale of 1-5, how would you rate the psychological well-being of your household after the flood?	Scale (1-5)

C3. Cultural Losses	
16. Did you lose any cultural/religious items (e.g., idols, religious books, artifacts)?	• Yes • No
17. If yes, provide details and estimated value	Text, Number (NPR)
18. Did the flood affect any cultural or religious practices in your community?	• Yes • No
19. If yes, please specify	Text
20. Did the flood affect any cultural sites?	• Yes • No
21. Did the flood affect your daily rituals?	• Yes • No
22. If yes, please explain	Text
C4. Education Disruption	
23. Were any children in the household unable to attend school after the flood?	• Yes • No
24. How many days of school were missed due to the flood?	Number
25. Were any school materials lost (books, uniforms, etc.)?	• Yes • No
26. If yes, provide estimated cost of replacing school materials	Number (NPR)
C5. Social Connectivity	
27. Did you lose contact with any family members or community members due to displacement?	• Yes • No
28. On a scale of 1-10, how would you rate the support from your community after the flood?	Scale (1-10)

Ecosystem Services Loss	
C6. Environmental Impact	
29. Did the flood affect any natural resources in your area?	• Yes • No
30. If yes, specify the type of resource	Select: • Forest • Wetland • River • Agricultural Land • Other
31. Was there a noticeable change in the quality of water sources (e.g., wells, rivers)?	• Yes • No
32. If yes, Text the type of change	Text (e.g., contamination, reduced flow, sediment deposit)
33. Did the flood cause soil erosion or land degradation?	• Yes • No
34. Was any local wildlife or biodiversity impacted by the flood?	Yes No
35. If yes, specify how	Text (e.g., loss of habitat, reduced population, habitat shift)
C7. Ecosystem Services Disruption	
36. Were there any disruptions to ecosystem services you relied on?	• Yes • No
37. If yes, specify the services impacted	Select all that apply: • Clean Drinking Water • Natural Waste Treatment • Flood Control • Pollination • Other
38. How did the loss of ecosystem services impact your household's livelihood or well-being?	Text
39. On a scale of 1-5, how would you rate the importance of ecosystem services to your household before the flood?	Scale (1-5)

C8. Mobility and Transportation Impact	
40. Access to Transport	
41. Was your access to transportation disrupted by the flood?	• Yes • No
42. How many days/weeks were you unable to use transportation services?	Number
43. What type of transport was primarily affected?	Select all that apply: • Roads • Bridges • Public Transport • Private Vehicles • Others
44. Were there any damage to roads, bridges, or paths you regularly use?	• Yes • No
45. If yes, Text the type and extent of the damage	Text
46. Did the flood cause damage to your personal vehicle(s)?	• Yes • No
47. If yes, specify the type of damage and estimated repair cost	Text, Number (NPR)
48. Impact on Mobility	
49. Were there challenges in accessing essential services (markets, schools, hospitals)?	• Yes • No
50. How far did you need to travel to access essential services after the flood?	Number (km)
51. On a scale of 1-5, how would you rate the impact of transport disruptions on your daily life?	Scale (1-5)

Community Well-Being and Social Impact	
C9. Community Cohesion and Support	
52. How has the flood affected social cohesion in your community?	Text
53. Did it bring the community together or create divisions?	Text
54. Was there an organized community effort for flood recovery?	• Yes • No
55. On a scale of 1-5, how would you rate the level of community support available to your household?	Scale (1-5)
C10. Social Displacement	
56. Were any members of your household displaced due to the flood?	• Yes • No
57. If yes, how long were they displaced?	Number (days/weeks/months)
58. Did the flood lead to displacement of other community members?	• Yes • No
59. If yes, specify the number of displaced families	Number
60. What impact did displacement have on your social life and access to networks?	Text
61. Sense of Security and Safety	
62. Did the flood affect your sense of safety and security?	• Yes • No
63. If yes, Text how	Text (e.g., fear of future floods, increase in thefts)
64. On a scale of 1-5, how safe do you feel in your community post-flood?	Scale (1-5)
C11. Community Well-Being Programs	
65. Have local organizations/government provided post-flood support?	• Yes • No
66. If yes, what type of support?	Select all that apply: • Counseling • Rebuilding • Livelihood Programs • Others
67. On a scale of 1-5, how satisfied are you with the recovery efforts?	Scale (1-5)

Annex-III:
Government Decisions to form Committee and Committee Listing

नेपाल सरकार
गृह मन्त्रालय
राष्ट्रिय विपद् जोखिम न्यूनीकरण तथा व्यवस्थापन प्राधिकरण

नेपाल सरकार
गृह मन्त्रालय
राष्ट्रिय विपद् जोखिम न्यूनीकरण तथा व्यवस्थापन प्राधिकरण
सिंहदरबार, काठमाडौं

पत्र संख्या: २०८१/०८२
चलानी नं.: १५४

सिंहदरबार, काठमाडौं
नेपाल ।

मिति: २०८१/५/६

सहसचिव (प्रा) श्री अर्जुन कुमार बम
राष्ट्रिय विपद् जोखिम न्यूनीकरण तथा व्यवस्थापन प्राधिकरण,

विषय :- समिति गठन सम्बन्धमा ।

यस प्राधिकरणको मिति २०८१ श्रावण २१ गते बसेको विपद् जोखिम न्यूनीकरण तथा व्यवस्थापन प्राधिकरण कार्यकारी समितिको २४ औं बैठकको निर्णय नं २ बमोजिम जल तथा मौसम विज्ञान विभागले वर्षा मापन गर्न शुरु भएपश्चात सबैभन्दा बढी वर्षा मापन भएको कञ्चनपुर जिल्लामा वर्षाका कारण भएको क्षति र नोक्सानीको आँकलन आकलन गरी प्रारम्भिक प्रतिवेदन ७ दिन भित्र र पूर्ण प्रतिवेदन ४५ दिन भित्र पेश गर्न तपाईंको संयोजकत्वमा तपशिल बमोजिमको समिति गठन गरिएको व्यहोरा मिति २०८१।५।५ को कार्यकारी प्रमुखस्तर निर्णयानुसार अनुरोध छ ।

तपशिल

सहसचिव (प्रा) श्री अर्जुन कुमार बम (NDRRMA)	- संयोजक
प्रतिनिधि श्री उर्जा जलस्रोत तथा सिंचाई मन्त्रालय	-सदस्य
प्रतिनिधि श्री कृषि तथा पशुपंक्षी मन्त्रालय	-सदस्य
प्रतिनिधि श्री शहरी विकास मन्त्रालय	-सदस्य
प्रतिनिधि श्री वन तथा वातावरण मन्त्रालय	-सदस्य
प्रतिनिधि श्री भौतिक पूर्वाधार विकास मन्त्रालय सुदुर पश्चिम प्रदेश	- सदस्य
इन्जिनियर श्री स्वाती आचार्य (NDRRMA)	-सदस्य सचिव

आमन्त्रित सदस्य

प्रदिप खतिवडा	youth Innovation
प्रताप महर्जन	Mercy Corps

बोधार्थ

श्री उर्जा जलस्रोत तथा सिंचाई मन्त्रालय	प्रतिनिधि तोकि पठाईदिनुहुन ।
श्री कृषि तथा पशुपंक्षी मन्त्रालय	प्रतिनिधि तोकि पठाईदिनुहुन ।
श्री शहरी विकास मन्त्रालय	प्रतिनिधि तोकि पठाईदिनुहुन ।
श्री वन तथा वातावरण मन्त्रालय	प्रतिनिधि तोकि पठाईदिनुहुन ।
श्री भौतिक पूर्वाधार विकास मन्त्रालय सुदुर पश्चिम प्रदेश	प्रतिनिधि तोकि पठाईदिनुहुन ।
इन्जिनियर श्री स्वाती आचार्य (NDRRMA)	
श्री प्रदिप खतिवडा	youth Inovation
श्री प्रताप महर्जन	Mercy Corps

गंगा सुवेदी भट्टराई
शाखा अधिकृत

फोन: ०१-४२११११६, ४२११११२, ४२११४८३, टोल फ्रि: ११४९, इमेल : info@bipad.gov.np, ndrrma@gmail.com, URL: www.bipad.gov.np

Annex-IV:
Human Casualties Details from District Administration Office, Kanchanpur

आज मिति २०८१ साल श्रावण ०३ गते बिहान ८ बजे जिल्ला विपद व्यवस्थापन समितिको बैठक समितिका संयोजक एवं प्रमुख जिल्ला अधिकारी श्री नारायण प्रसाद सापकोटाज्यूको अध्यक्षतामा गृह मन्त्रालय, राष्ट्रिय विपद् जोखिम न्यूनीकरण तथा व्यवस्थापन प्राधिकरणका कार्यकारी प्रमुख श्री अनिल पोखरेलज्यू, गृह मन्त्रालय विपद् तथा द्वन्द्व व्यवस्थापन महाशाखा प्रमुख सहसचिव डा.भीष्मकुमार भुषालज्यूको बिशेष उपस्थिति एवं स्थानीय तहका प्रमुखज्यूहरु तथा देहाय बमोजिमका महानुभावहरुको उपस्थितिमा यस जिल्लाको बाढीको प्रभाव र बाढी प्रभावित आमनागरिकहरुको बाढीबाट सृजित समस्याको समाधानार्थ छलफल गरी देहाय बमोजिम निर्णय गरियो।

बिशेष उपस्थिति:

क. श्री अनिल पोखरेल, कार्यकारी प्रमुख, राष्ट्रिय विपद् जोखिम न्यूनीकरण तथा व्यवस्थापन प्राधिकरण

ख. डा. भीष्मकुमार भुषाल, सह-सचिव, गृह मन्त्रालय, विपद् तथा द्वन्द्व व्यवस्थापन महाशाखा, सिंहदरबार,काठमाडौं

उपस्थिति:

सि.नं	पदाधिकारी	कार्यालय	नाम थर	हस्ताक्षर
१.	प्र.जि.अ.	जिल्ला प्रशासन कार्यालय कञ्चनपुर	श्री नारायण प्रसाद सापकोटा	
२.	नगर प्रमुख	भीमदत्त नगरपालिका, कञ्चनपुर	श्री पदम बोगटी	
३.	नगरप्रमुख	बेदकोट नगरपालिका, कञ्चनपुर	श्री भोजराज बोहरा	
४.	नगरप्रमुख	पुनर्बास नगरपालिका, कञ्चनपुर	श्री तोयाप्रसाद शर्मा	
५.	नगरप्रमुख	कृष्णपुर नगरपालिका, कञ्चनपुर	श्री हेमराज ओझा	
६.	नगरप्रमुख	शुक्लाफाँटा नगरपालिका, कञ्चनपुर	श्री रण बहादुर महरा	
७.	नगरप्रमुख	दोधारा चाँदनी नगरपालिका, कञ्चनपुर	श्री किशोर कुमार लिम्बु	
८.का.वा.	नगर प्रमुख	बेलौरी नगरपालिका	श्री पोतीलाल चौधरी जोगा राम-चौधरी	
९.	अध्यक्ष	बेलडाँडी गाउँपालिका, कञ्चनपुर	श्री हरिश चन्द्र राना	
१०.	अध्यक्ष	लालझाडी गाउँपालिका, कञ्चनपुर	श्री निर्मल राना	
११.	उप-प्रमुख	जिल्ला समन्वय समिति	श्री रामकुमारी राना	

१२.	प्रहरी उपरीक्षक	जिल्ला प्रहरी कार्यालय कञ्चनपुर	श्री चक्रराज जोशी	
१३.	स.प्र.उपरीक्षक	स.प्र.ब.३५ नं. गण शैलेश्वरी	श्री खगेन्द्र बहादुर चन्द	
१४.	उ.अ.निर्देशक	रा.अ.जि.का. कञ्चनपुर	श्री नरेश बहादुर बम	
१५.	सेनानी	श्रीदल गुल्म, भगतपुर ब्यारेक	श्री राजिव जंग ठकुरी	
१६.	प्र.प्र.अधिकृत	भीमदत्त नगरपालिका, कञ्चनपुर	श्री महेश बहादुर बम	
१७.	प्र.प्र.अधिकृत	बेदकोट नगरपालिका, कञ्चनपुर	श्री पदमराज भट्ट	
१८.	प्र.प्र.अधिकृत	कृष्णपुर नगरपालिका, कञ्चनपुर	श्री खगेन्द्र प्रसाद भट्ट	
१९.	प्र.प्र.अधिकृत	शुक्लाफाँटा नगरपालिका, कञ्चनपुर	श्री खेमराज विष्ट	
२०.	प्र.प्र.अधिकृत	पुनर्वास नगरपालिका, कञ्चनपुर	श्री लक्ष्मी दत्त भट्ट	
२१.	प्र.प्र.अधिकृत	दोधारा चाँदनी नगरपालिका, कञ्चनपुर	श्री गणेश दत्त मिश्र	
२२.	प्र.प्र.अधिकृत	बेलौरी नगरपालिका,कञ्चनपुर	श्री तारादत्त अवस्थी	
२३.	प्र.प्र.अधिकृत	बेलडाँडी गाउँपालिका, कञ्चनपुर	श्री किर बहादुर शेर	
२४.	प्र.प्र.अधिकृत	लालझाडी गाउँपालिका, कञ्चनपुर	श्री राज बहादुर सार्की	
२५.	अध्यक्ष	नेपाल रेडक्रस सोसाईटी	श्री टेकराज उप्रेती	
२६.	निर्देशक	NNSWA	श्री अशोक कुमार जैरु	
२७.	का.संयोजक	निड्स नेपाल, कञ्चनपुर	श्री इश्वर उपाध्याय	
२८.	का.संयोजक	नेपाल रेडक्रस सोसाईटी	श्री विनायक पौडेल	
२९.	[illegible]	[illegible]	[illegible] राज भट्ट	
३०.	विषय सम्पर्क व्यक्ति	निड्स नेपाल	किरन [illegible]	
२५		,,	[illegible]	

३२.	इन्जिनियर	महाकाली बेसिन कार्यालय	श्री संजय कु. शाह
३३.	फिल्ड फोकल	भीमदत्त नगरपालिका कञ्चनपुर	श्री मनोज प्रसाद भट्ट
३४.	का.अधिकृत	एन.एन.एस.डब्लु.ए.	श्री प्रभुनिधि पन्त
३५.	वा. अध्यक्ष ५नं.	पुर्नवास नगरपालिका	
३६.	सचिव	नेपाल पत्रकार महासंघ	राजेन्द्र [illegible]
३७.	अध्यक्ष	" " "	भवानी भट्ट
३८.	[illegible]	[illegible]	[illegible]
३९.	[illegible]	[illegible]	[illegible]

निर्णय नं. १

मिति २०८१/०३/२३ र २४ गतेको भारी वर्षाको कारण जिल्लाका विभिन्न स्थानमा भएको घर क्षति, कृषि बाली क्षति, उद्योग व्यवसाय क्षति, घर बासको पूर्ण र आंशिक क्षतिको विवरण सबै स्थानीय तहहरुबाट माग गरी क्षतिपूर्ति उपलब्ध गराई दिनका लागि तालुक गृह मन्त्रालय मार्फत नेपाल सरकारमा अनुरोध गर्ने।

निर्णय नं. २

भौतिक पूर्वाधारहरुको क्षतिको विवरण र क्षति भएको मूल्याङ्कित रकम सम्बन्धित विषयगत कार्यालयहरुबाट माग गरी सम्बन्धित निकायमा पठाई क्षतिपूर्ति उपलब्ध गराई दिन अनुरोध गर्ने। साथै क्षति भएको भौतिक पूर्वाधार (सडक, नहर लगायत) तत्काल मर्मत गराई सञ्चालन गरिएकोमा मनसुन अवधिभर सञ्चालन गराई मनसुन पश्चात पक्की संरचना निर्माण गर्न सम्बन्धित सडक डिभिजन कार्यालय कञ्चनपुर, महाकाली सिंचाई व्यवस्थापन कार्यालय, कञ्चनपुर र महाकाली सिंचाई आयोजना तेस्रो चरणलाई लगाउने।

निर्णय नं. ३

मिति २०८१/०३/२३ र २४ को भारी वर्षाको कारण आएको बाढी र डुबानमा परी मृत्यु हुने कृष्णपुर नगरपालिका वडा नं. १ पहरिया बस्ने सागर धानुकको छोरा वर्ष सात महिनाका सोनु धानुक, बेलडाँडी गाउँपालिका वडा नं. ३ बस्ने वर्ष २३ का डम्मर चौधरी, पुनर्वास नगरपालिका वडा नं. ५ नयाँ कटान बस्ने वर्ष १५ का लक्ष्मी बोगटी, पुनर्वास नगरपालिका वडा नं. ५ नयाँ कटान बस्ने वर्ष १५ का सृजना बोगटी, पुनर्वास नगरपालिका वडा नं. ५ नयाँ कटान बस्ने वर्ष १६ का आरती बोगटीको चीर शान्तिको कामना गर्दै मृतकहरु प्रति भावपूर्ण श्रद्धाञ्जली व्यक्त गर्दछौं। यस शोक सन्तप्त दुखद: घडीमा मृतकका शोकाकुल परिवारहरुलाई धैर्यधारण गर्ने शक्ति मिलोस भनी ईश्वरसँग प्रार्थना गर्दछौं।

नारायण प्रसाद सापकोटा
प्रमुख जिल्ला अधिकारी
तथा
अध्यक्ष
जिल्ला विपद् व्यवस्थापन
समिति, कञ्चनपुर

नारायण प्रसाद सापकोटा
प्रमुख जिल्ला अधिकारी

नेपाल सरकार
गृह मन्त्रालय

जिल्ला प्रशासन कार्यालय

कञ्चनपुर

फोन नं. : ५२११०९
५२११७८
५२१११०
फ्याक्स नं. :५२४९९०

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प.सं. २०८०/०८१
च.नं. 6633

मिति २०८१।०३।२३
ने.सं. ११४४ दिल्लाथ्व २, आईतबार

विषयः- जाहेरी ।

श्री गृह मन्त्रालय,
शान्ति सुरक्षा तथा अपराध नियन्त्रण शाखा
सिंहदरबार, काठमाडौं ।

१. मिति २०८१।०३।२२ गते बेलुकी ४:०० बजे देखि निरन्तर रुपमा वर्षा भैरहेकोमा मिति २०८१/०३/२३ गते बेलुकी १८:०० बजेको महाकाली नदीमा पानीको बहाव २,१२,५६० क्यु. सेक. रहेको र पानीको बहाव बढीरहेको ।
२. जिल्ला कञ्चनपुर बेल्डाँडी गाउँपालिका ३ बैबाह बस्ने वर्ष २१ को डम्मर राना चौधर नदीमा माछा मार्ने क्रममा अ. १६:०० बजे बगेर बेपत्ता भएको ।खोज तलाश जारी रहेको ।
३. प्रहरी चौकी झोलेझे कार्यालय भवनमा पानी पसेपछि उक्त कार्यालय महाकाली पुलको टहरामा स्थानान्तरण गरिएको ।
४. जिल्लाको शान्ति सुरक्षाको अवस्था समान्य रहेको व्यहोरा सादर जाहेर ।

२०८१।०३।२३
धर्मराज जोशी
सहायक प्रमुख जिल्ला अधिकारी

बोधार्थ
श्री आन्तरिक मामिला तथा कानून मन्त्रालय,
धनगढी, कैलाली।

नेपाल सरकार

गृह मन्त्रालय

जिल्ला प्रशासन कार्यालय

कञ्चनपुर

फोन नं. : ५२११०१
५२११७८
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फ्याक्स नं. :५२४९९०

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प.सं. ०८०/०८१
च.नं. ६८३६

मिति: २०८१।०३।२४
ने.सं. ११४४ दिल्लाथ्व तृतिया

विषय: जाहेरी ।

श्री गृह मन्त्रालय,
शान्ति सुरक्षा तथा अपराध नियन्त्रण शाखा
सिंहदरबार, काठमाण्डौं।

१. हिजो विहान ४ बजे देखी शुरू भएर अहिलेसम्म जिल्लाभर निरन्तर अविरल वर्षा भैरहेको ।
२. आज विहान १:०० बजे कृष्णपुर न.पा. मोतिपुरमा घरभित्र पलङबाट पानीमा खस्दा ७ महिनाको बच्चाको मृत्यु भएको जानकारी प्राप्त भएको ।
३. हाल(विहान ६:०० बजे)सम्म प्राप्त जानकारी अनुसार बाढी/डुवानका कारण यस जिल्लामा ५११ घर/परिवारबाट १७३८ जनालाई सुरक्षित स्थानमा स्थानान्तरण गराइएको ।
४. रातिको समयमा नेपाल प्रहरी र सशस्त्र प्रहरीको संयुक्त टोलीले मोटरबोट प्रयोग गरी डुब्ने लागेको अवस्थामा रहेका २९ जना व्यक्तिलाई जिल्लाको विभिन्न स्थानहरूबाट उद्धार गरिएको ।
५. पूर्व-पश्चिम राजमार्ग चेनेज १०१६+६०८ (वेदकोट न.पा. स्थित सुडा बजार) कल्भर्ट पूर्ण रुपमा क्षति भए पश्चात वैकल्पिक मार्ग मार्फत सवारी आवागमन सुचारु गरिएको । उक्त स्थानमा तत्काल आवश्यक मर्मत सुधार गर्न सडक डिभिजन कार्यालय, कञ्चनपुरलाई निर्देशन गरिएको ।
६. आज विहान ६:०० बजे महाकाली नदीमा पानीको वहाव ३ लाख ५० हजार क्यू.सेक रहेकोमा अहिले ९:०० बजे २ लाख ५५ हजार ३८० क्यू.सेक. रहेको । महाकाली पुल परियोजनाका अनुसार ५ लाख ४७ हजार क्यू. सेक थान्न सक्ने महाकाली बाँध र पुलको Capability रहेको ।

थप जाहेरी निरन्तर हुनेछ ।

२०८१।०३।२४

नारायण प्रसाद सापकोटा
प्रमुख जिल्ला अधिकारी

सादर अवगतार्थ:

श्रीमान् सचिव ज्यू,
गृह मन्त्रालय, सिंहदरबार, काठमाडौं ।

बोधार्थ

श्री विपद् व्यवस्थापन महाशाखा, सिंहदरबार, काठमाडौं
श्री आन्तरिक मामिला तथा कानून मन्त्रालय, धनगढी, कैलाली ।
श्री राष्ट्रिय आपतकालिन कार्य सञ्चालन केन्द्र, काठमाडौं ।

नेपाल सरकार
गृह मन्त्रालय

जिल्ला प्रशासन कार्यालय

कञ्चनपुर

फोन नं. : ५२११०९,
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फ्याक्स नं. :५२४९९०

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प.सं. ०८०/०८१
च.नं. ६६४९

मिति: २०८१।०३।२४
ने.सं. ११४४ दिल्लाथ्व तृतिया

विषय: थप जाहेरी।

श्री गृह मन्त्रालय,
शान्ति सुरक्षा तथा अपराध नियन्त्रण शाखा
सिंहदरबार, काठमाण्डौं।

१. बेलडाँडी गाउँपालिका वडा नं. ५ इमलिया बस्तीमा चौधर नदीको वहाव बढेकाले बेलडाँडी झिलमिला नहरको तटबन्ध फुटाई नदी गाउँतर्फ पसेकोमा बस्तीमा रहेका स्थानीय वासिन्दाहरुलाई सुरक्षित स्थानमा स्थानान्तरण गरिएको। बेलडाँडी खैरीघाटबाट ३ जना सहित थप २२ जनाको सशस्त्र प्रहरी टोलीबाट उद्धार गरी सुरक्षित स्थानमा पुर्‍याइएको। अहिले उक्त स्थानमा पानीको सतह घट्दै गएकोले जनजीवन सामान्य अवस्थामा आएकोले खतरामुक्त रहेको।
२. जिल्लामा ४२३५ जना व्यक्तिलाई सुरक्षित स्थानमा स्थानान्तरण गराइएकोमा हाल अधिकांश व्यक्ति आफ्नो वासस्थानमा फर्किने क्रममा रहेको।
३. पुनर्वास नगरपालिकाको वडा नं. ५ नयाँ कटानमा बेपत्ता भएका तीनै जना बालिकाको शव फेला परेको।
४. जिल्लामा अधिकांश स्थानीय तहहरु जलमग्न भएकोमा मनसुनी वर्षा क्रमश कम भई दिउसो १:३० बजे रोकिइ हाल जमिनमा पानीको सतह घटी सकेको।
५. महाकाली नदीमा पानीको वहाव १,८५,४५० क्युसेक रहेको। पानीको वेग घटदो क्रममा रहेको।
६. जिल्लाको अन्य अवस्था सामान्य रहेको व्यहोरा सादर जाहेर।

२०८१।०३।२५
नारायण प्रसाद सापकोटा
प्रमुख जिल्ला अधिकारी

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